

國立臺灣大學
National Taiwan University

3D FinFET: Device Architecture, Design, and Characteristics

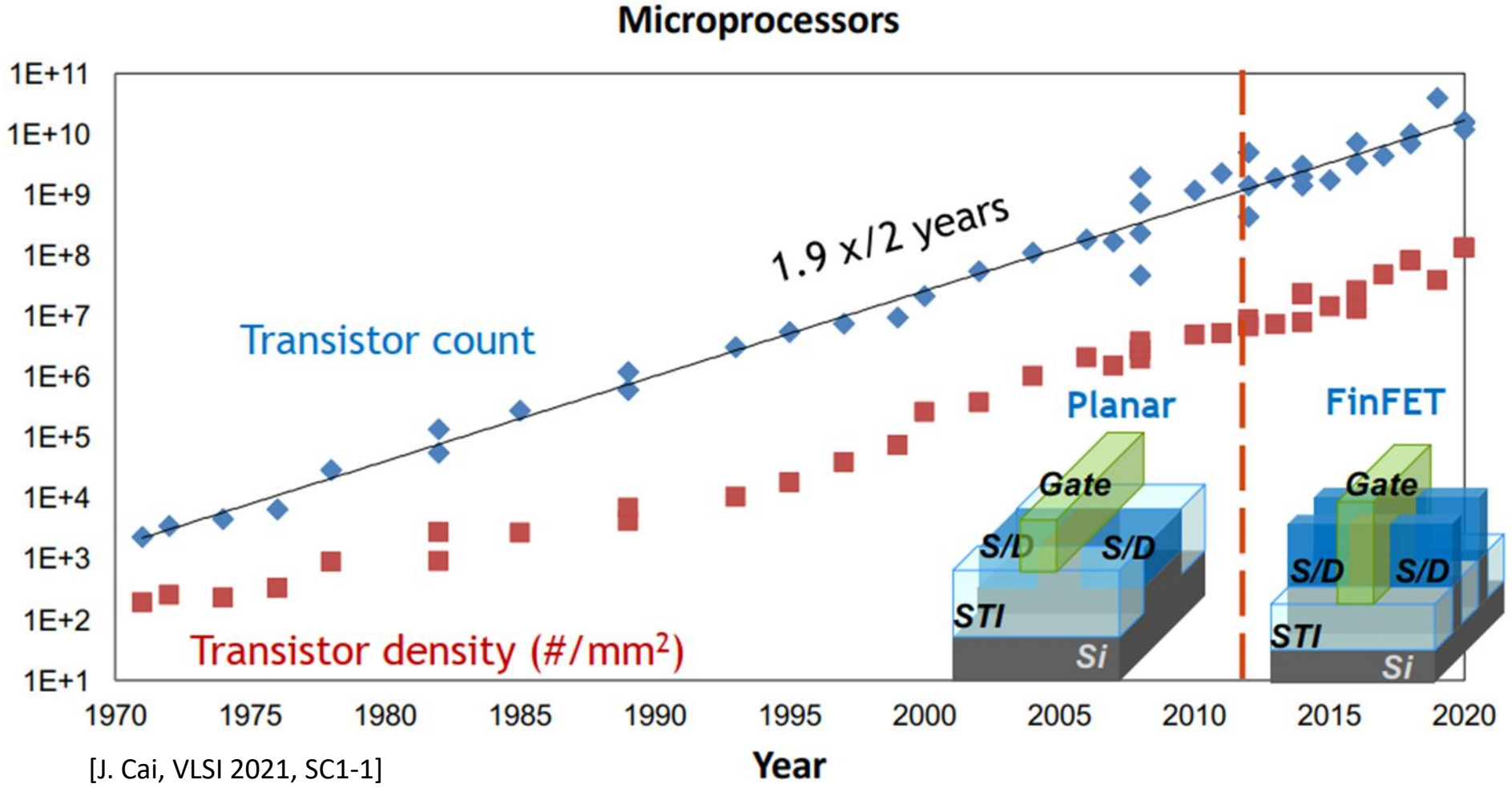
Vita Pi-Ho Hu 胡璧合

Department of Electrical Engineering

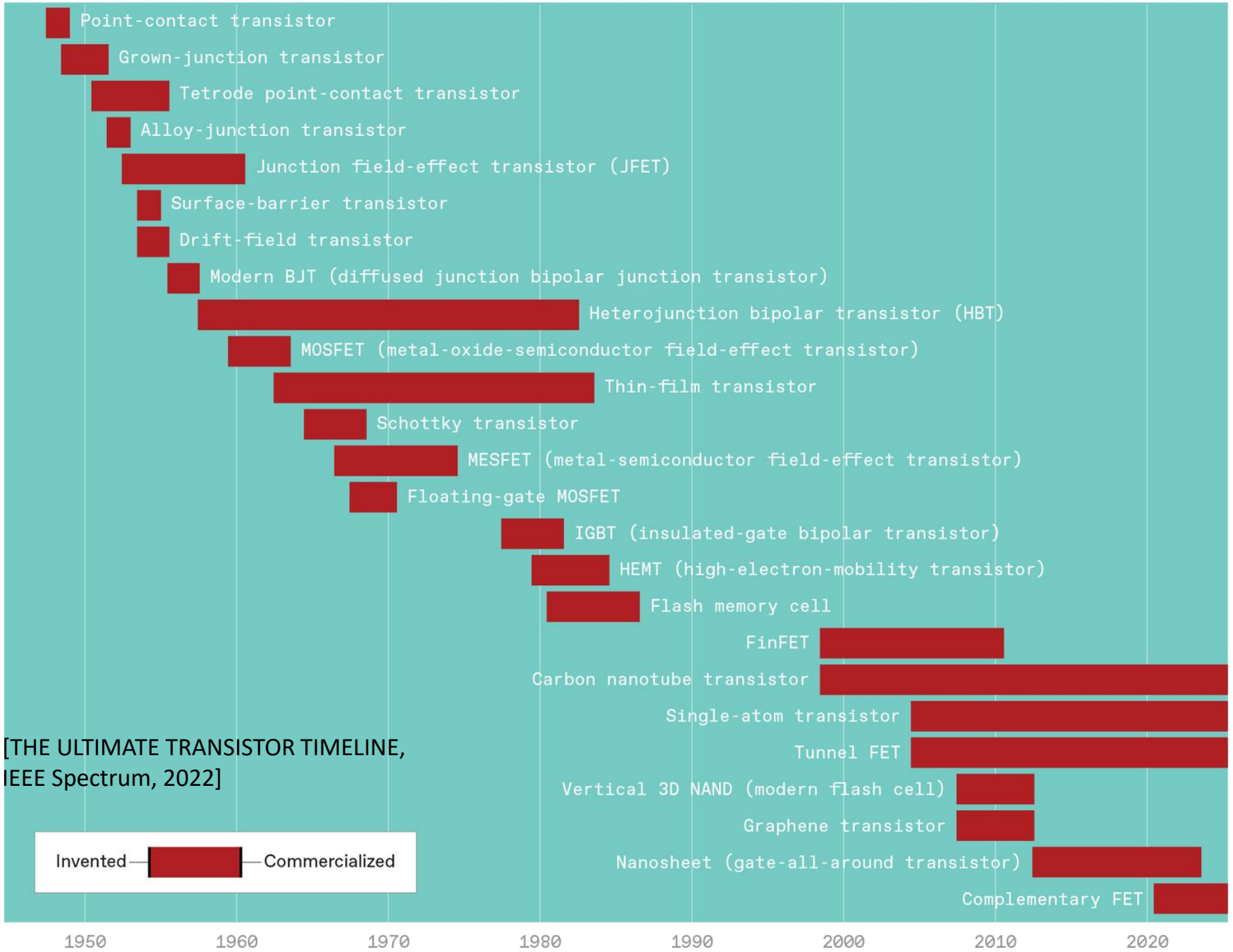
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Moore's Law

- FinFET has been commercially introduced for all leading-edge technologies since 2011.

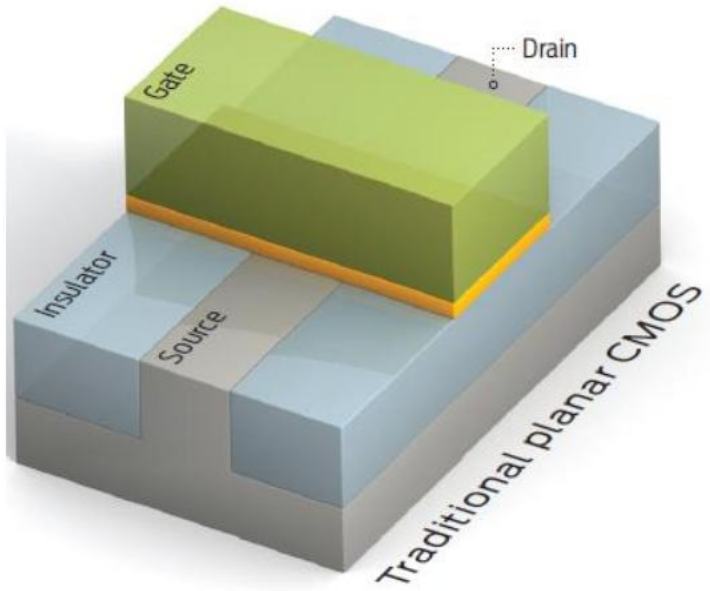


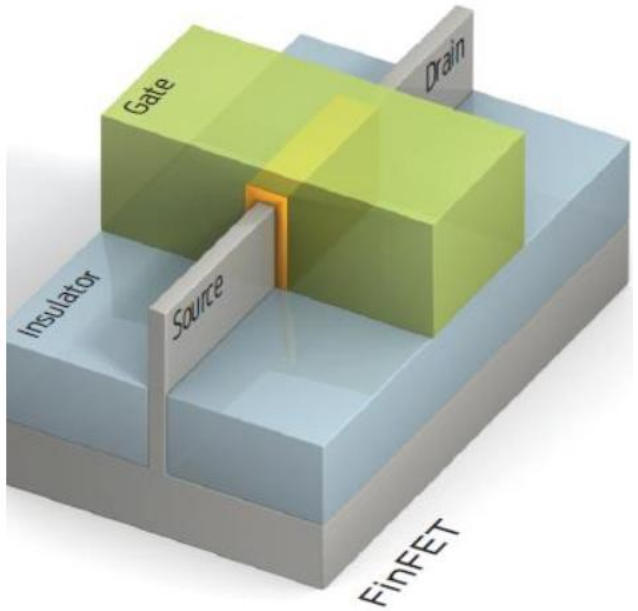
[J. Cai, VLSI 2021, SC1-1]



[THE ULTIMATE TRANSISTOR TIMELINE, IEEE Spectrum, 2022]

FinFET (Fin Field Effect Transistor)





Planar CMOS:

- **Poor gate control**
- **Worse short channel effect (SCE)**
- **Higher Ioff**

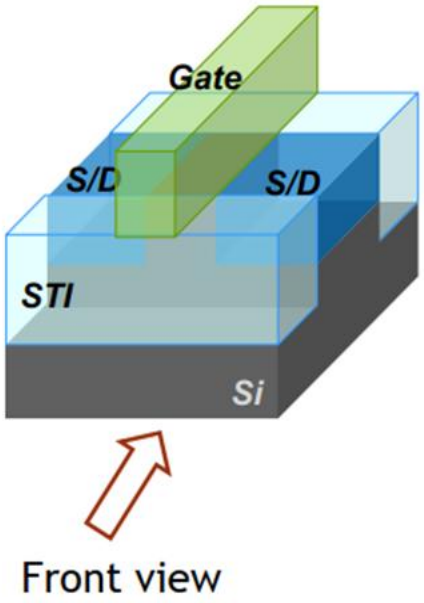
FinFET:

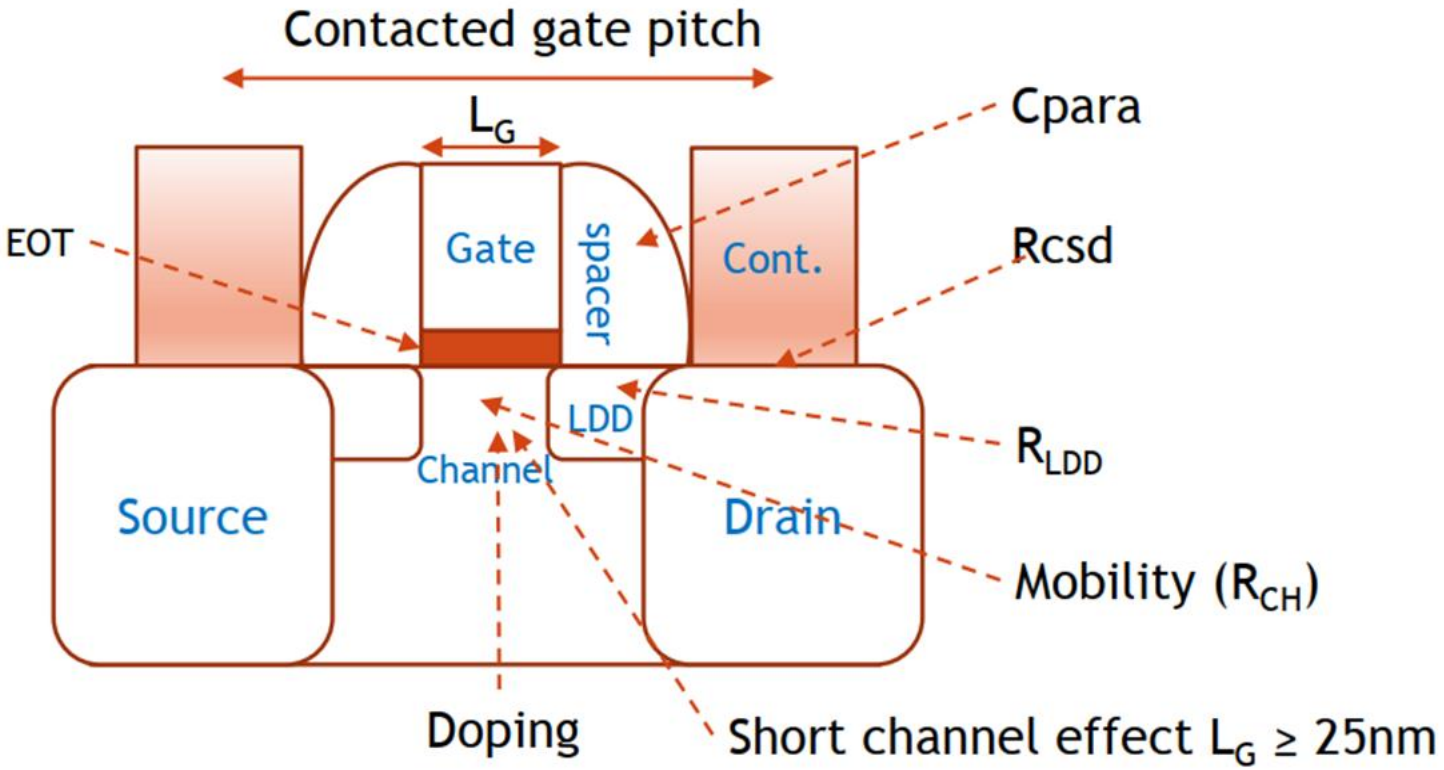
- **Smaller foot-print**
- **Better gate control (less SCE)**
- **Higher Ion and lower Ioff**

[K. Ahmed et al., “Transistor Wars”, IEEE Spectrum, 2011]

Bulk Transistor (Planar MOSFET)

- Contacted gate pitch (CPP): $L_G + 2T_{spacer} + L_{Cont}$

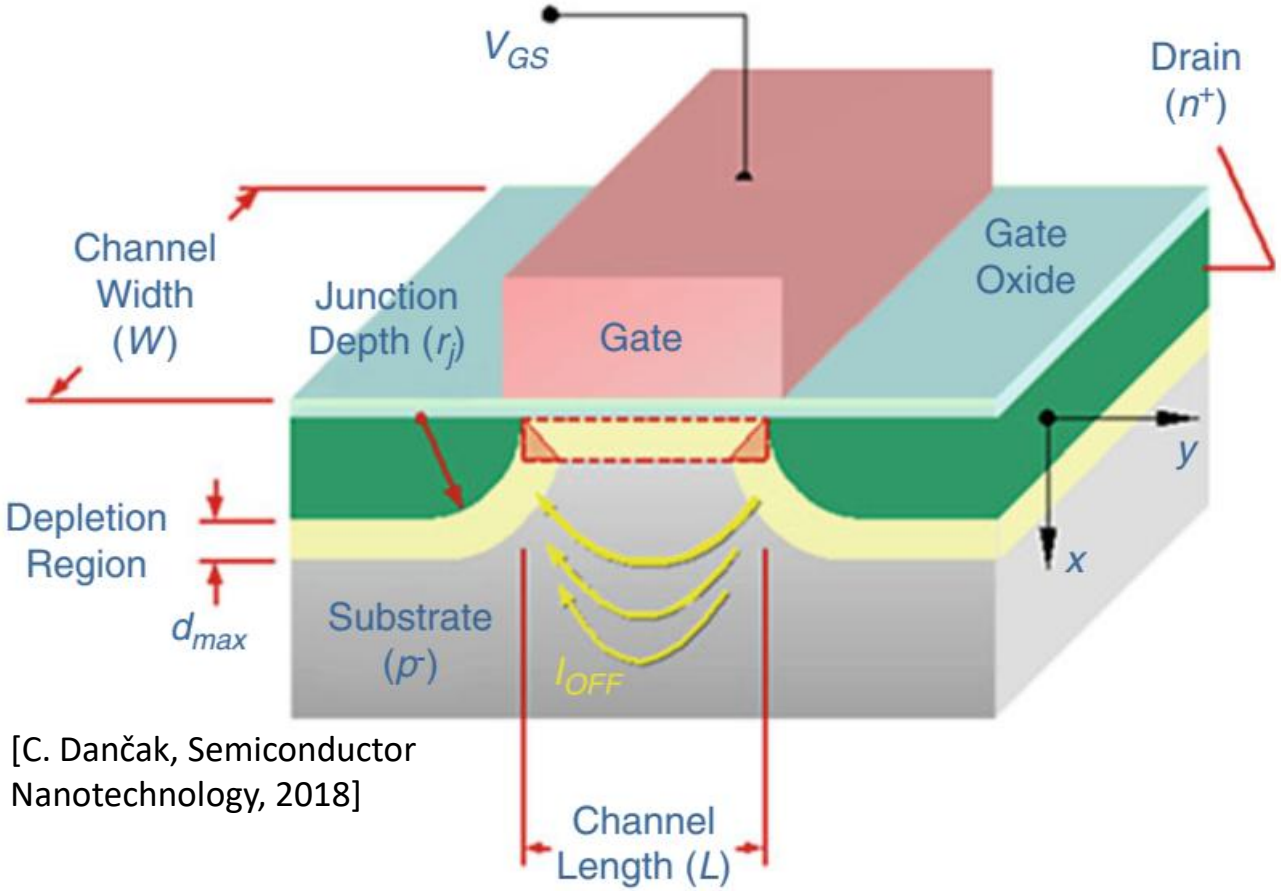




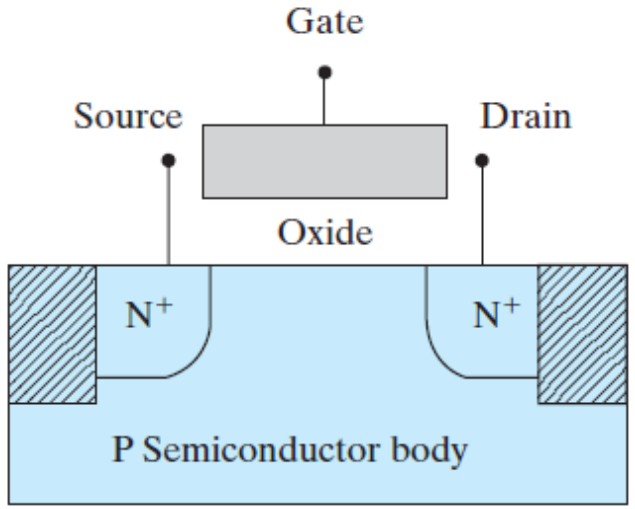
[J. Cai, VLSI 2021, SC1-1]

Short-Channel N-type MOSFET

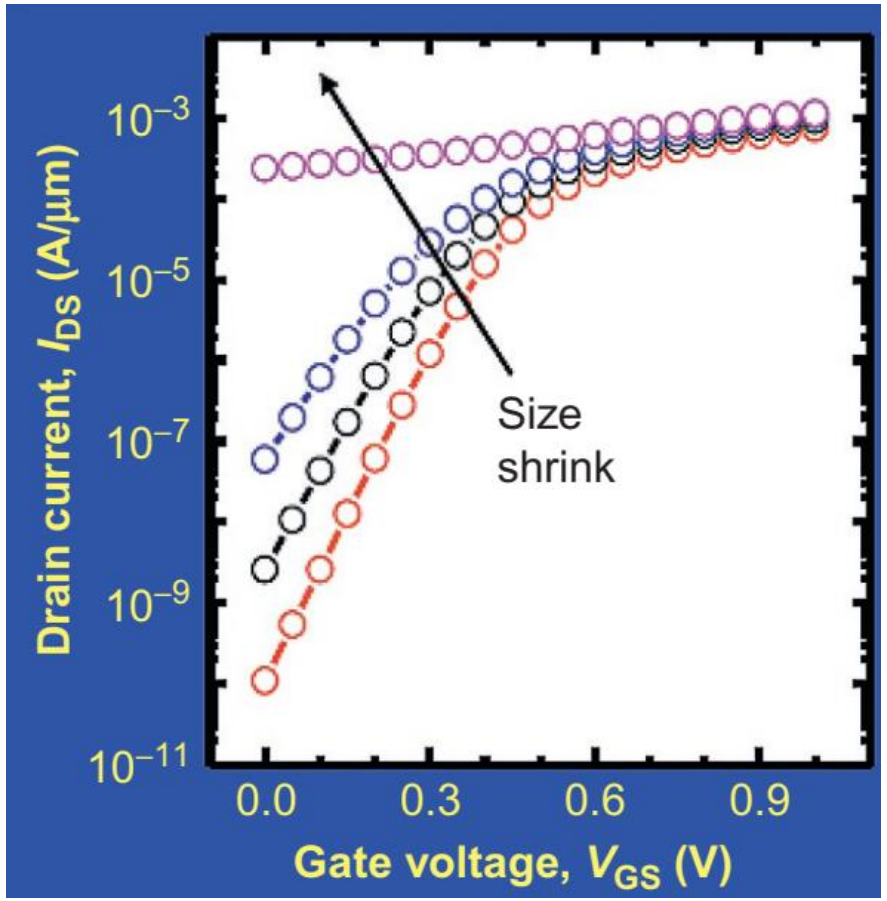
- Planar MOSFET
- The depth of the source-drain islands (r_j) is comparable to the channel length L.



[C. Dančak, Semiconductor Nanotechnology, 2018]



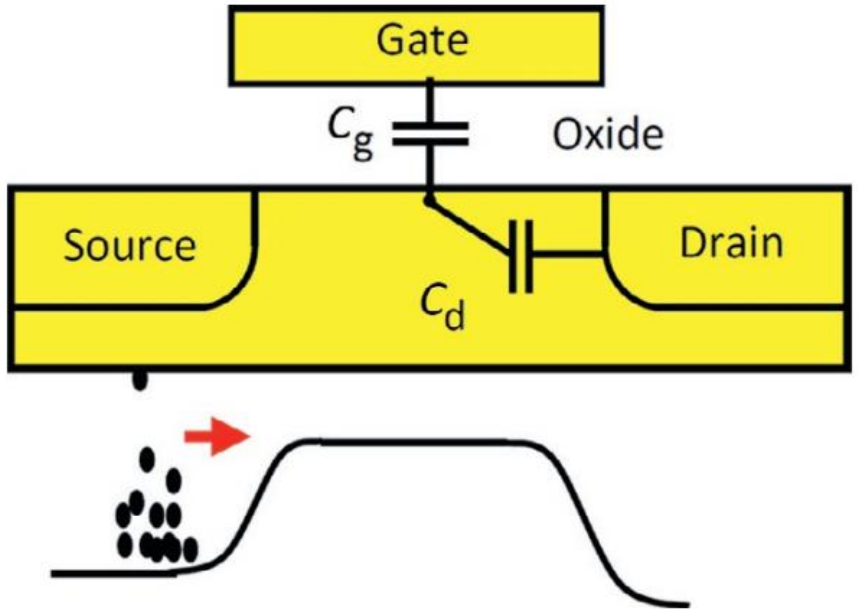
Short-Channel Effects (SCE)



- As the gate length shrinks, the MOSFET's Ids-Vgs characteristics degrade in two major ways.
  - The subthreshold swing (*S*) degrades and *V*t decreases—the device cannot be turned off easily by lowering *V*g.
  - Second, *S* and *V*t become increasingly sensitive to *L*g variations— device variations become more problematic.

[Y. S. Chauhan et al., "FinFET Modeling for IC Simulation and Design", Elsevier 2015]

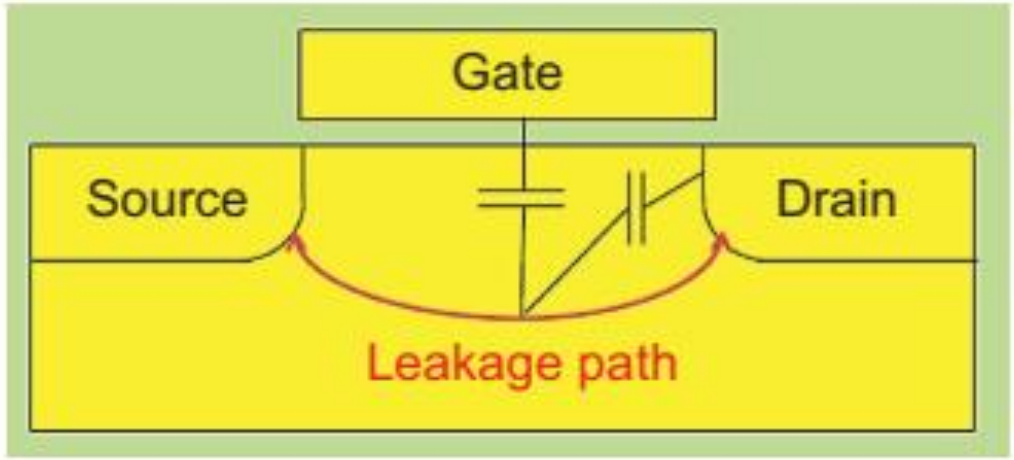
Root Cause of Short-Channel Effects



- In an ideal transistor, the channel potential is controlled only by Vg and Cg. In a real transistor, the channel potential is also subjected to the influence of Vd through Cd.
- With decreasing *L*, rising *C*d allows *V*d to control the channel potential just as *V*g.
- **Scaling the gate oxide thickness** was a good solution for this problem.

[Y. S. Chauhan et al., “FinFET Modeling for IC Simulation and Design”, Elsevier 2015]

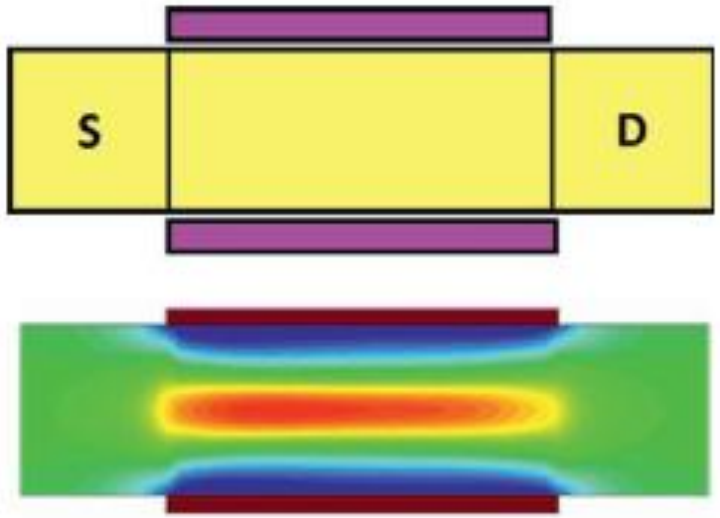
Beyond 20 nm **even a 0 nm thick gate dielectric** cannot stop Vd from lowering the potential barrier along leakage paths a few nanometers below the interface.



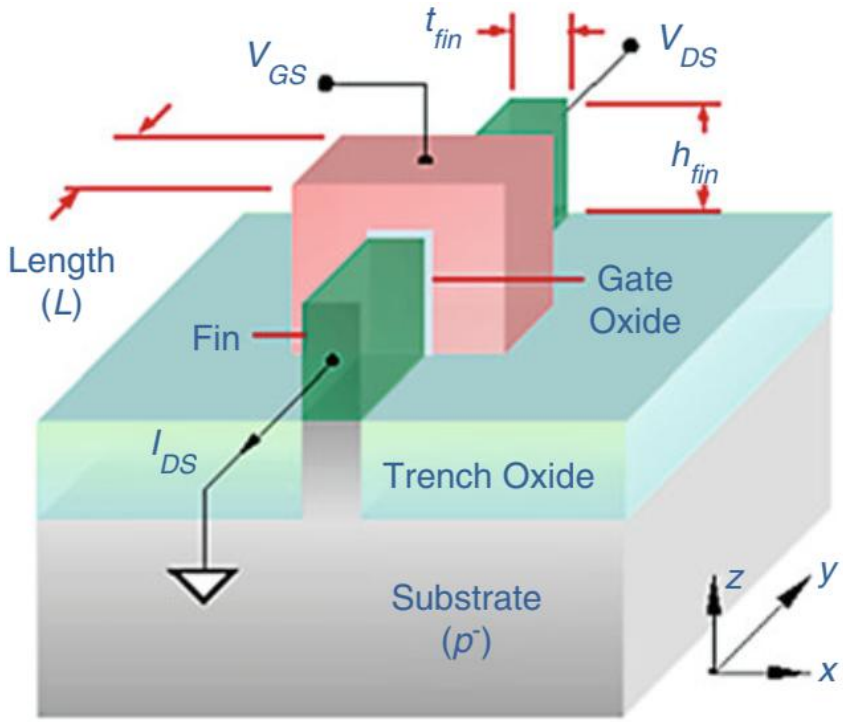
- Leakage paths far from the gate, they are only weakly controlled by *V*g—that is, "*C*g" is small even with zero oxide thickness.
- As a result, the potential barriers along these weakly controlled paths can be easily lowered by *V*d through the large *C*d in a small-*L*g device.

[Y. S. Chauhan et al., "FinFET Modeling for IC Simulation and Design", Elsevier 2015]

Thin-Body MOSFET Concept



[Y. S. Chauhan et al., "FinFET Modeling for IC Simulation and Design", Elsevier 2015]

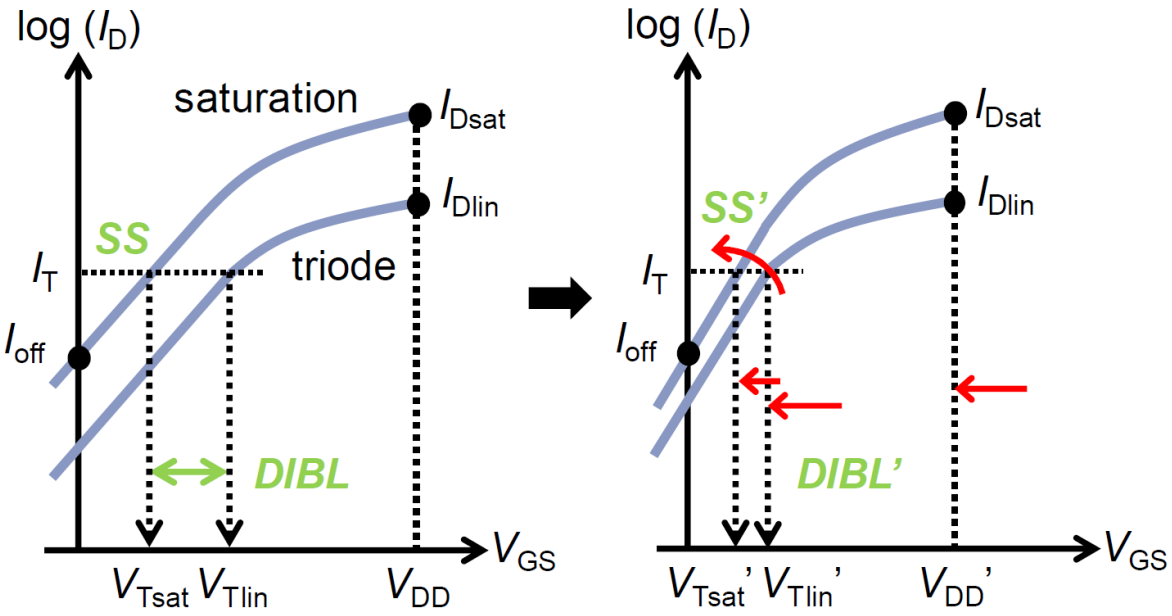


[C. Dančak, Semiconductor Nanotechnology, 2018]

- If the body is thin, any lines drawn between the source and the drain (any potential leakage paths) would not be far from one or the other gate.
- FinFET, tri-gate FET, SOI, Gate-All-Around Nanowire/Nanosheet

FinFETs: Better Short Channel Effect

Lower dynamic power:
smaller VT, and reduced supply V_{DD}



Ref: [3] A. L. S. Loke et al., *2018 IEEE Custom Integrated Circuits Conference (CICC)*, 2018 [4] J.-H. Ju et al., ECS Trans., 60, 745, 2014.

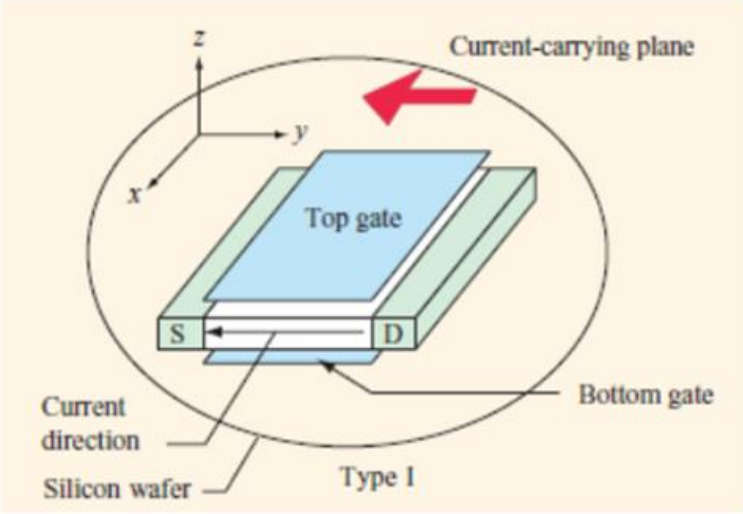
- **Performance comparison**

Vdd=0.9V		Constant Ioff 3nA/um								
		Planar		SegFET				FinFET		FD-SOI
AA width (nm)		1000	1000	70	70	20	20	13	7	1000
AA Height (nm)		0	0	15	15	15	15	40	40	0
AA Pitch (nm)		1000	1000	90	90	40	40	48	48	1000
Lgate (nm)		24	24	24	24	24	24	24	24	24
Doping Process		PKT	SSRW	PKT	3D SSRW	PKT	3D SSRW	Uniform	Uniform	Uniform
Channel doping (/cm^2)		4.5E+18	1E+18	4E+18	1E+18	2.00E+18	1E+18	1.00E+18	1.00E+18	1.00E+18
Vtsat (V)		0.307	0.305	0.263	0.279	0.258	0.255	0.247	0.248	0.284
DIBL (mV/0.85V)		122	85	83	65	68	53	50	48	59
SSsat (mV/Dec)		93	92	79	82	76	76	73	74	85
Idsat (uA/um)		549	605	532	629	706	674.6	764	710	724
Ieff(uA/um)		236	286	237	302	342	346	404	379	361
Ioff (pA/um)		3000	3060	2991	2940	2981	3038	3000	3000	2947
Iboff (pA/um)		13	329	18	371	4.37	361	0	0	0
Cgg (fF/um)		1.04	1.04	1.09	1.07	1.11	1.08	1.06	1.00	1.01
Cov (fFum)		0.256	0.249	0.230	0.217	0.253	0.222	0.237	0.225	0.230
Vtsat_ RDF (mV.um$^{1/2}$)		9.46	6.99	7.39	3.9	6.6	3.6	4.2	3.8	3.16
DeltaVti (mV,Vb=0,-Vdd/2)		21	62	10	41	1	20	2	2	69
Lgate 15nm	Vtsat	0.128	0.099	0.207	0.160	0.189	0.159	0.130	0.117	0.180
	DIBL	257	208	143	138	137	111	124	134	128
	SSsat	156	140	105	123	105	95	129	141	115

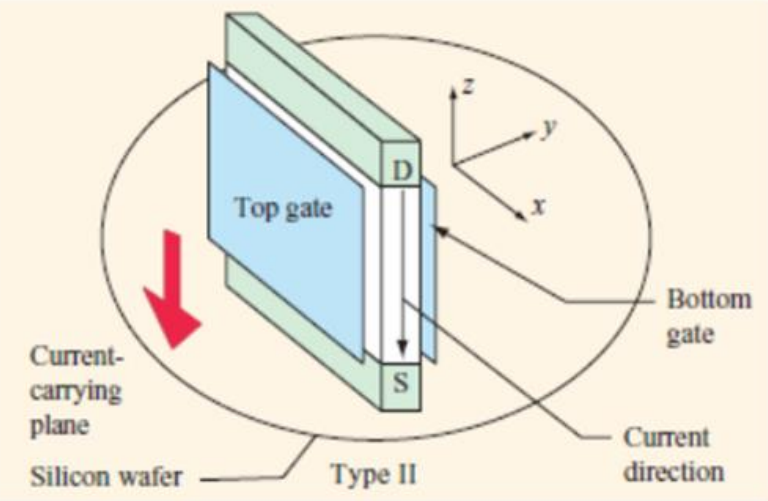
- At same dynamic power, FinFET has 20% less delay.
- At same delay, FinFET has 46% less dynamic power consumption.

Possible Orientations of a Double-Gate MOSFET

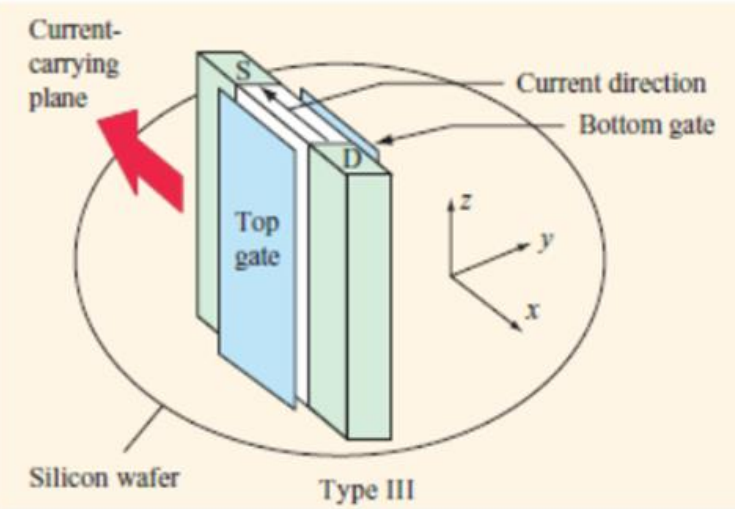
Type I: Planar double gate



Type II: Vertical pillar



Type III: Vertical Delta

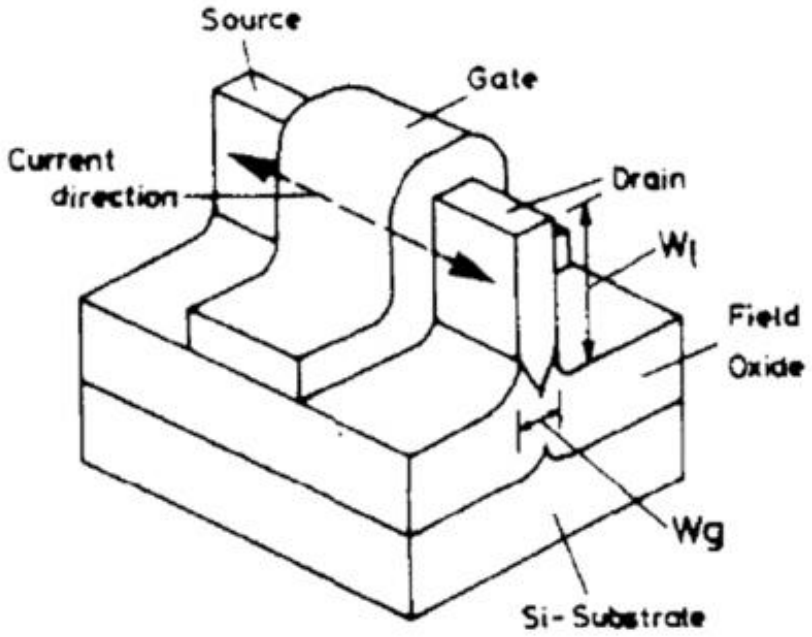


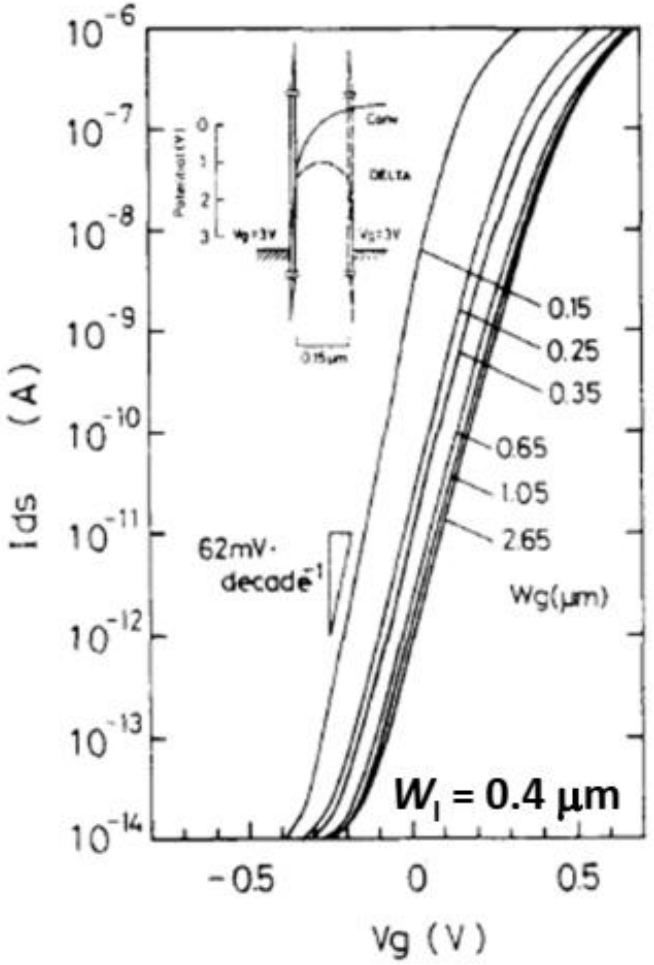
[H.-S. P. Wong et al., Proceedings of IEEEE, vol. 87, no. 4, pp. 537-570, 1999]

- Type I: Planar double gate [J.-P. Colinge et al., IEDM, 1990]→ Nanosheet
- Type II: Vertical pillar [H. Takato et al., IEDM, 1988]→ Vertical CMOS
- Type III: Vertical Delta device [D. Hisamoto et al., IEDM, 1989]→ FinFET

Delta MOSFET

D. Hisamoto, T. Kaga, Y. Kawamoto, and E. Takeda (Hitachi Central Research Laboratory),
"A fully depleted lean-channel transistor (DELTA) – a novel vertical ultrathin SOI MOSFET,"
IEEE Electron Device Letters Vol. 11, pp. 36-39, 1990

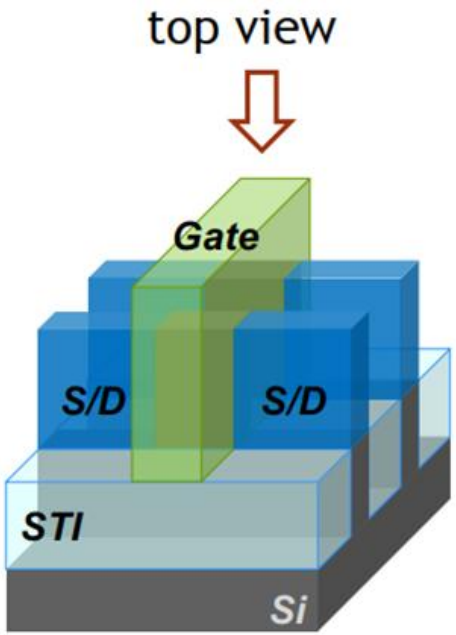




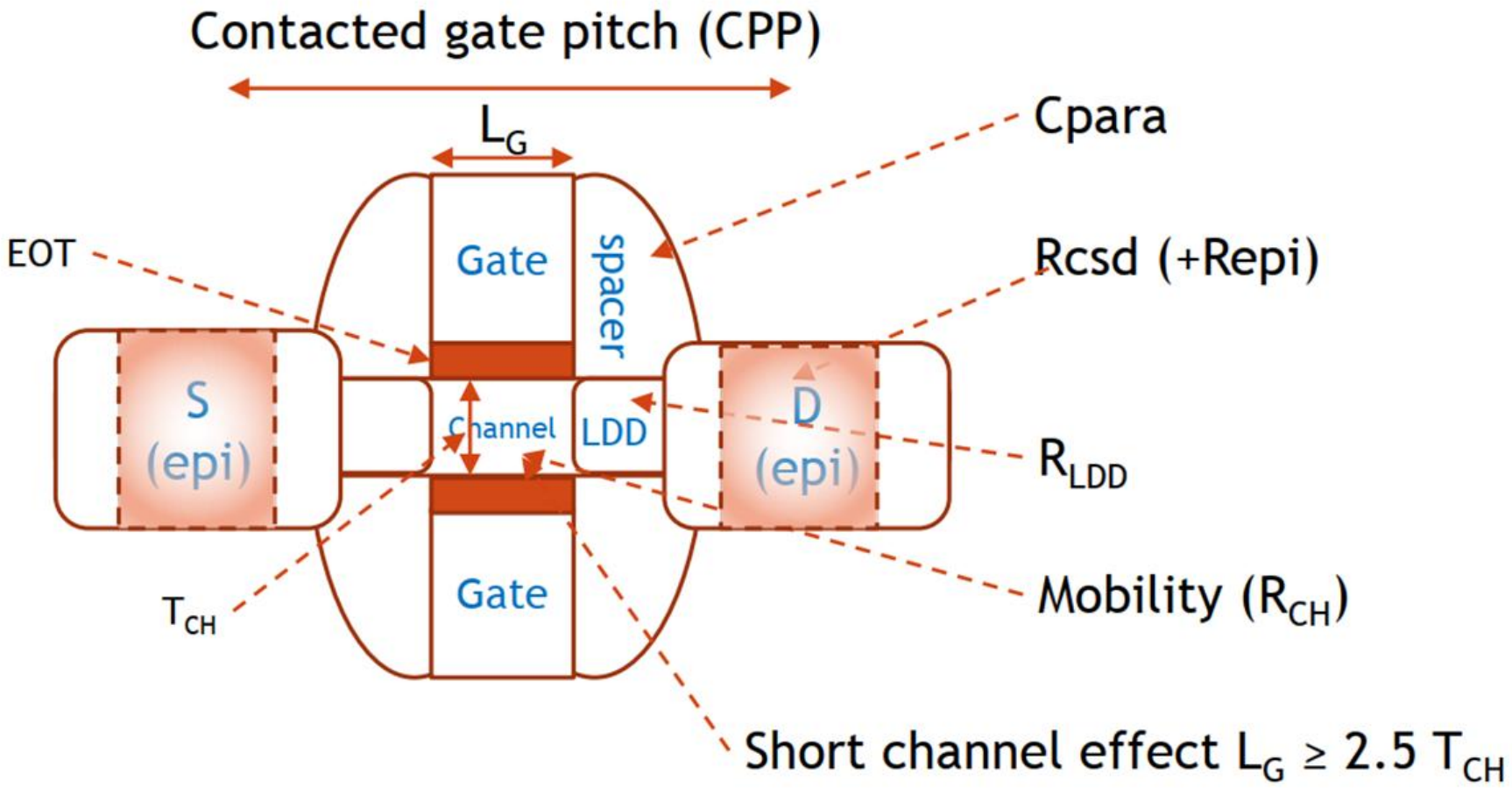
- **Improved gate control observed for W_g < 0.3 μm**
  - **L_{eff}= 0.57 μm**

[T.-J. K. Liu, VLSI, 2012, SC]

3D FinFET



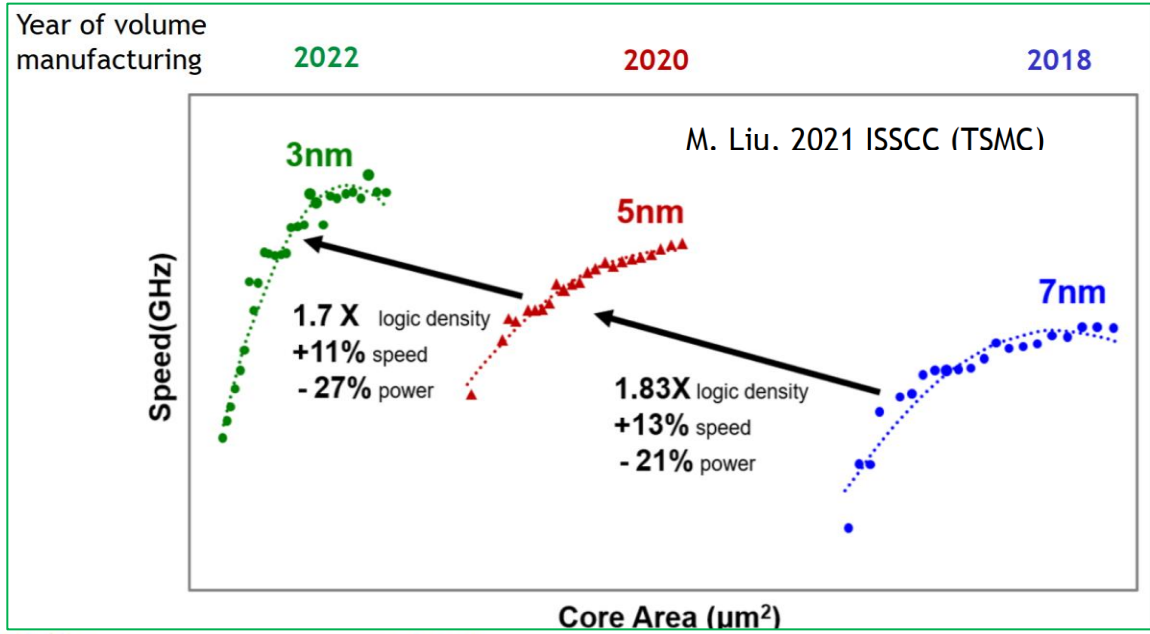
[J. Cai, VLSI 2021, SC1-1]



- **Fin width quantization**: the effective finFET transistor width (Weff) is quantized and can be adjusted higher only by laying out more fins connected in shunt.

FinFET Evolution

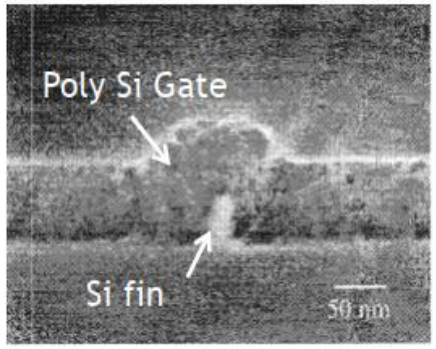
- Increased fin height; reduced fin width; reduced fin pitch
- Tapered fin → Vertical straight fin, rectangular Fin profile improves SCE for gate length scaling



Prototype

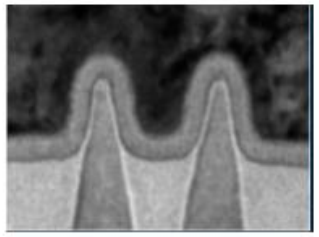
Production

1998



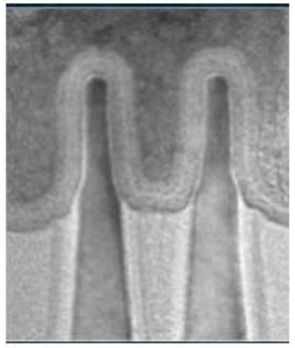
D. Hisamoto et al., 1998 IEDM

2012



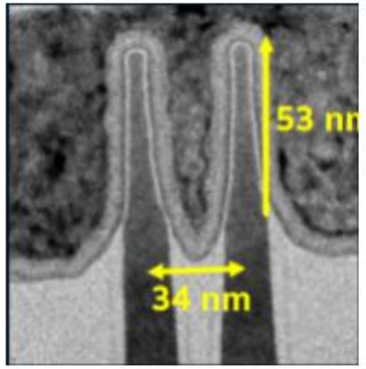
C. Auth et.al., 2012 VLSI (Intel 22nm)

2014



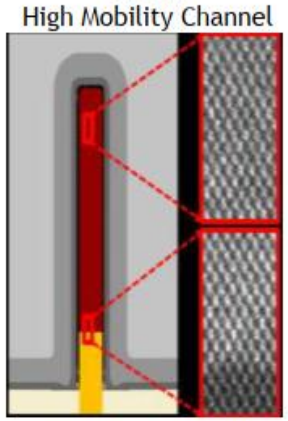
S. Natarajan et.al., 2014 IEDM (Intel 14nm)

2018



K. Mistry et.al., 2017 IEDM (Intel 10nm)

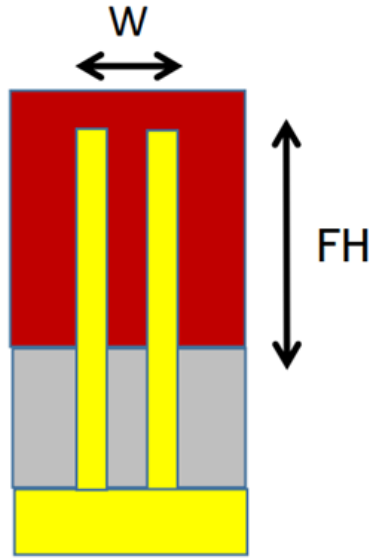
2019



G. Yeap et.al., 2019 IEDM (TSMC 5nm)

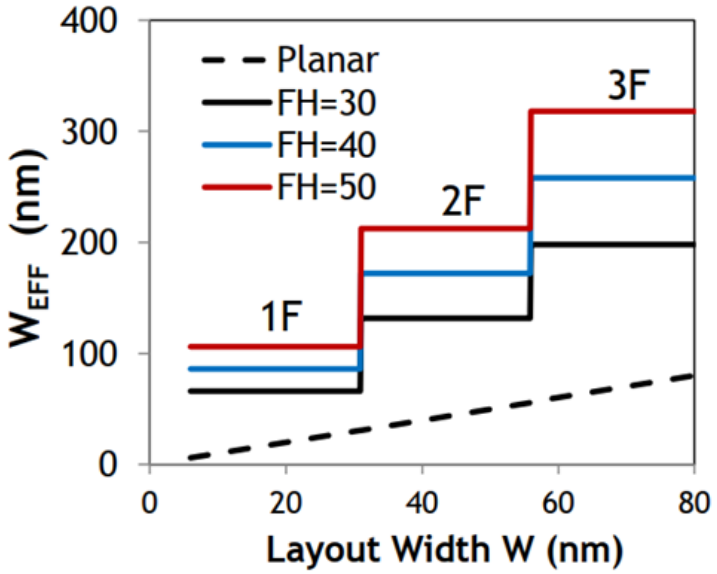
[J. Cai, VLSI 2021, SC1-1]

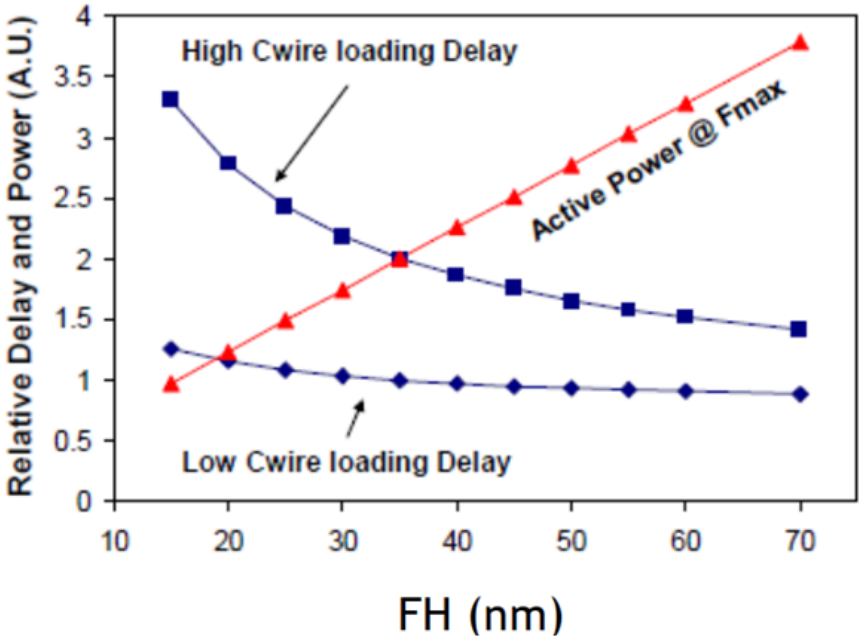
Fin Height Design



2 Fins (2F)

[J. Cai, VLSI 2021, SC1-1]

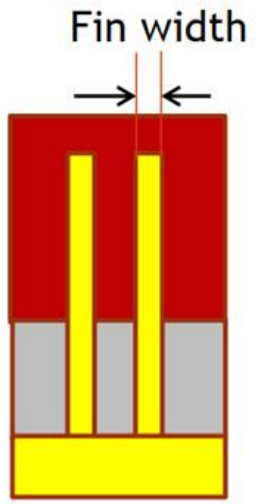




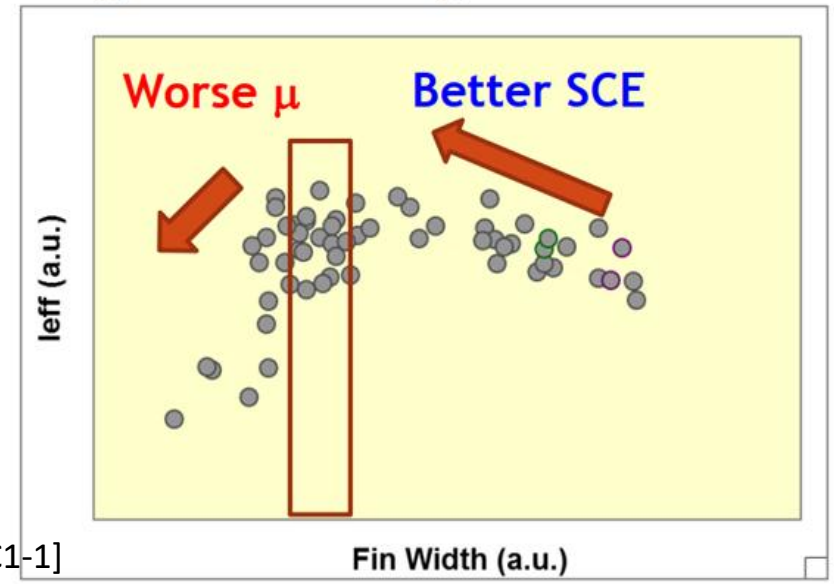
[C.-H. Lin et al., IEDM, 2014]

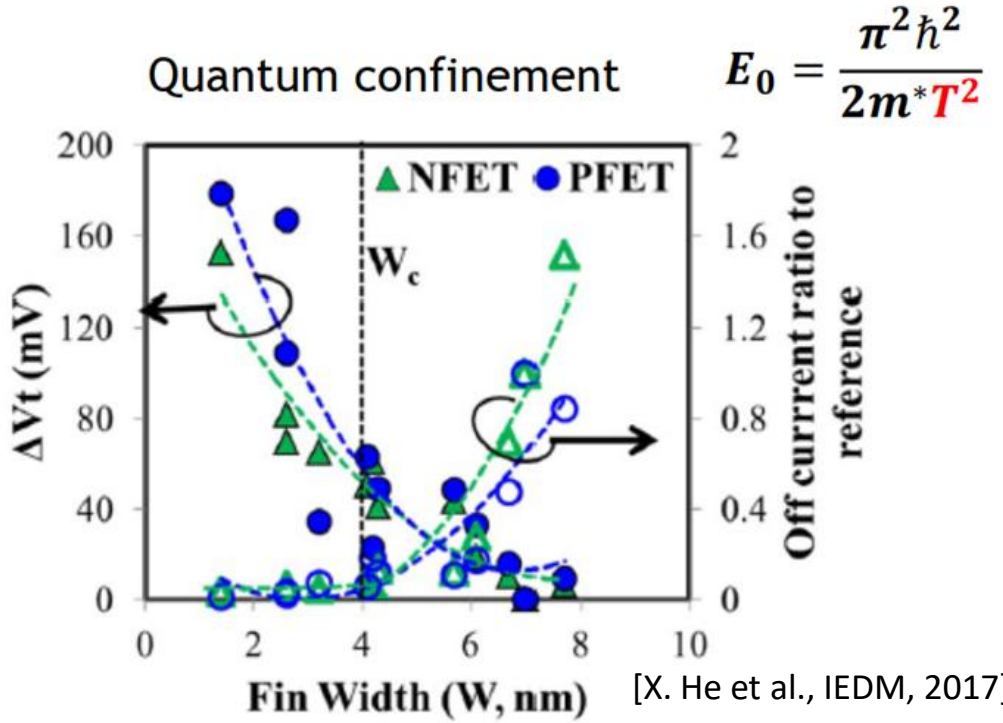
- FH: fin height; FW: fin width
- Tall fin increases Weff = 2xFH+FW per layout width (W)
- Good for heavily wire loaded circuits $\tau_{wire} = C_{wire} \cdot V_{DD}/I(W_{eff})$
- Excessively tall fins waste active power

Fin Width and Shape Optimization

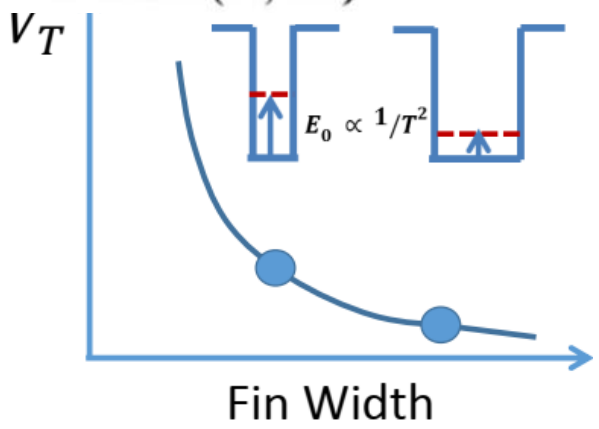


[J. Cai, VLSI 2021, SC1-1]



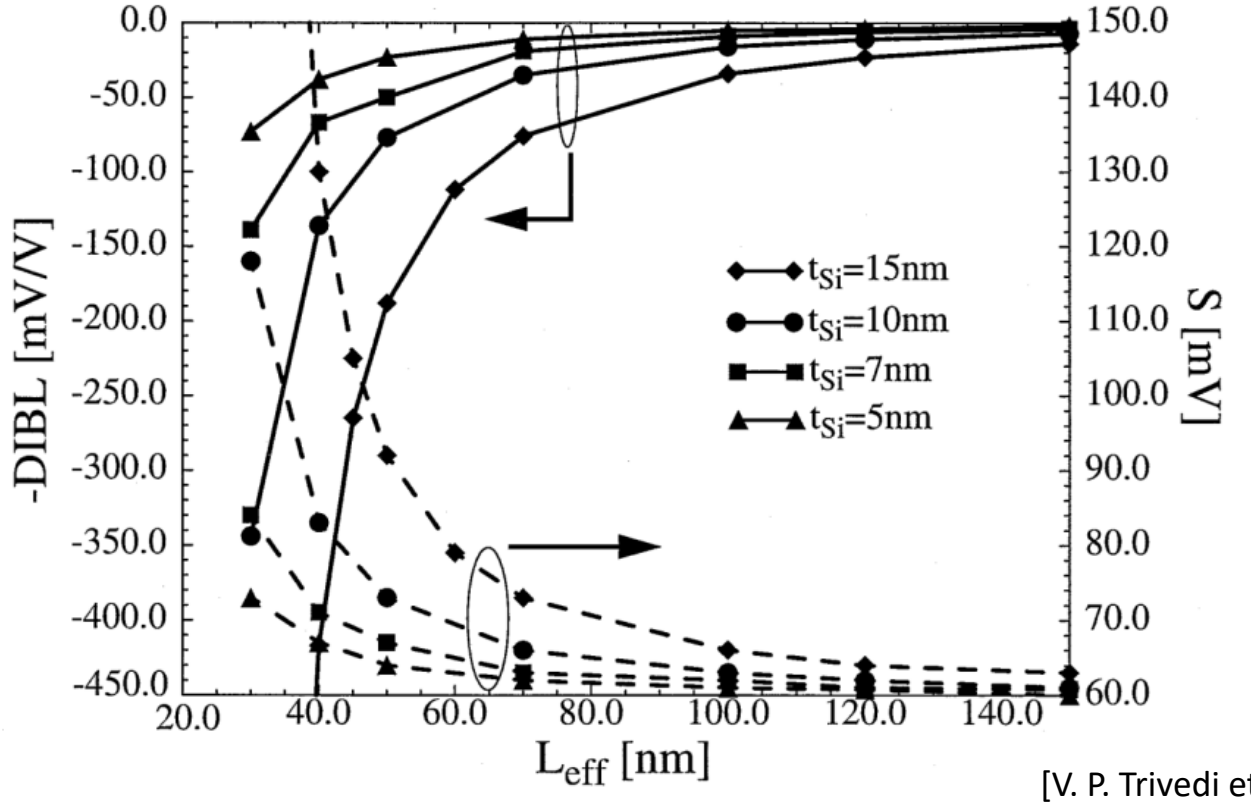


[X. He et al., IEDM, 2017]



- Straight and skinny fin good for short-channel effects
- Fin width reduction limited by mobility (μ) degradation and threshold voltage (Vt) variation

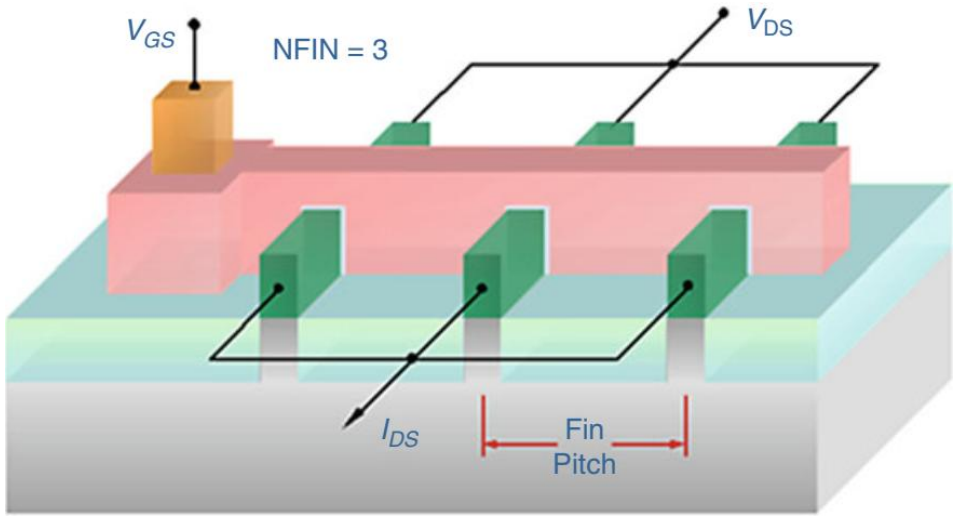
SCE Improvement at Thinner Film

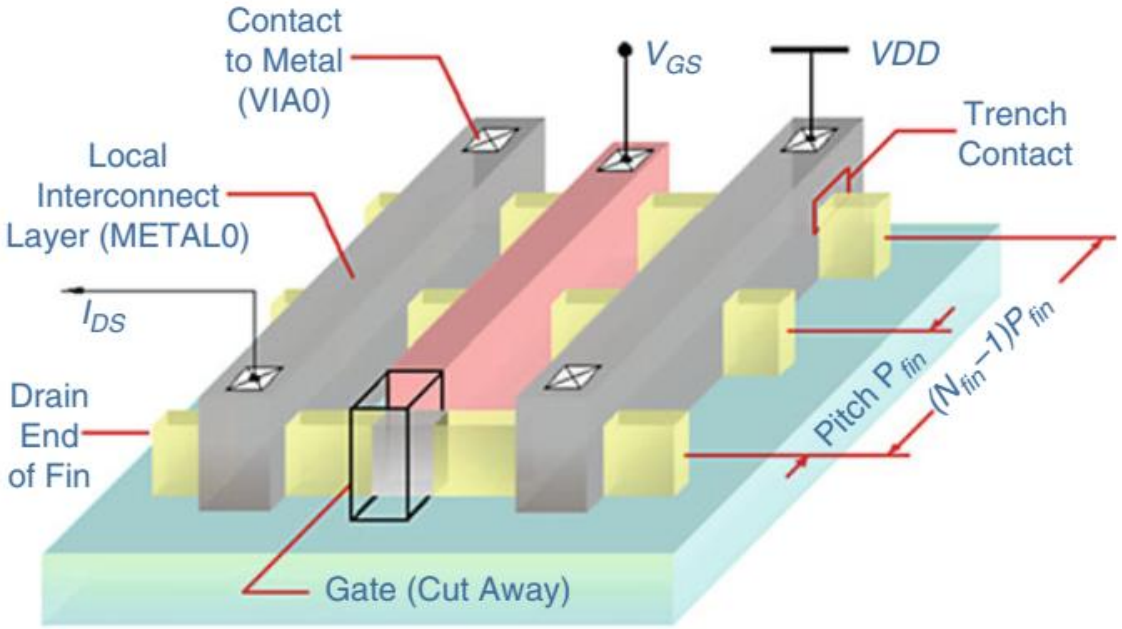


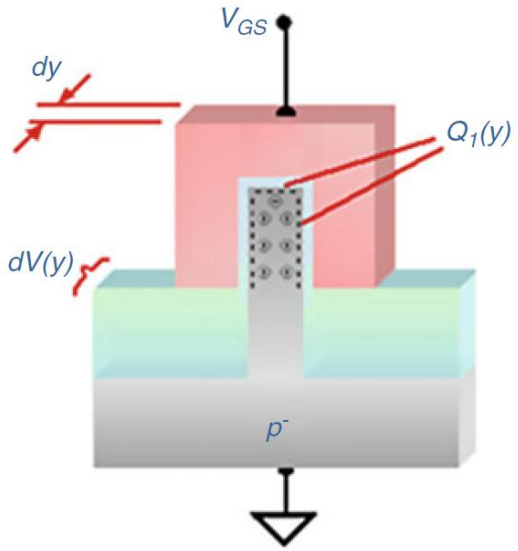
[V. P. Trivedi et al., TED, 2003]

- Thinner channel thickness (fin width)
  - Smaller drain-induced barrier lowering (DIBL)
  - Smaller subthreshold swing (S)

Multiple Fins



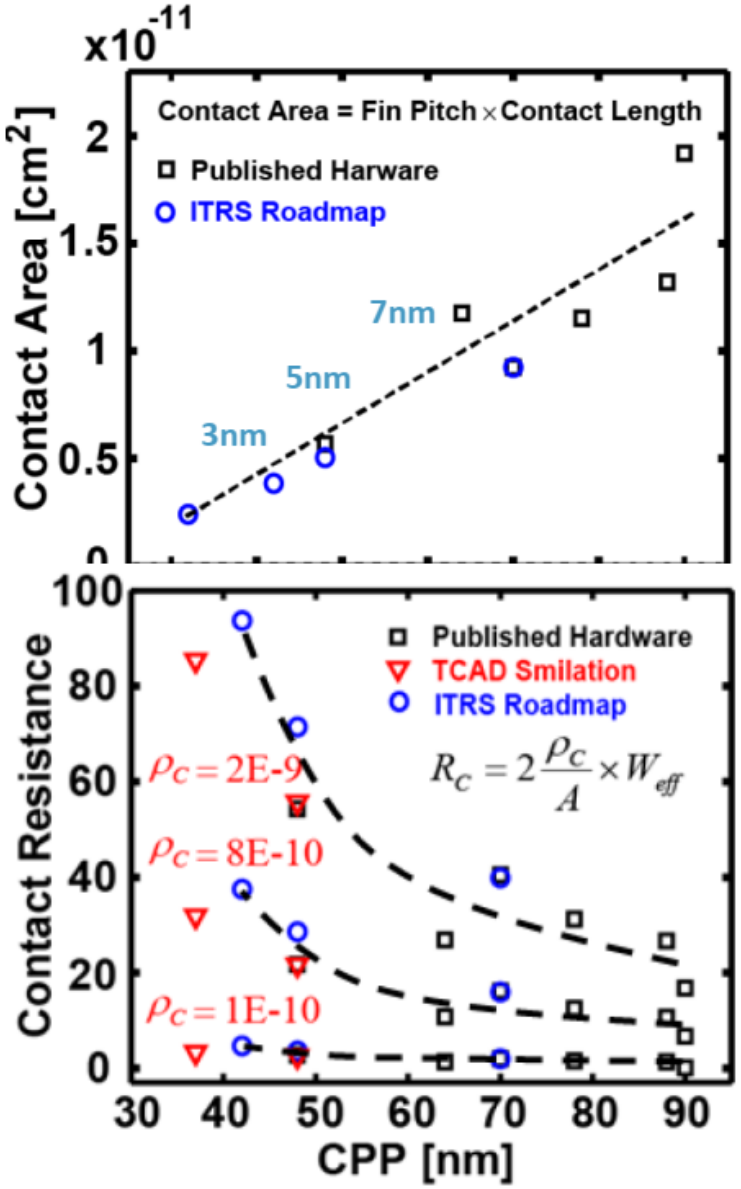




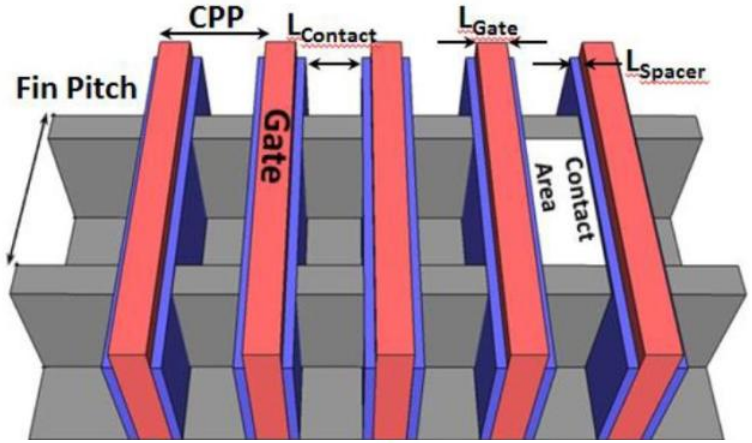
- The effective width W_{eff} of a finFET device is quantized.
- Multiple parallel fins increase the FinFET current drive
- $W_{eff} = N_{fin} \times (2H_{fin}+W_{fin})$
- Local interconnect layer, middle-of-line (MEOL): Tungsten (W) (METAL0)
  - parallel to the gate
  - fill in tiny spaces more easily than copper (Cu)
  - Localized, not extend beyond the ACTIVE area of a single transistor
  - square contact cut (VIA0) between METAL1 and METAL0

[C. Dančak, Semiconductor Nanotechnology, 2018]

Contact Resistance Increases with Scaling



- ~25% reduction of the contact area per node driving up contact resistance (Rc, ohm-μm)
- Fin pitch and gate pitch reduction reduces contact area
  - Contact resistivity (ρ_c) reduction necessary to keep Rc low
  - Wrap-around contact (WAC) maximizing S/D contact area (A) improves access resistance

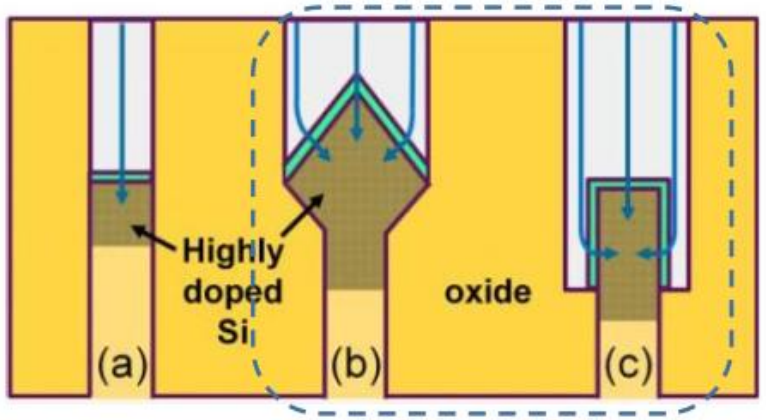


$$R_c \sim 2(\rho_C/A)W_{eff} = \frac{2\rho_C}{L_{CON}FP}(2FH + W_{FIN})$$

FP=fin pitch

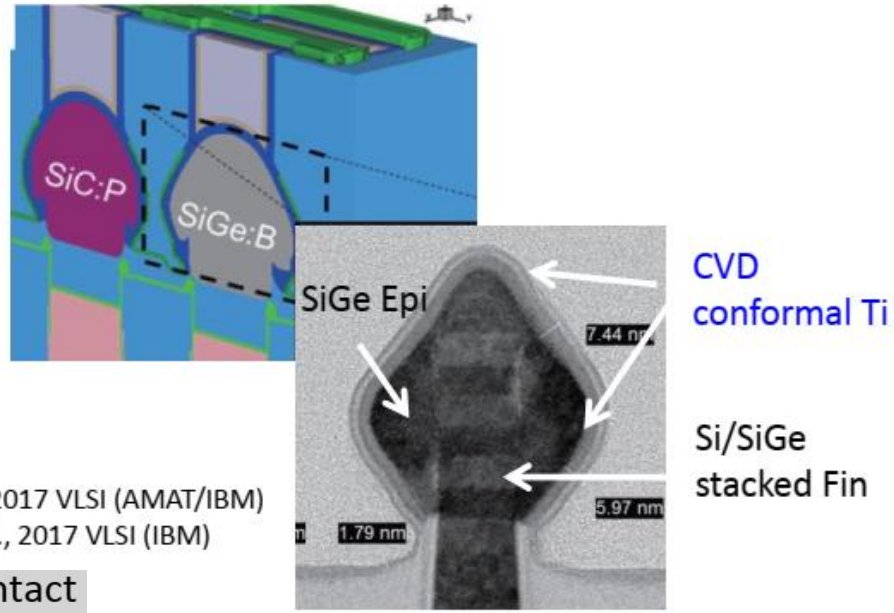
Maximize Contact Area

- Nominal contact reduces with CPP and fin pitch scaling
- Wrap-around contact (WAC) can maximize S/D contact area

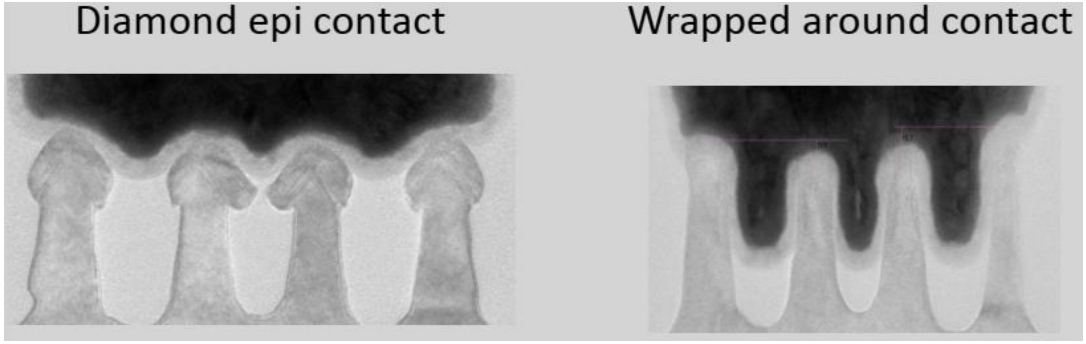


$$R_{con} \propto \frac{\rho_C}{Acon}$$

H. Yu et al., 2016 TED (IMEC)



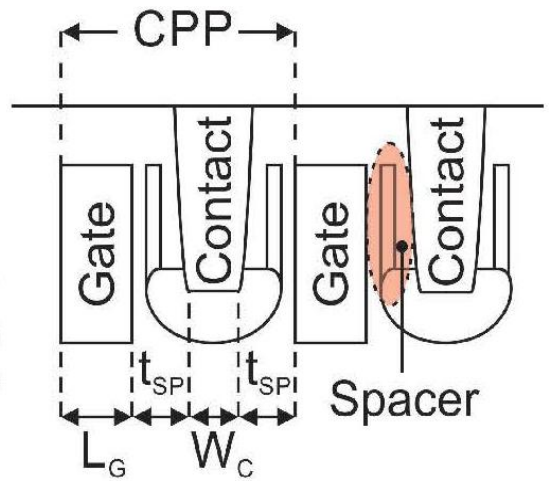
N. Breil et al., 2017 VLSI (AMAT/IBM)
N. Loubet et al., 2017 VLSI (IBM)



Contacted Poly Pitch CPP Scaling Challenges

- Assuming a 5nm fin silicon thickness limit (due to mobility collapse), the gate length limit for a Trigate FinFET is ~16nm and for HNW ~13nm [1].
- Shrinking contacts increase contact resistance.
- Thinner spacer increases capacitance.

L_G = *gate length*
t_{SP} = *spacer thickness*
W_C = *contact width*



$$CPP = L_G + 2t_{SP} + W_C$$

TSMC Actual/Forecast

Node	Device	CPP	L_G	t_{SP}	W_C
16	FF	90	30	18	24
10	FF	66	25	10.5	20
7	FF	57	18	10	16
5	FF	45	16	9	11
3.5	HNW	37	13	7?	10

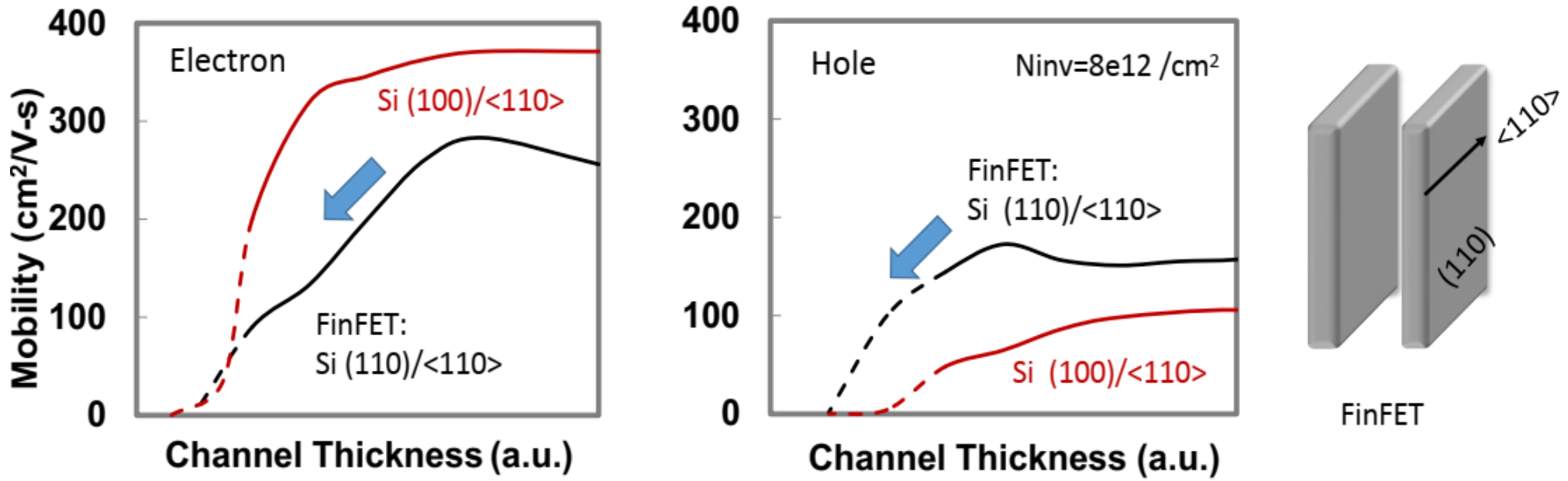
Actual Dimensions [2]

Node	CPP	L_G	t_{SP}	W_C
10	64	25	10	18
7	44	17	8.5	10

IC KNOWLEDGE LLC

[1] J.P. Colinge, p 313, SISPAD (2014),
[2] K.-I. Seo, et. al. VLSIT (2015),R. Xie, et.al IEDM (2016)

Mobility vs Fin Width and Orientation

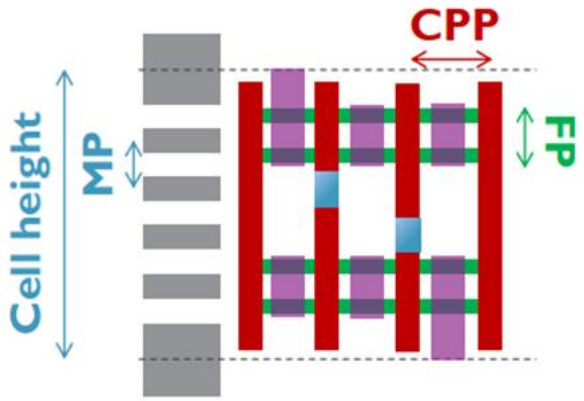


Model: 2x2 and 6x6 k.p band structures for electrons and holes; Self-consistent Schrödinger solver for carrier distribution over multi-subbands; Acoustic and optical phonon scattering based on deformation potential; Surface roughness scattering based on exponential auto-correlation function with calibrated roughness and correlation length.

[J. Cai, IEDM, 2019, SC]

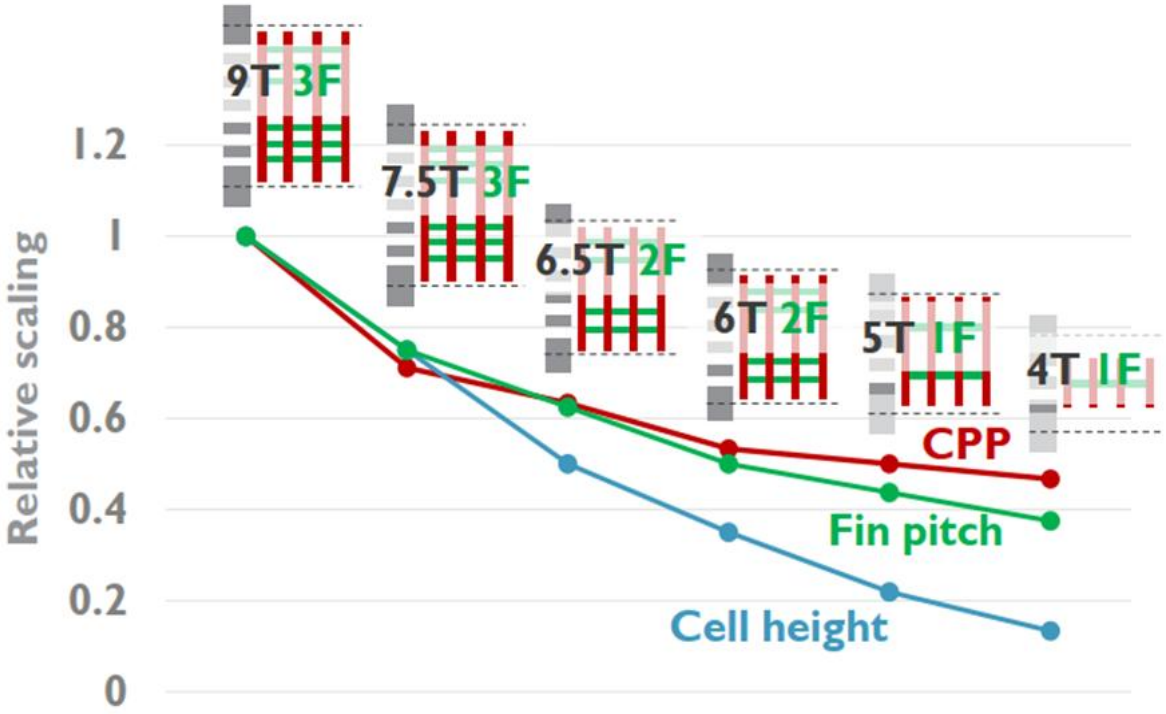
- Mobility reduces with fin width due to increased quantum confinement and surface roughness scattering

Standard Cell Scaling Trend



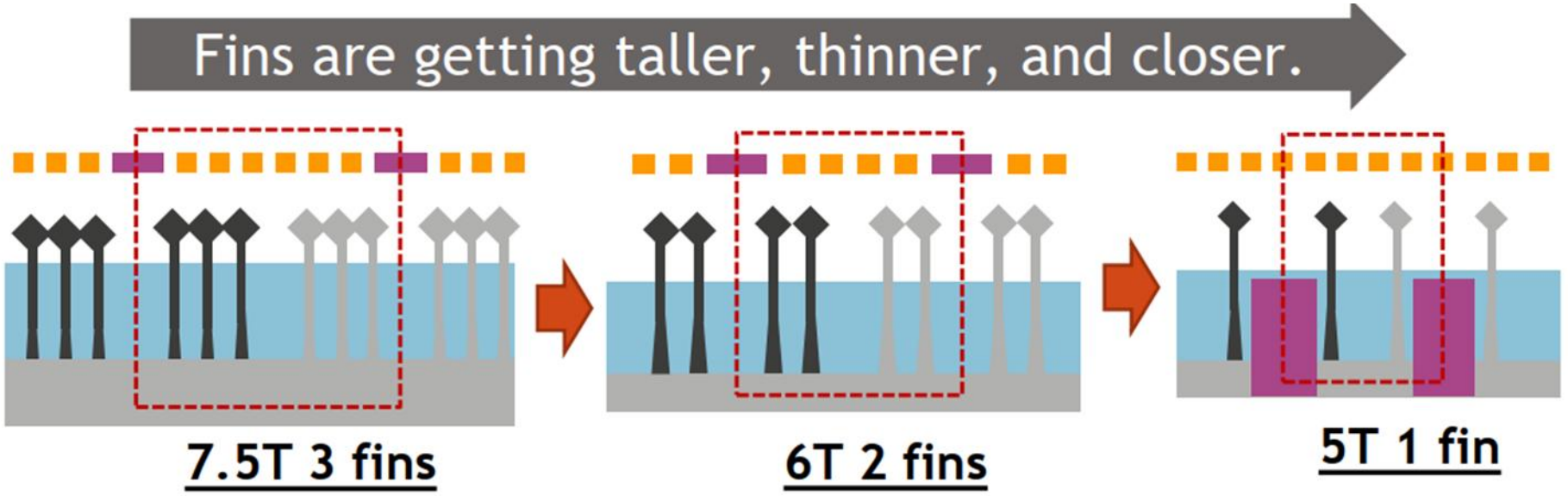
CPP: Contacted poly pitch
FP: Fin pitch
MP: Metal pitch
Cell height: # of metal lines/cell x MP

[N. Horiguchi, VLSI 2021, SC1-2]



- Cell height scaling drives standard cell scaling by reducing # of metal lines (track)/cell

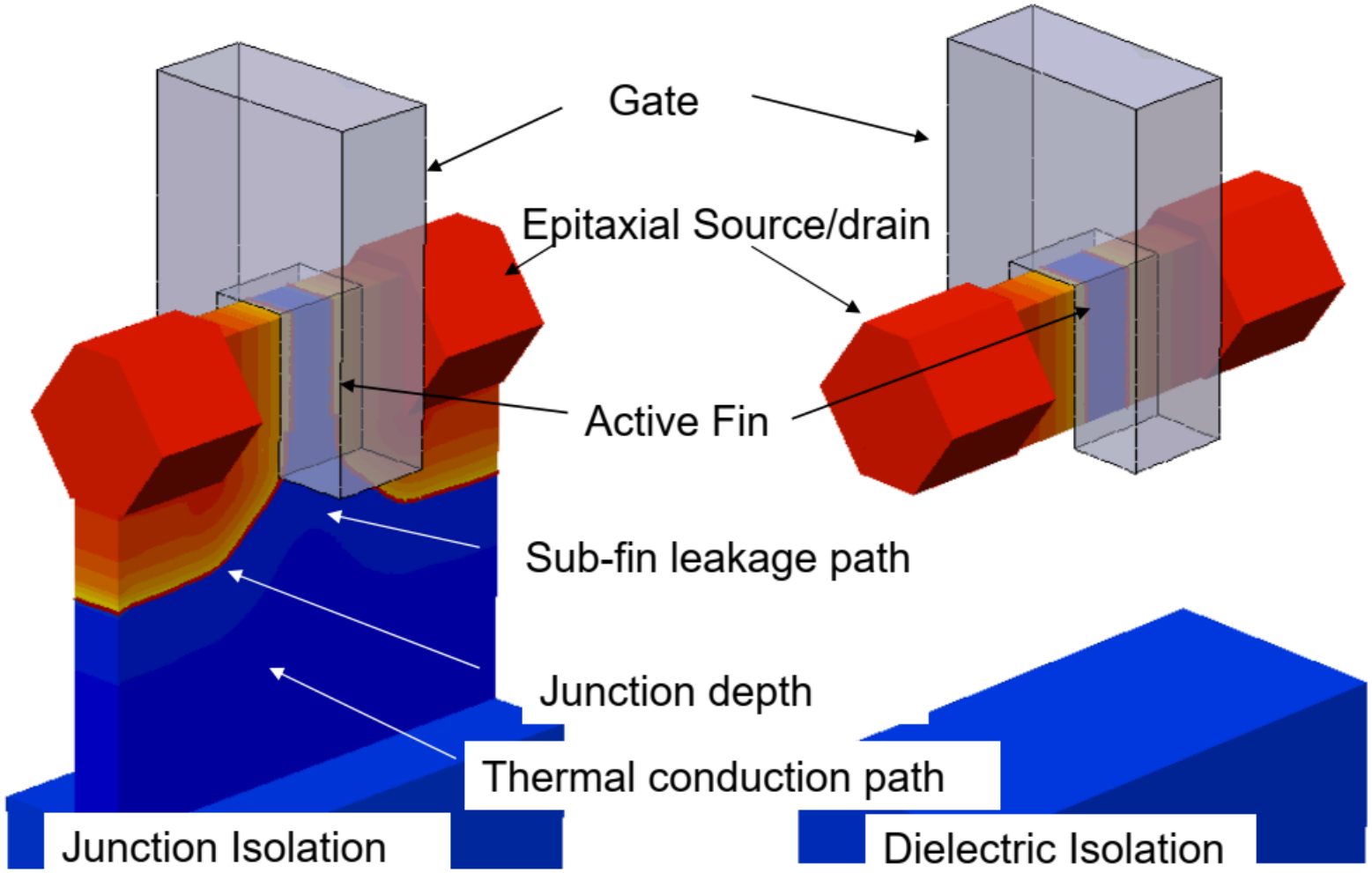
Cell Height Scaling Challenge



[N. Horiguchi, VLSI 2021, SC1-2]

- **Fin de-population** is required for standard cell scaling → leads to reduced drive strength and increased variability.
- Capacitance reduction

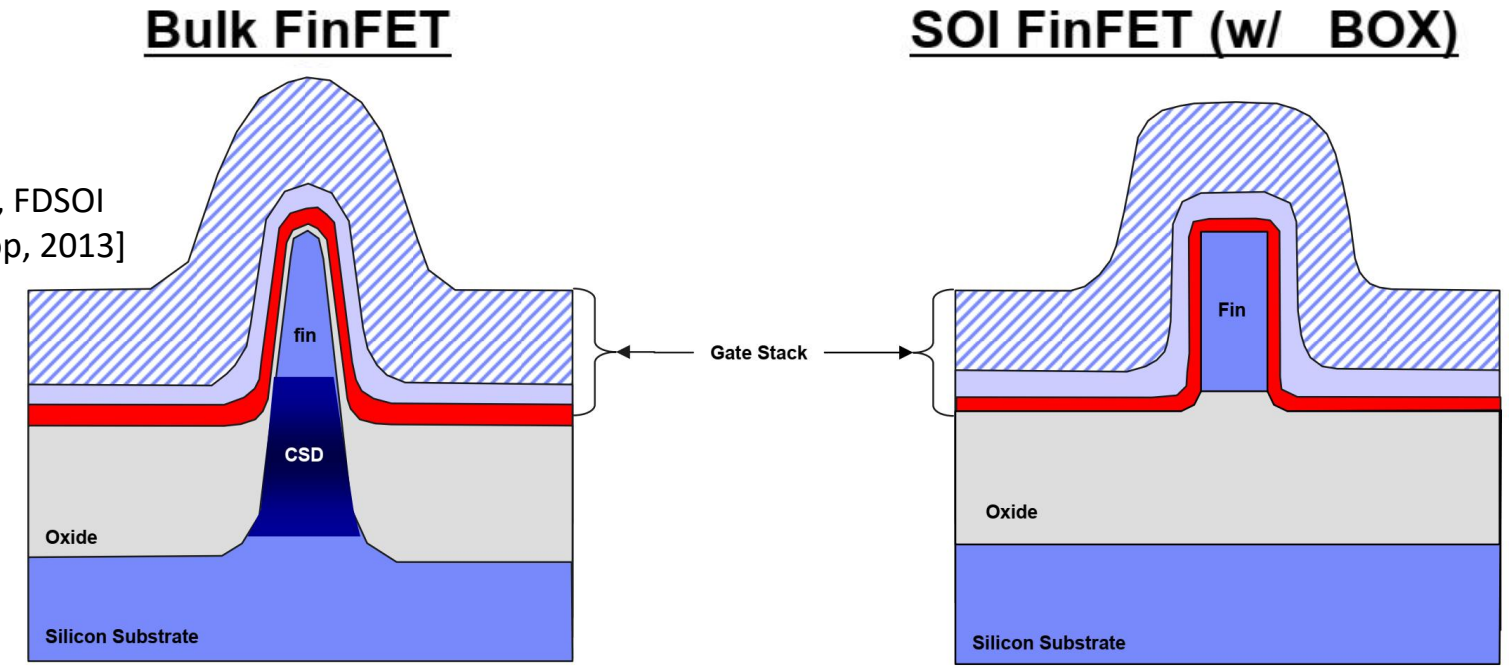
FinFETs on Bulk and SOI substrates



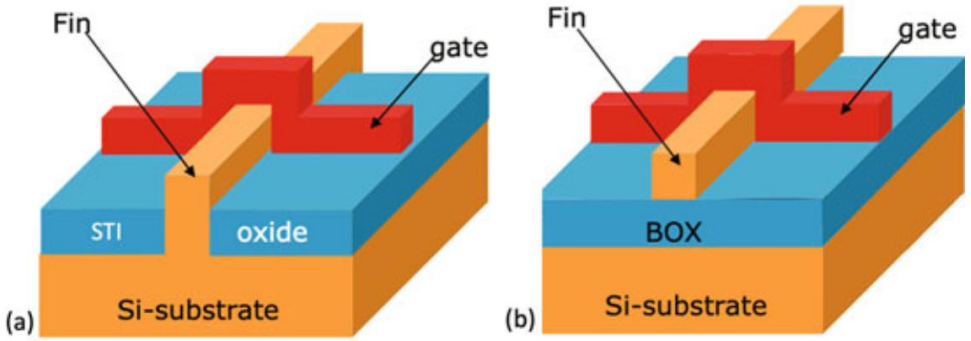
[T. Hook, FDSOI workshop, 2013]

SOI vs. Bulk FinFET: Isolation

[T. Hook, FDSOI workshop, 2013]

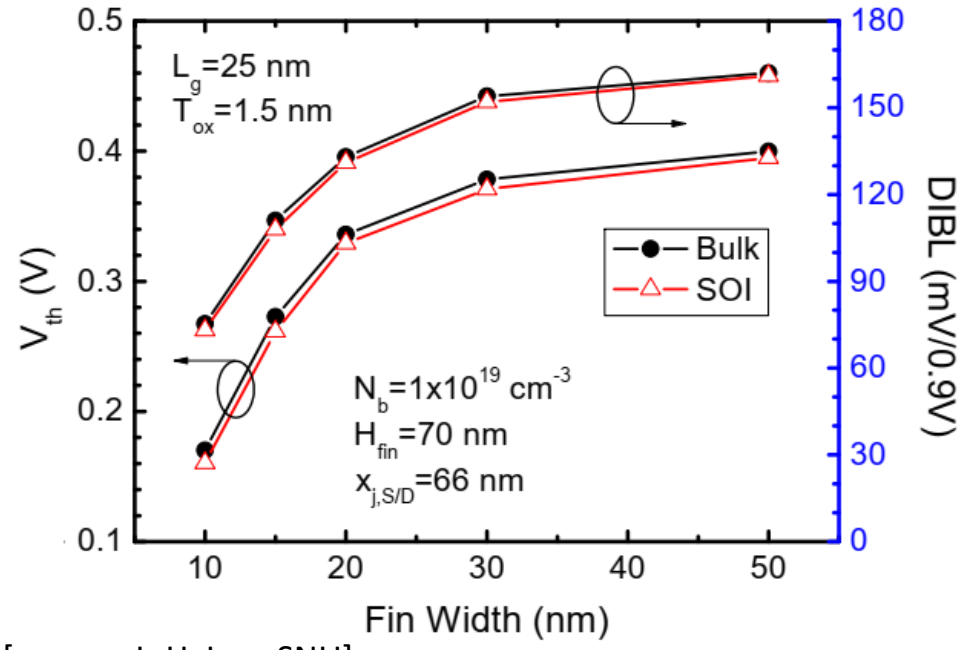


- Retrograde-well doping required as punch through-stop (PTS) layer.
- HALO is also often adopted.
- Tapered fin shape due to STI process.

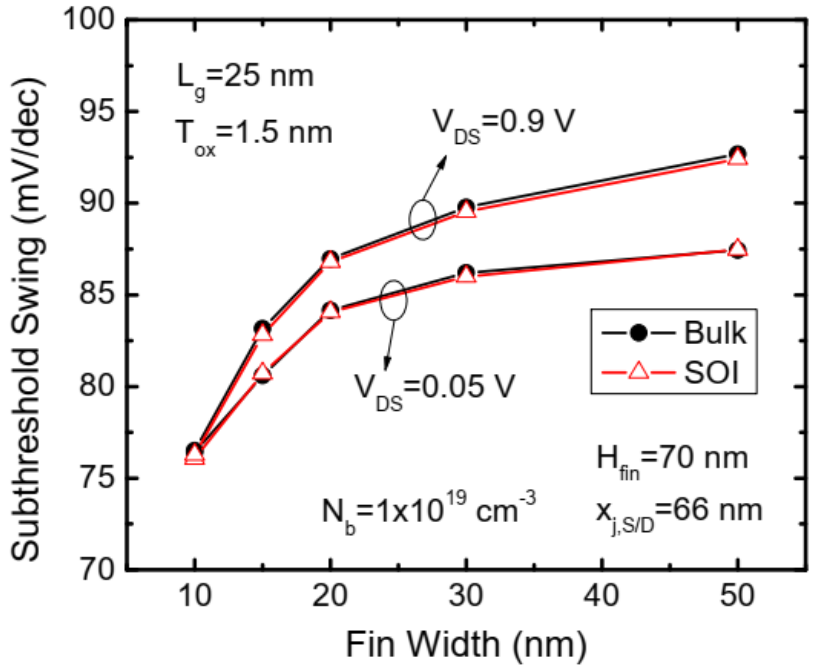


- No doping process needed to avoid PT.
- Rectangular fin shape.
- Lower thermal conductivity in buried oxide
- Silicon-on-insulator (SOI)

Bulk FinFET vs. SOI FinFET

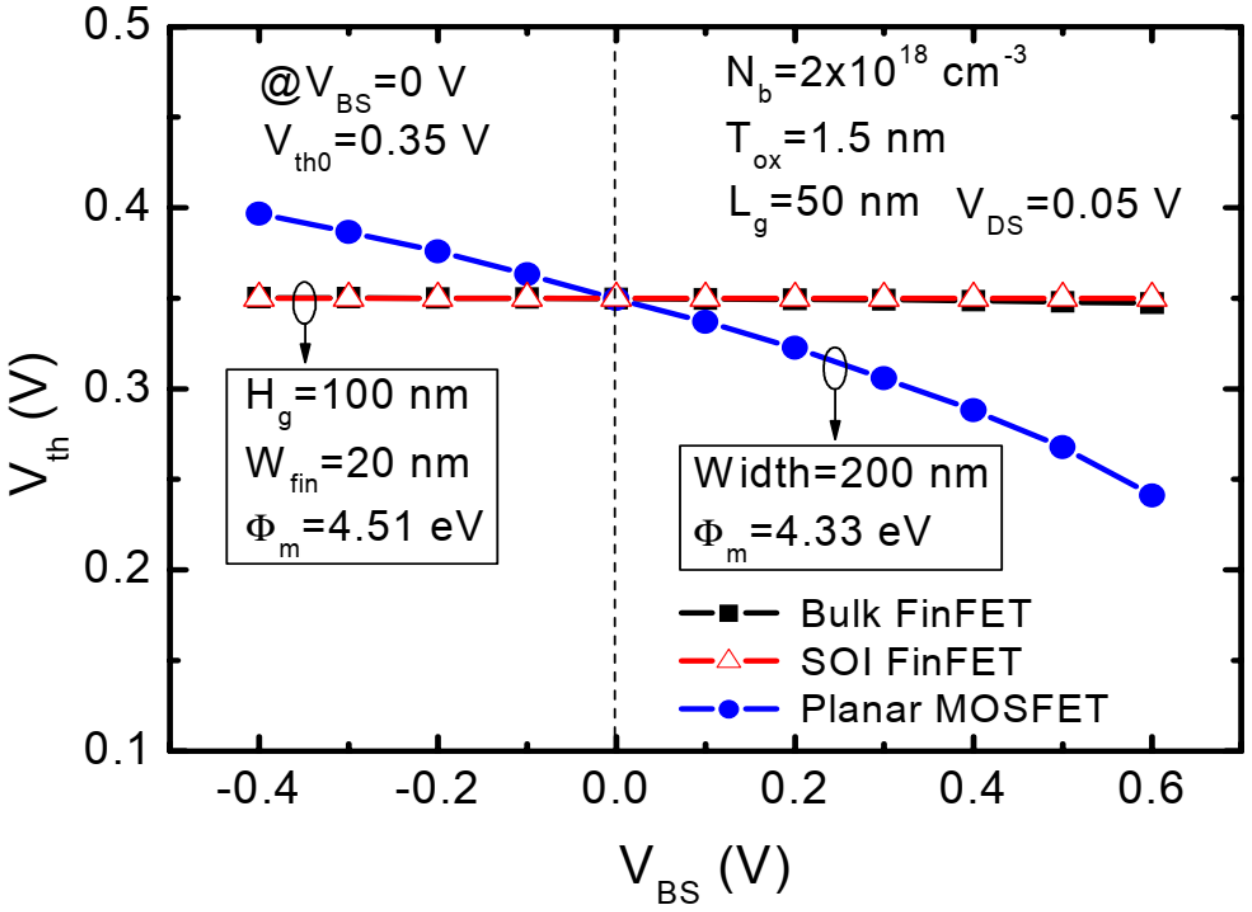


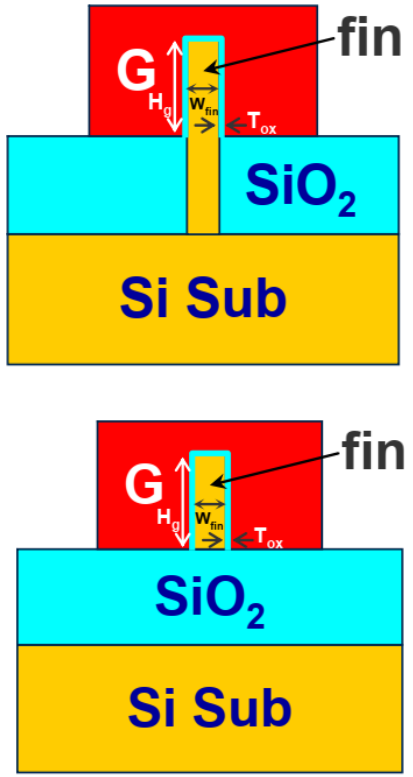
[source: J.-H. Lee, SNU]



- The bulk FinFETs have nearly the same threshold voltage (Vth), DIBL, and SS characteristics as those of SOI FinFETs.

Back-Bias Effect

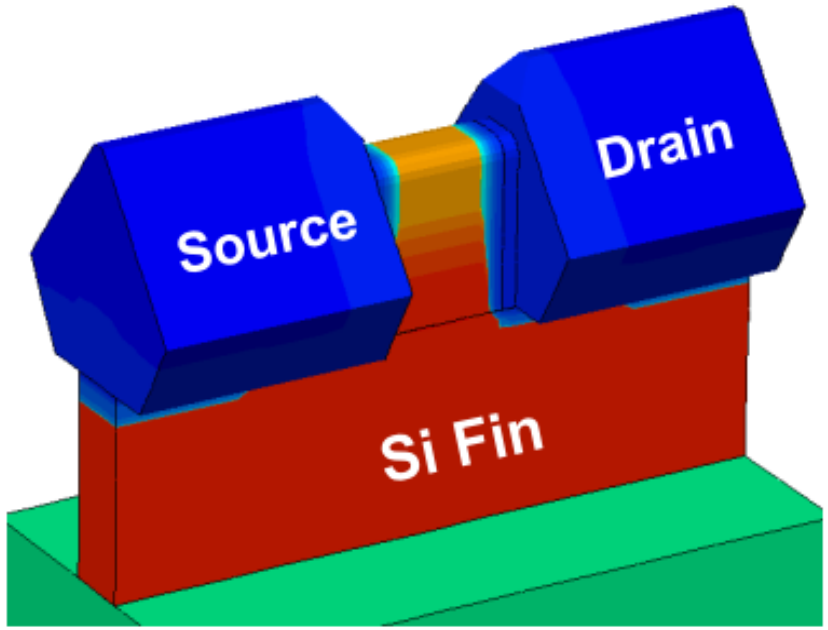




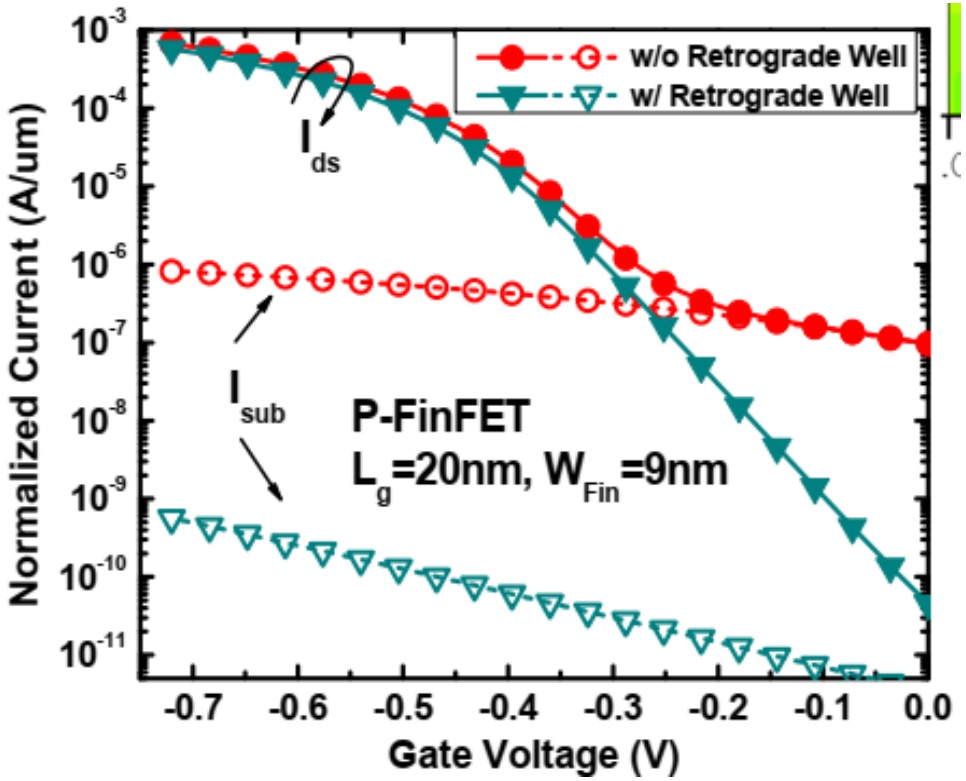
- Bulk and SOI FinFETs have an ignorable back bias effect compared to planar MOSFET.

Jong-Ho Lee et al., "Comparison of SOI FinFETs and Bulk FinFETs, " ECS Transactions, Volume 19, 101, 2009

Bulk FinFET with Retrograde Well



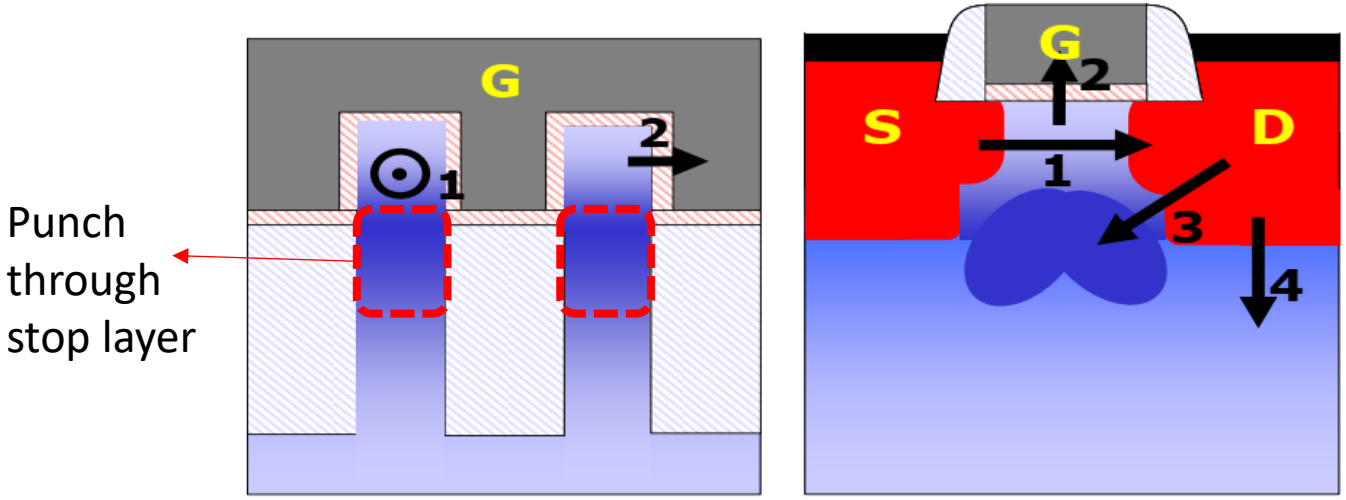
[N. Xu, UC Berkeley]



- With retrograde well (punch through stop layer), leakage current is suppressed.

Leakage Paths in Bulk FinFET

Perpendicular to Gate **Parallel to Gate**



Bulk FinFET has similar leakage mechanisms as bulk planar

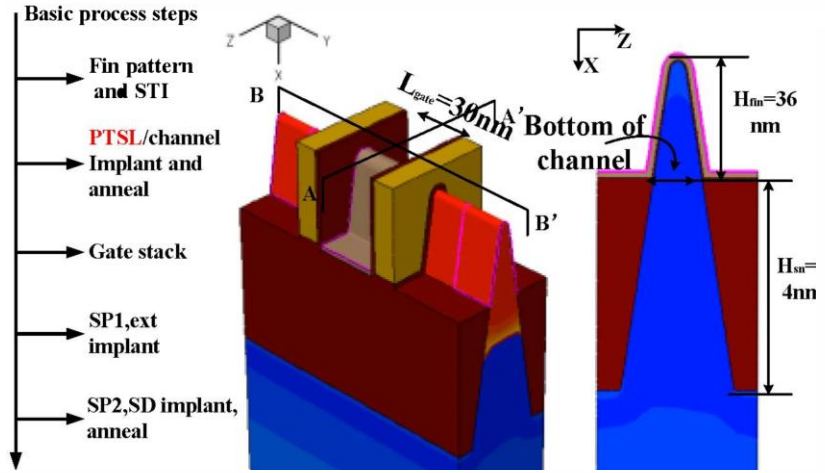
Bulk FinFET Leakage Mechanisms:

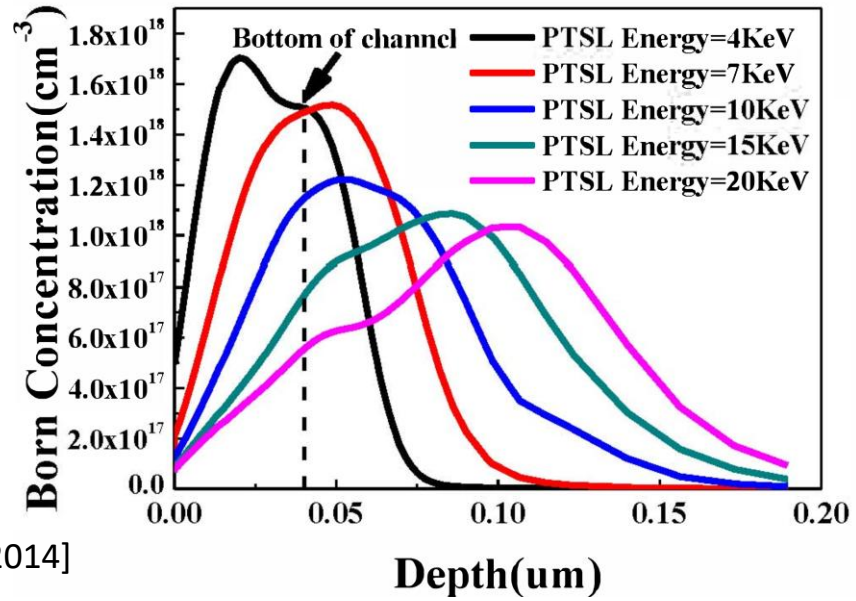
1) **Sub threshold leakage**
2) **Gate Leakage**
3) **Gate Induced Drain Leakage (GIDL)**
4) **Junction Leakage**

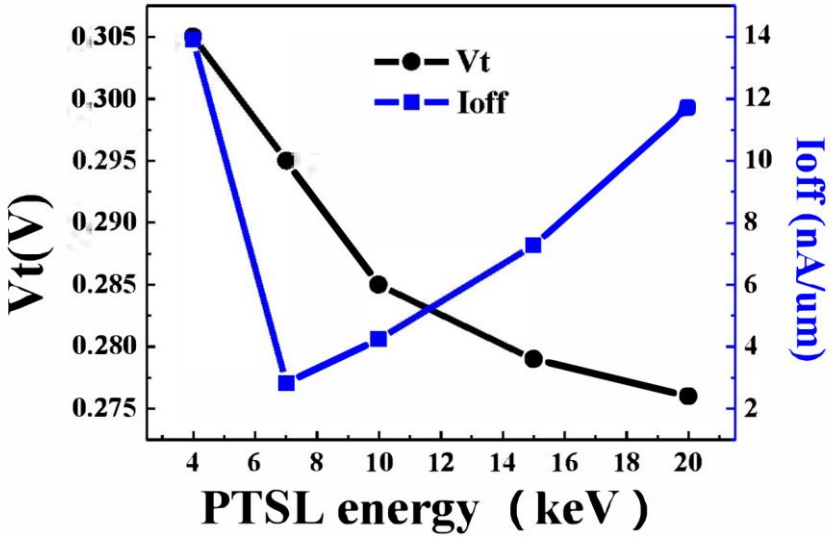
[B. B. Boris, IEDM, 2015, SC]

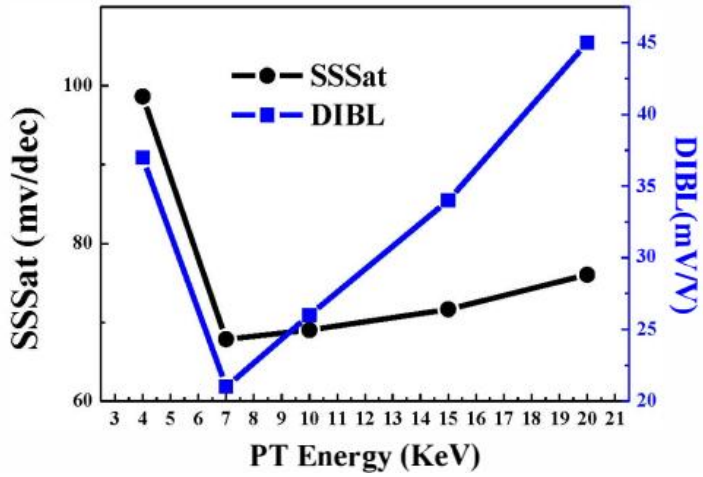
Punch Through Stop Layer (PTSL) in FinFET

- Positioning of the PTSL is great importance to suppress the punch-through leakage



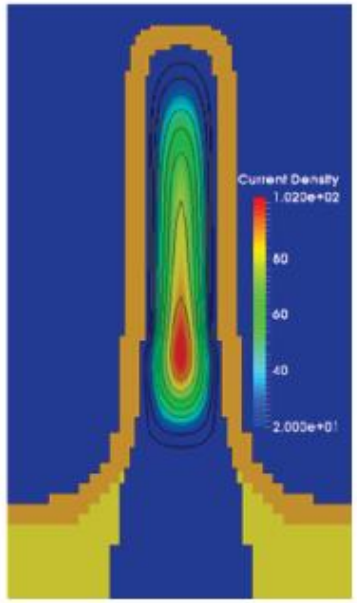




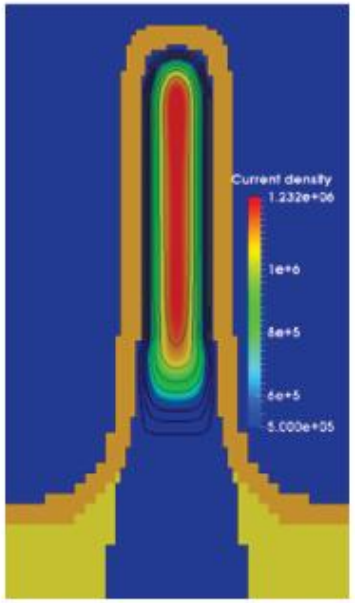


[R. Li et al., IEEE ICSICT, 2014]

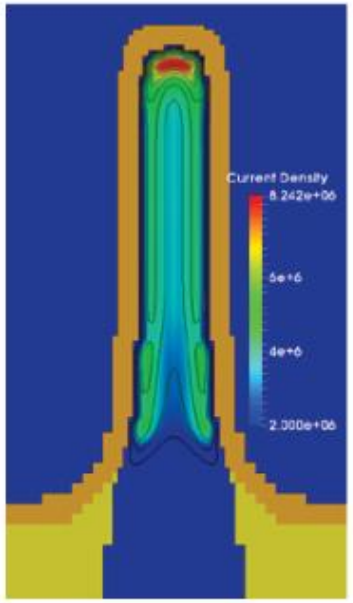
FinFET Conduction: Current Density Profile



V_G=0 V



V_G=0.35 V

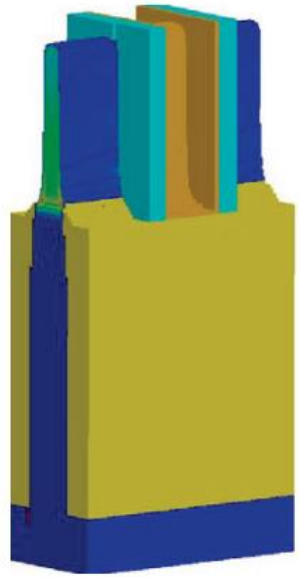


V_G=0.7 V

[A. Asenov, IEDM, 2015, SC]

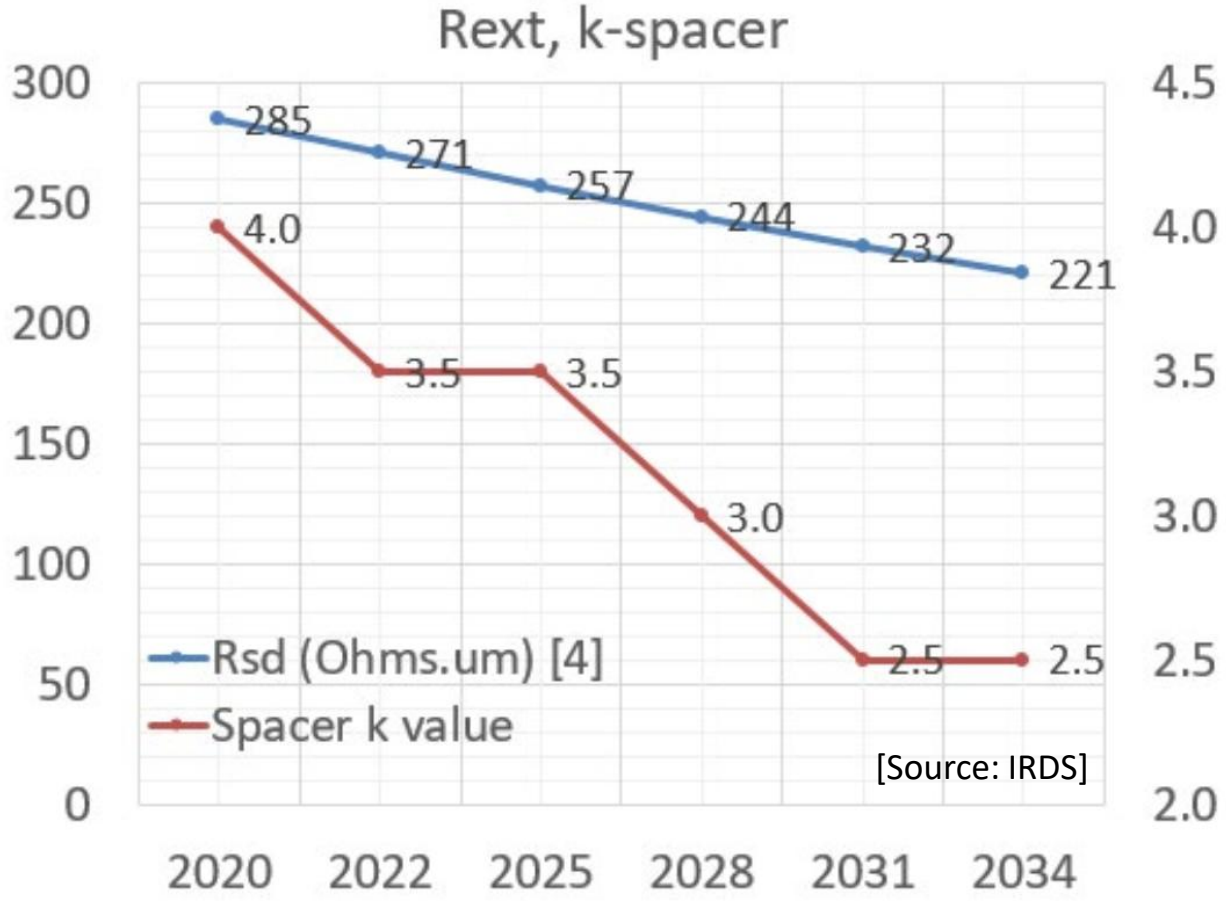
Intel 14 nm FinFET

Parameter	Value
H_{FIN}	42 nm
W_{FIN}	8 nm
L_G	20 nm
EOT	~ 0.9 nm
I_{DSAT}	1.04 mA/um
I_{DLIN}	0.237mA/um

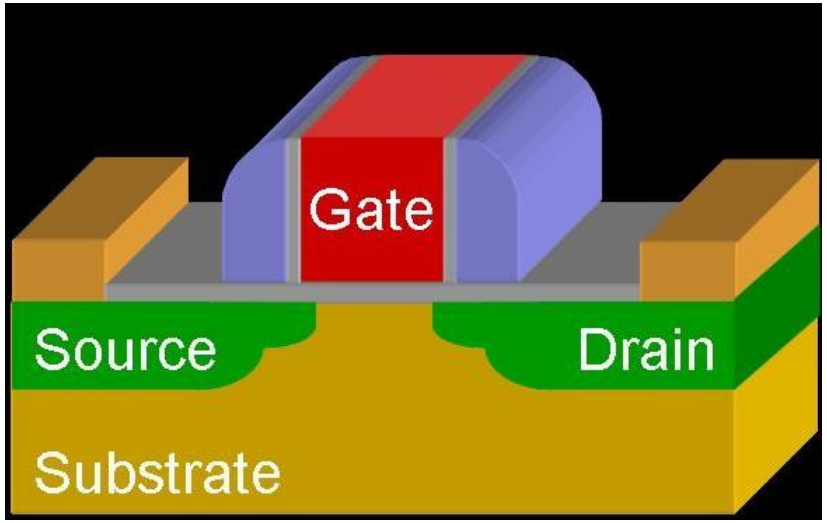


- Effective fin width: Weff = 2Hfin + Wfin
- Hfin: fin height
- Wfin: fin width

Scaling trend of device S/D resistance (Rsd) and k-value of device spacer



[Source: IRDS]

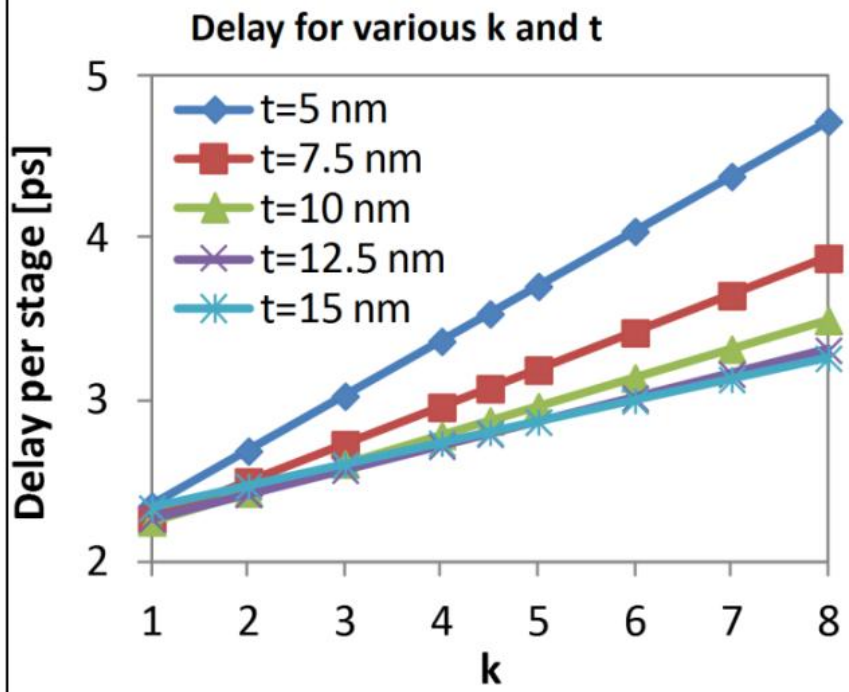


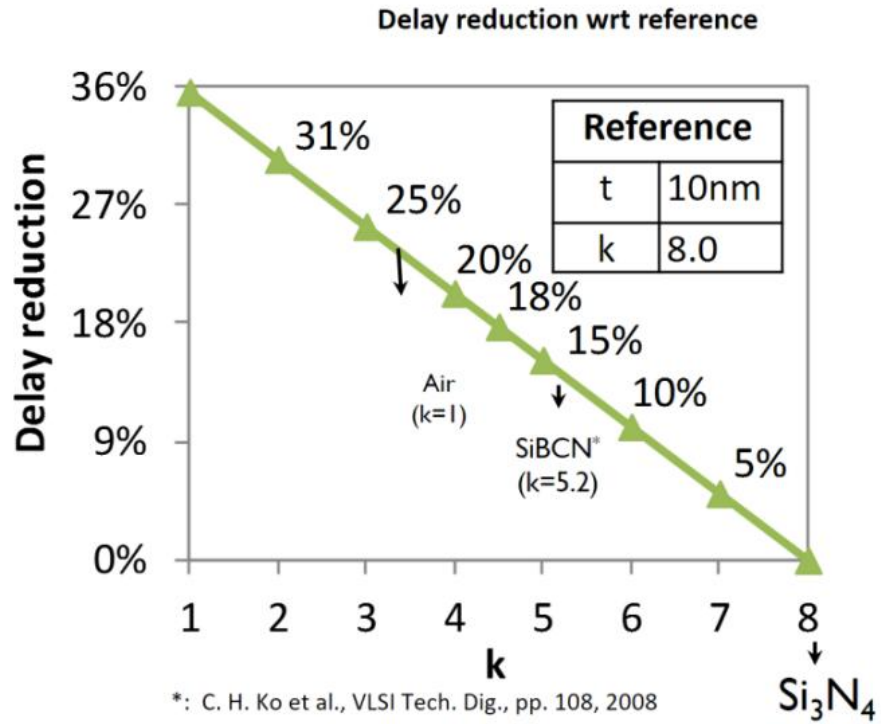
- Parasitic capacitance between gate and source/drain terminal of the device is expected to increase with technology scaling. → low-κ spacer materials and even air spacer needed (with good reliability and etch selectivity for S/D contact formation)

Impact of low-k spacers

- For 10nm spacer thickness (t), 14% reduction could be achieved by replacing Si_3N_4 spacers with k=5.2 spacers (36% with air spacers)



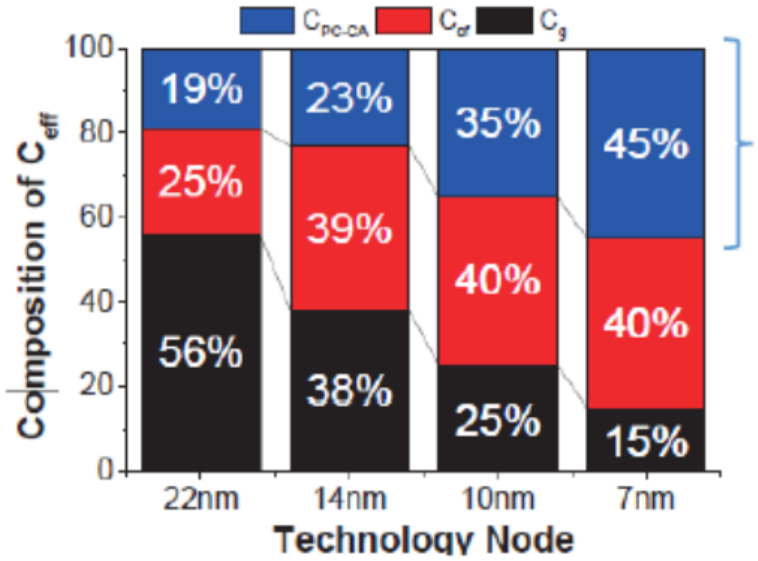




Zsolt Tokei, IEDM 2013

Capacitance reduction – Gate Spacer Evolution

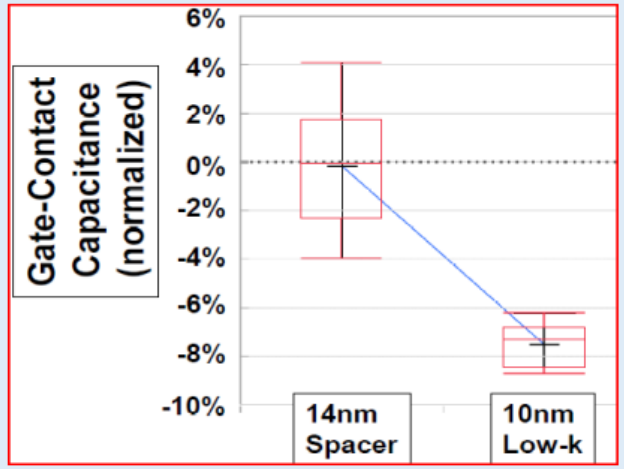
Gate to contact parasitic capacitance weighting increase from node to node



Gate to contact parasitic cap

Source: T. Yamashita et al., 2015 VLSI (IBM)

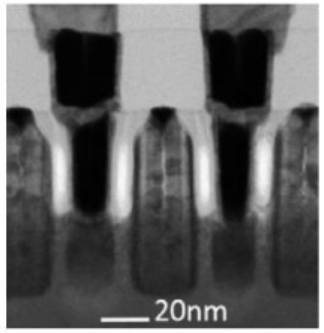
1 Lower K Spacer

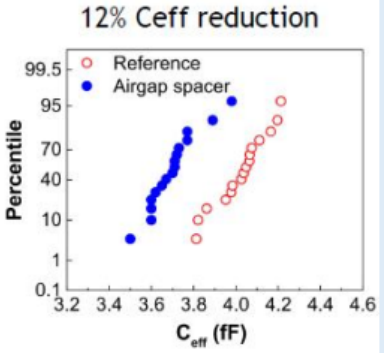


Source: C. Auth, Intel, IEDM2017

New Materials:
SiN → SiCON → SiCO

2 Air spacer



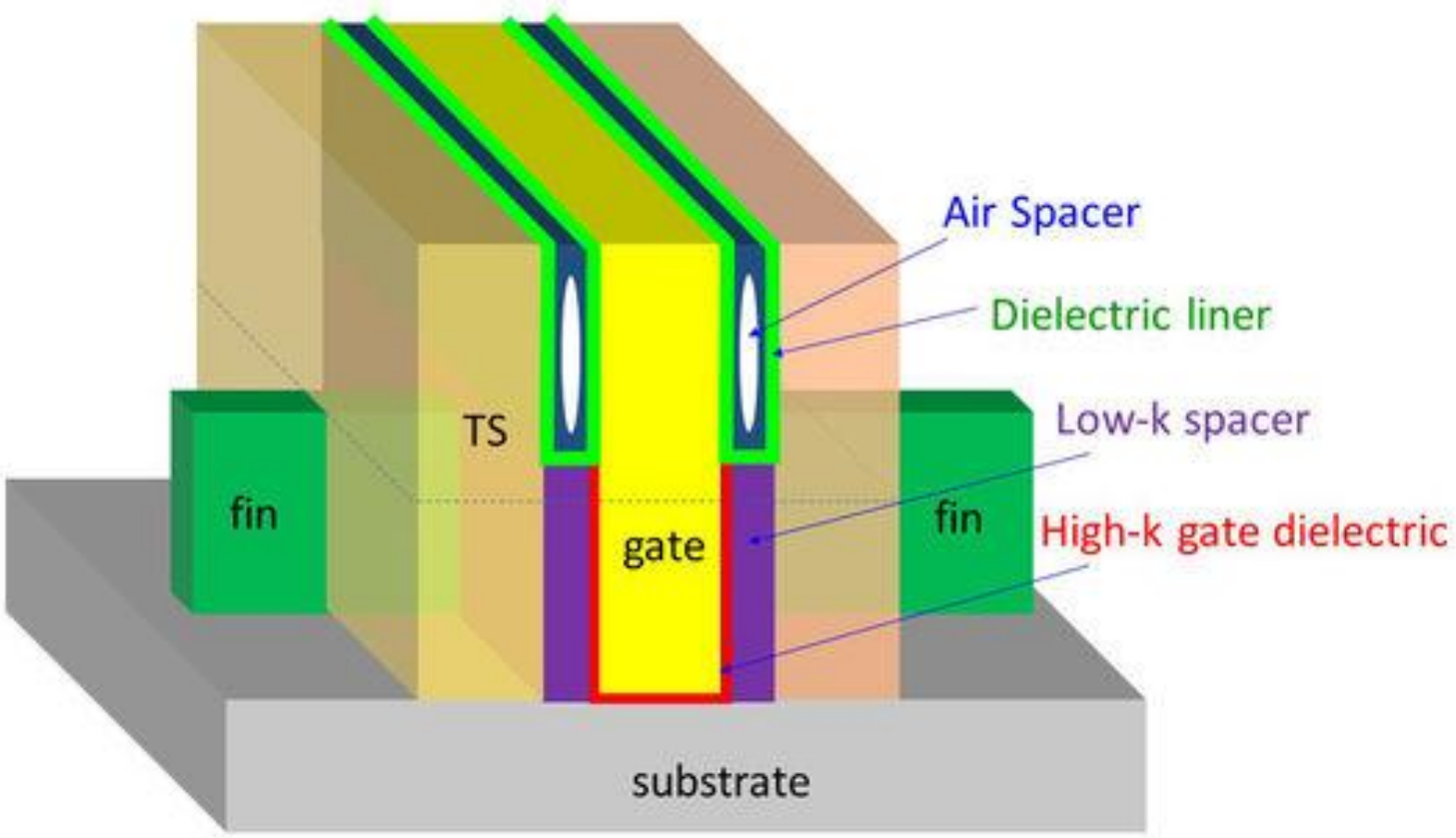


Source: K. Cheng, IBM, IEDM2016

Challenges: Selective removal of disposable spacer

[B. Colombeau, IEDM, 2021, SC1-6]

Low-k Spacer

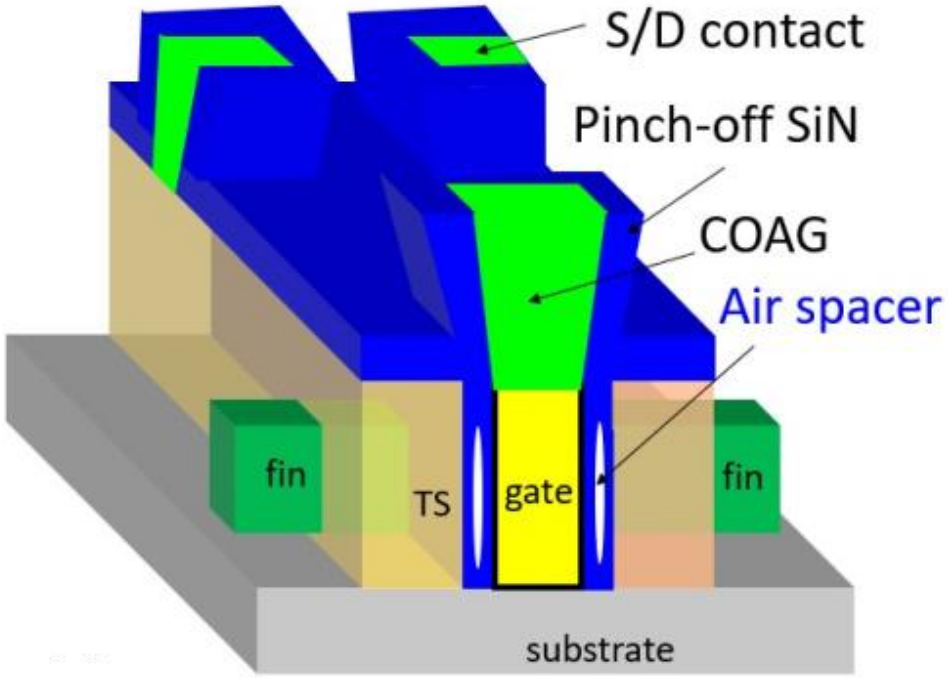


- Source/drain **trench silicide** metallization (**TS**)
- Silicide: compound of silicon with metal → low resistance

Improved Air Spacer Co-Integrated with Self-Aligned Contact (SAC) and Contact Over Active Gate (COAG) for Highly Scaled CMOS Technology

Kangguo Cheng, Chanro Park, Heng Wu, Juntao Li, Son Nguyen, Jingyun Zhang, Miaomiao Wang, Sanjay Mehta, Zuoguang Liu, Richard Conti, Nicolas Loubet, Julien Frougier, Andrew Greene, Tenko Yamashita, Bala Haran, Rama Divakaruni

IBM Research, Albany, NY 12203, USA
Email: cheng@us.ibm.com



[K. Cheng, VLSI, 2020]

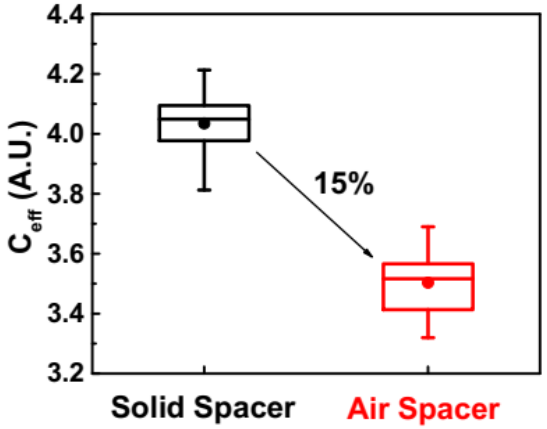


Fig. 12 15% reduction in the effective capacitance (C_{eff}) by full air spacer experimentally achieved in a fan-out3 (FO3) ring oscillator (RO)

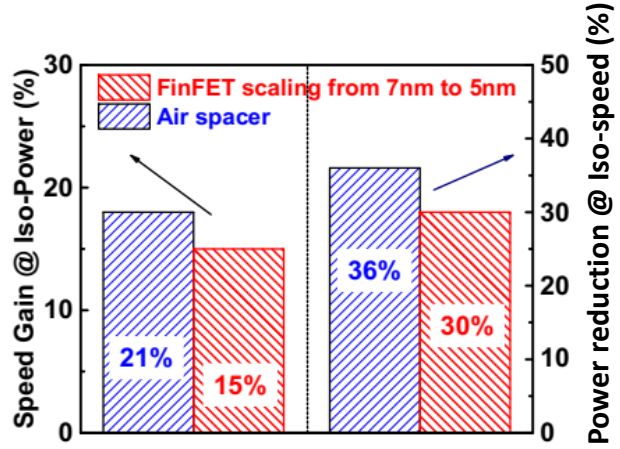


Fig. 13 Performance/power benefit comparisons between air spacer and scaling FinFET from 7nm to 5nm [9]. The benefits of air spacer exceed that of the classical node scaling

- FinFET transistors with Self-Aligned Contacts (SAC) and Contacts Over Active Gate (COAG)

FinFET: Strengths and Weaknesses

- **Enhanced scalability**: Due to improved electrostatic control of the gate over the channel, the FinFET is less prone to SCEs such as threshold roll-off and subthreshold conduction. This enhanced scalability potentially extends Moore's law.
- **Lower Ioff**: Below threshold, the FinFET is turned off more fully than a planar MOSFET of similar dimensions. This reduces standby leakage power. The figure of merit typically employed is low subthreshold swing, measured in mV/decade of leakage current. Intel's 22-nm low-power FinFET devices showed a swing of only 65 mV/decade, compared to 100 mV/dec for planar 32-nm devices.
- **Higher Ion in small-cell area**: Standard cells using FinFET transistors can pack several times the device width into the same cell footprint as planar standard cells—because the effective width Weff of nonplanar devices extends partly upward from the substrate.
- **Enhanced mobility**: The vertical fin may enable still other refinements. While the face of the standard silicon wafer lies in the (100) crystal plane, individual fins could be oriented along other axes, characterized by higher hole mobility μp (or by equal μn and μp). Such process refinements might reduce the need to widen p-type devices.
- **LP, HP, I/O device types**: The FinFET transistor has qualified as a platform SOC technology. It can support a mixed family of devices on one die, including low-power, high-performance, higher (I/O) voltage, SRAM, and analog/RF. Intel's 22-nm process employed longer (40-nm) finFETs for either ultra-low-power logic with a very low subthreshold current or higher-voltage devices to handle legacy 1.8 or 3.3V I/Os, voltage regulators, or transceivers.

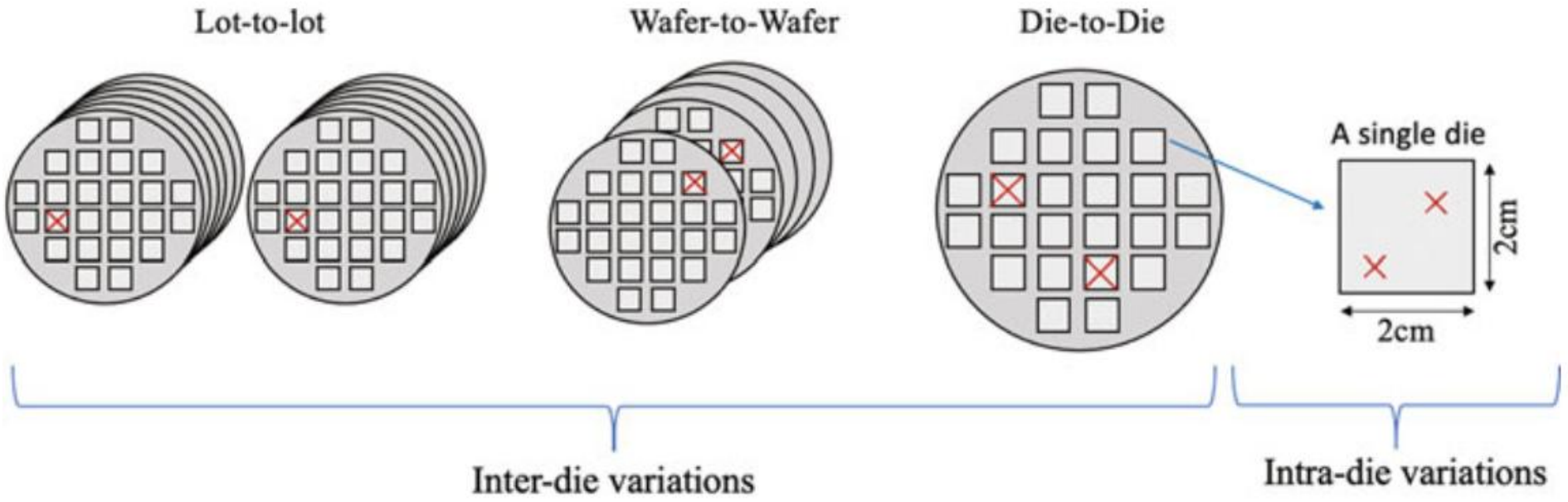
[C. Dančak, Semiconductor Nanotechnology, 2018]

FinFET: Strengths and Weaknesses

- **Quantized width**: The effective FinFET transistor width Weff is quantized and can be adjusted higher only by laying out more fins connected in shunt. Precise device sizing is harder, making it difficult to trade off performance and power.
- **Overheating in fin**: Localized overheating can result from high-drive currents flowing through thin fins. The heating can degrade I_{ON} and cause a reliability problem. Less of an issue for bulk FinFETs, whose fins are thermally coupled to the substrate, it may be more of an issue for SOI FinFETs, fabricated upon a buried-oxide layer. The thermal conductivity of a thin layer of SiO_2 can be two orders of magnitude lower than for bulk silicon.
- **Parasitic capacitance**: Parasitic values are process-specific. Because the fin is a vertical ridge, FinFETs tend to have less parasitic junction capacitance C_j than planar MOSFETs, with their wide, flat source and drain areas. But a tall fin and gate stack, in close proximity to local interconnect lines, can result in higher gate-to-source/drain parasitic capacitance C_{gs} between the gate and the exposed fins or METAL0 lines forming the common source and drain. In general, the 3-D structure of a finFET complicates the extraction of parasitic RC values from device geometry.
- **Parasitic resistance**: High parasitic resistance can arise from the source and drain terminals at both ends of the thin fin, compared to the planar MOSFET. High series R can impact on-chip RF circuits via lower transconductance g_m, which affects amplifier gain or frequency response.

[C. Dančak, Semiconductor Nanotechnology, 2018]

Intra-die and Inter-die Variations

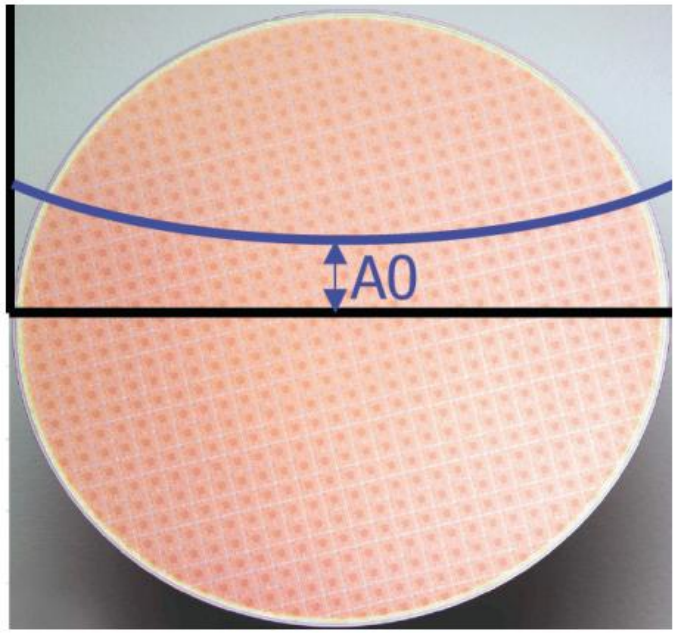


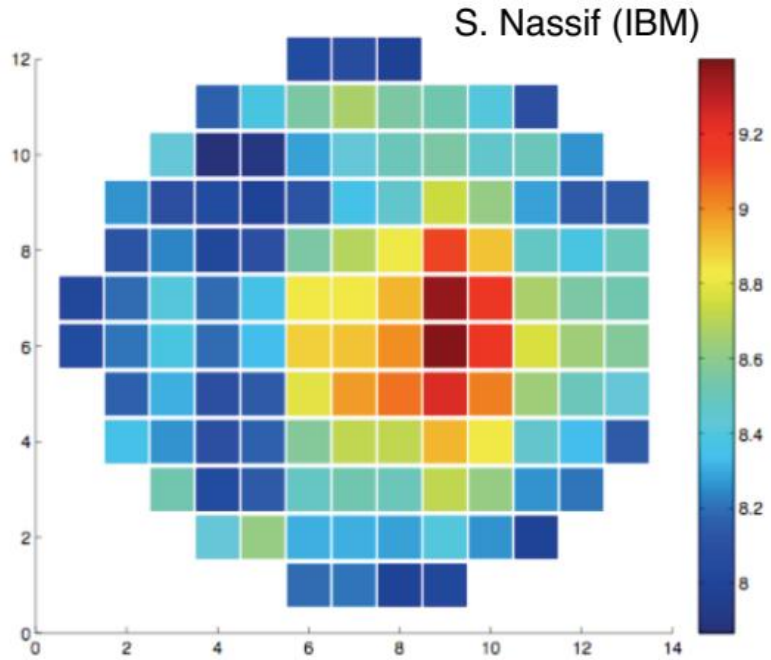
- The intra-die variations are deviations inside the same chip, in different locations of the same die.
- The inter-die variations are characterized by die-to-die, wafer-to-wafer, and lot-to-lot fluctuations.
- The die-to-die variations occur between two dies in the same wafer. The wafer-to-wafer variations happen between dies in different wafers, and the lot-to-lot variations appear between dies in different lots of production volume.

[A. Zimpeck et al., Mitigating process variability and soft errors at circuit-level for FinFETs, 2021]

Global Process-Induced Variability

Parameter distribution across a wafer



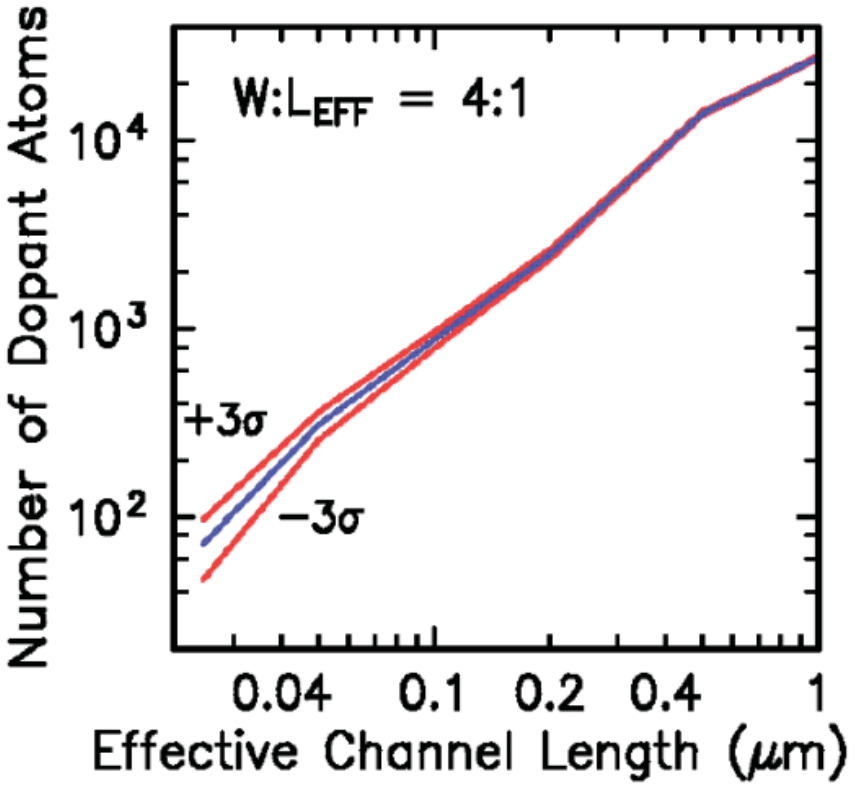


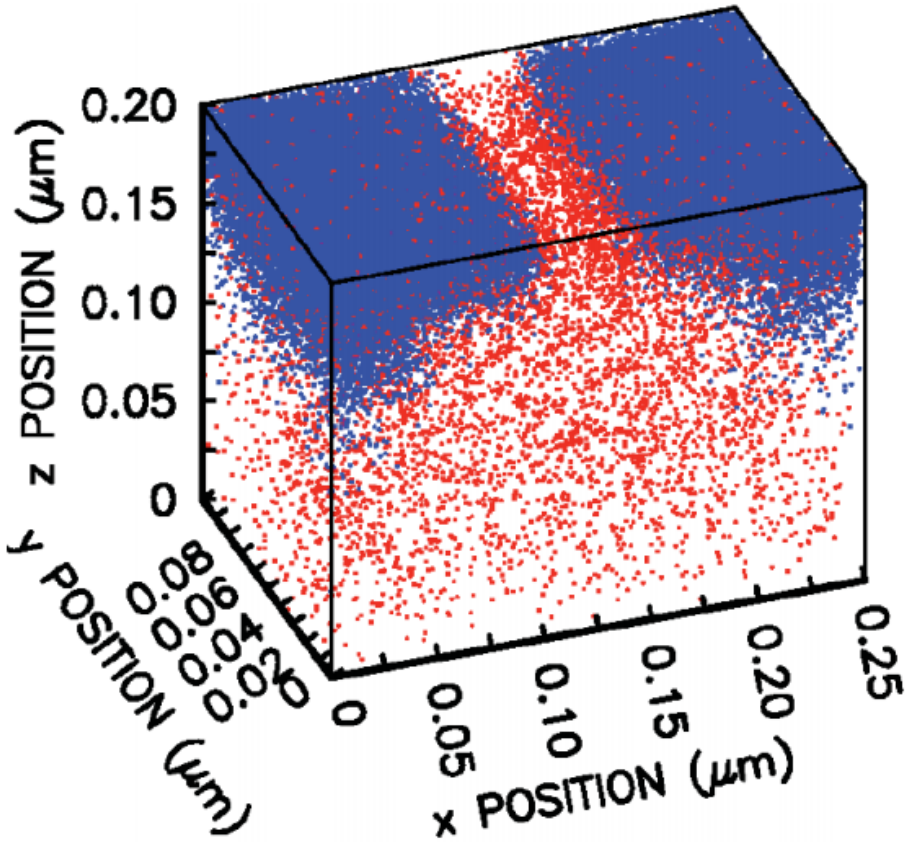
Equipment non-uniformity
Process variations
Across the wafer
From wafer to wafer

Dimensions, thicknesses, doping concentrations, shapes …

[A. Asenov, IEDM, 2015, SC]

Random Dopant Fluctuation (RDF)



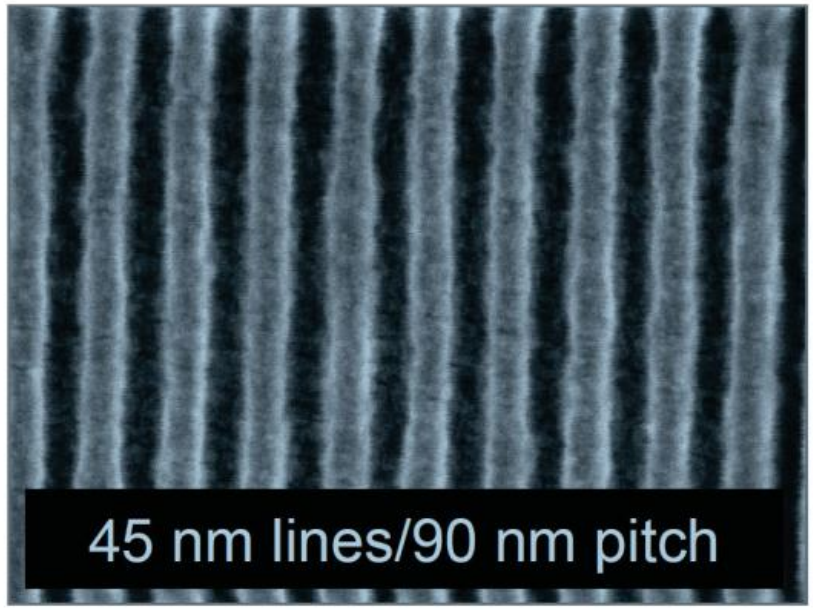


D.J. Frank, IBM

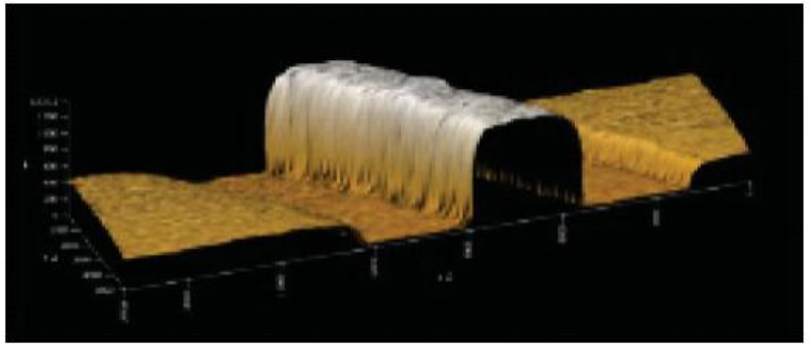
- Random discrete dopants (RDD)

[A. Asenov, IEDM, 2015, SC]

Line Edge Roughness (LER)



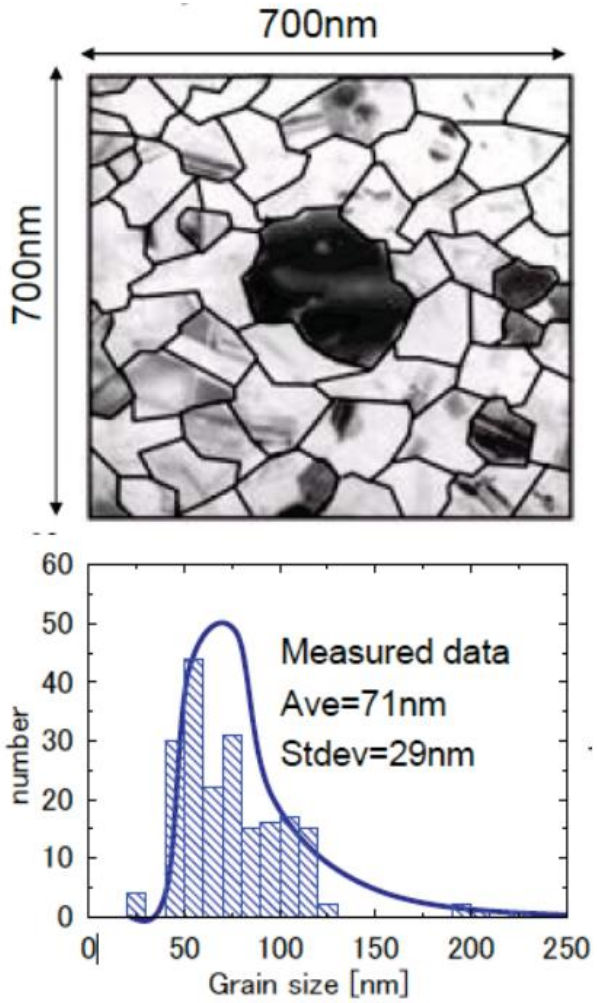
T. Brunner, ICP 2003



- Fluctuation in the total dose due to light quantization.
- Fluctuation in phonon absorption position
- Nanoscale non-uniformities in resist composition.

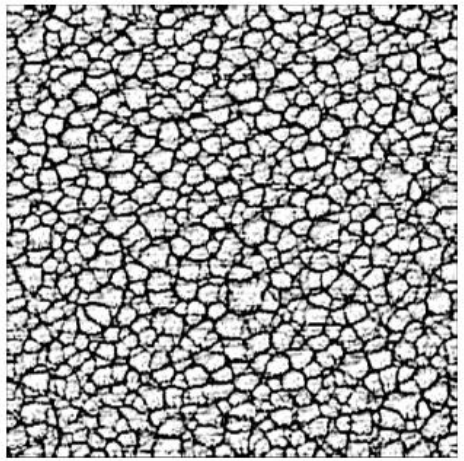
[A. Asenov, IEDM, 2015, SC]

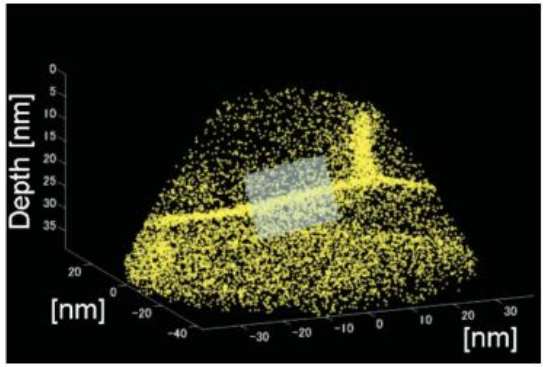
Poly-Silicon Granularity



A.T. Putra, ISDRS 2007

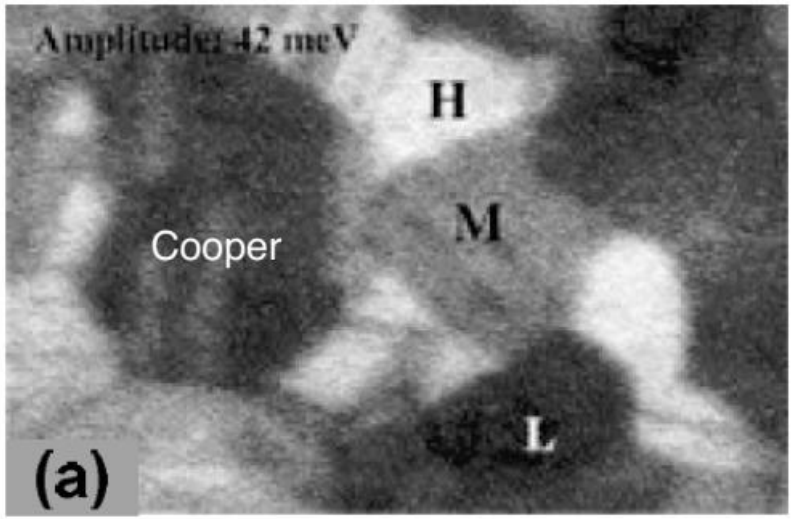
[A. Asenov, IEDM, 2015, SC]

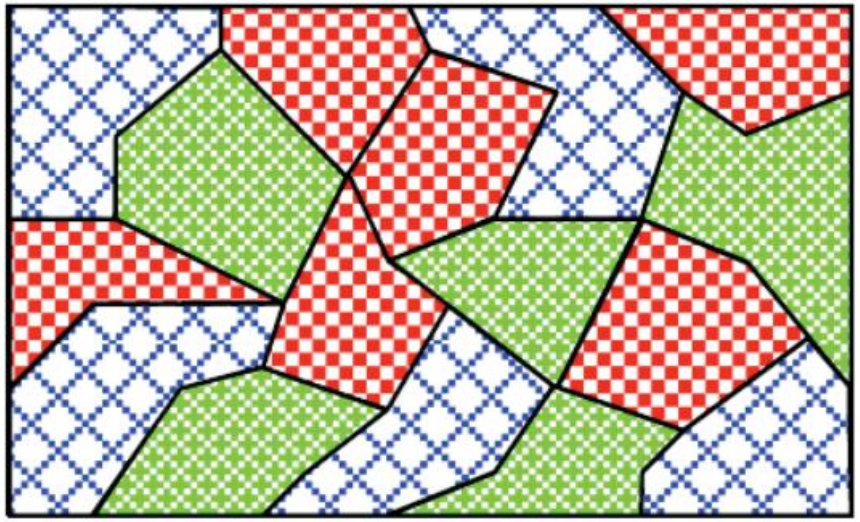




As dopant segregation
K. Inoue

Work Function Variation / Metal Gate Granularity

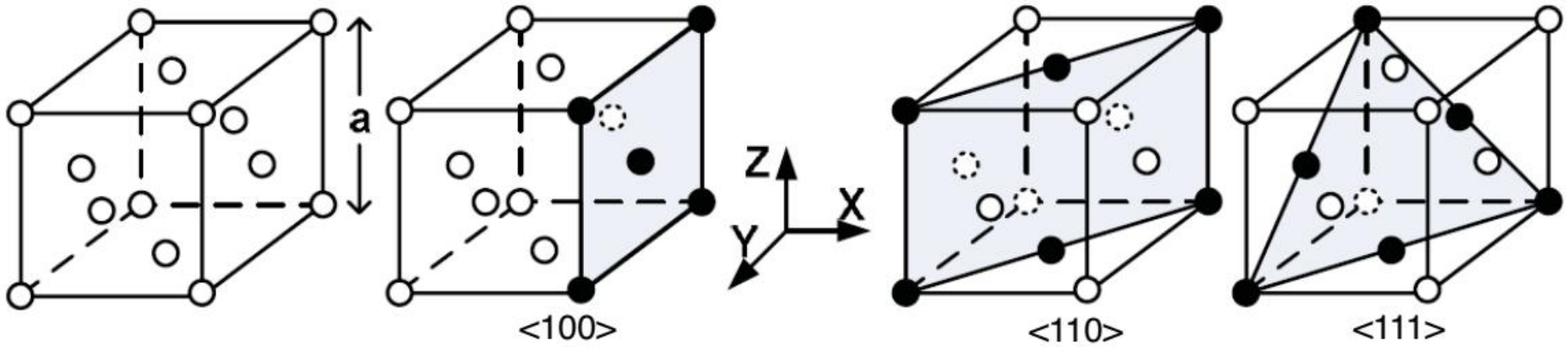




$\Phi_{111}(L)>\Phi_{100}(M)>\Phi_{110}(H)$

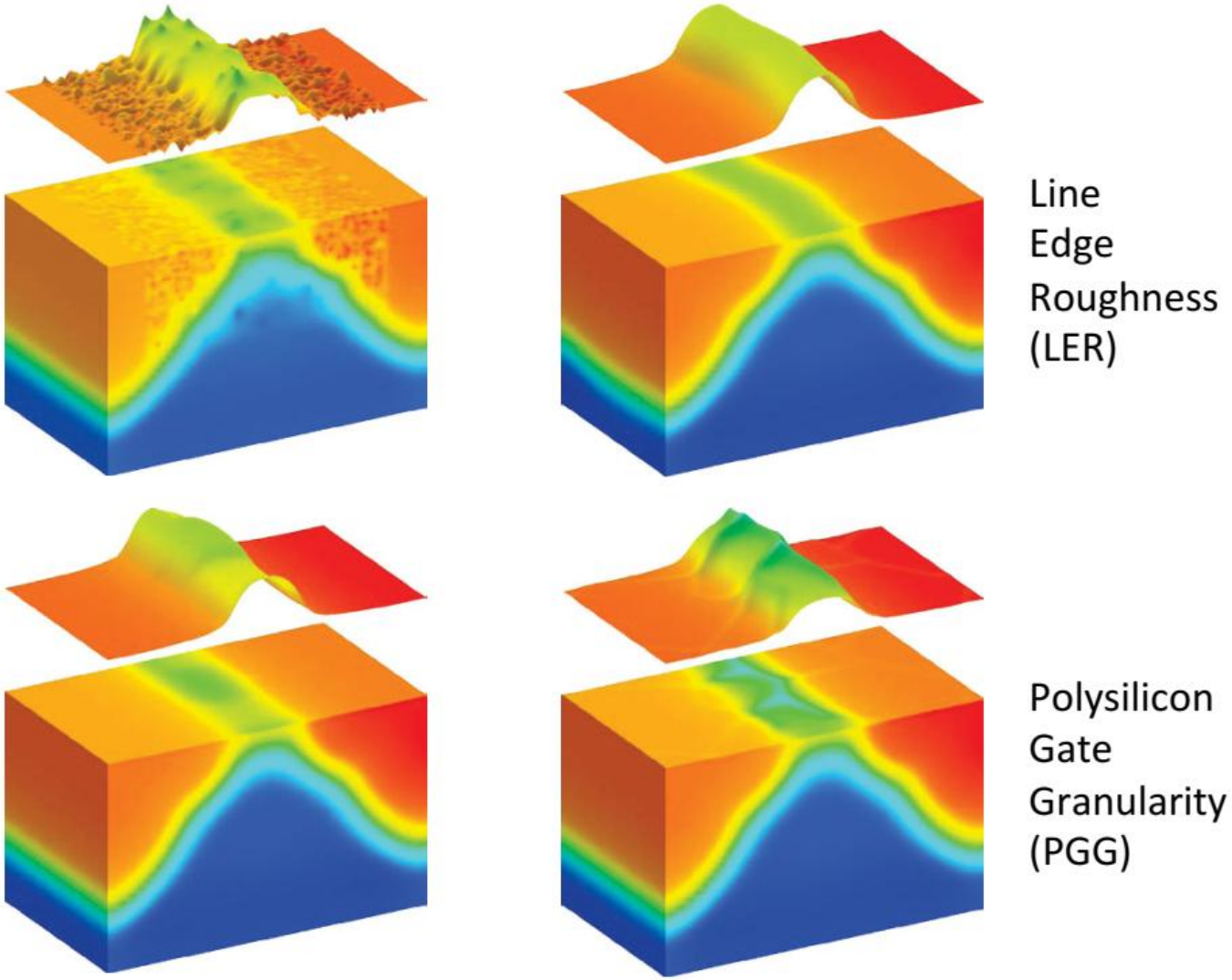
H. Dadgour et al.

Different surface density at different orientations



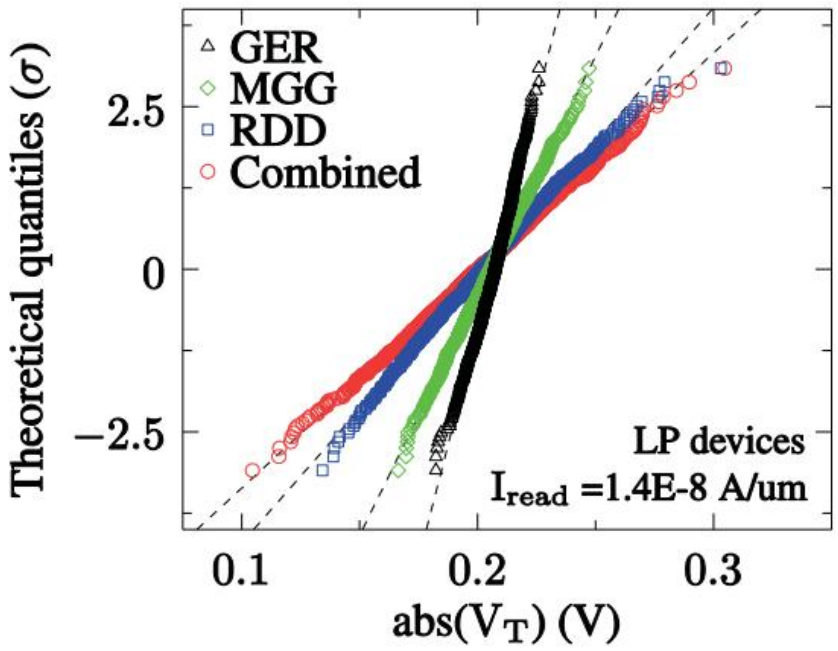
[A. Asenov, IEDM, 2015, SC]

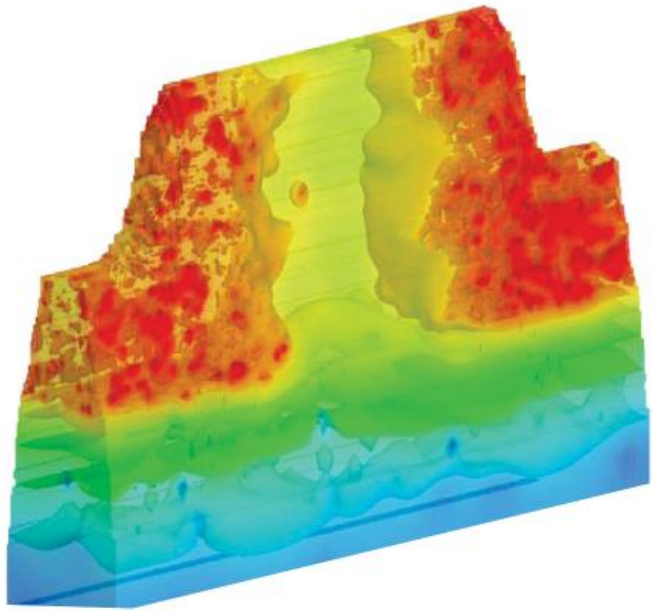
Physical Impact on Transistors



[A. Asenov, IEDM, 2015, SC]

Intel 22 nm FinFETs



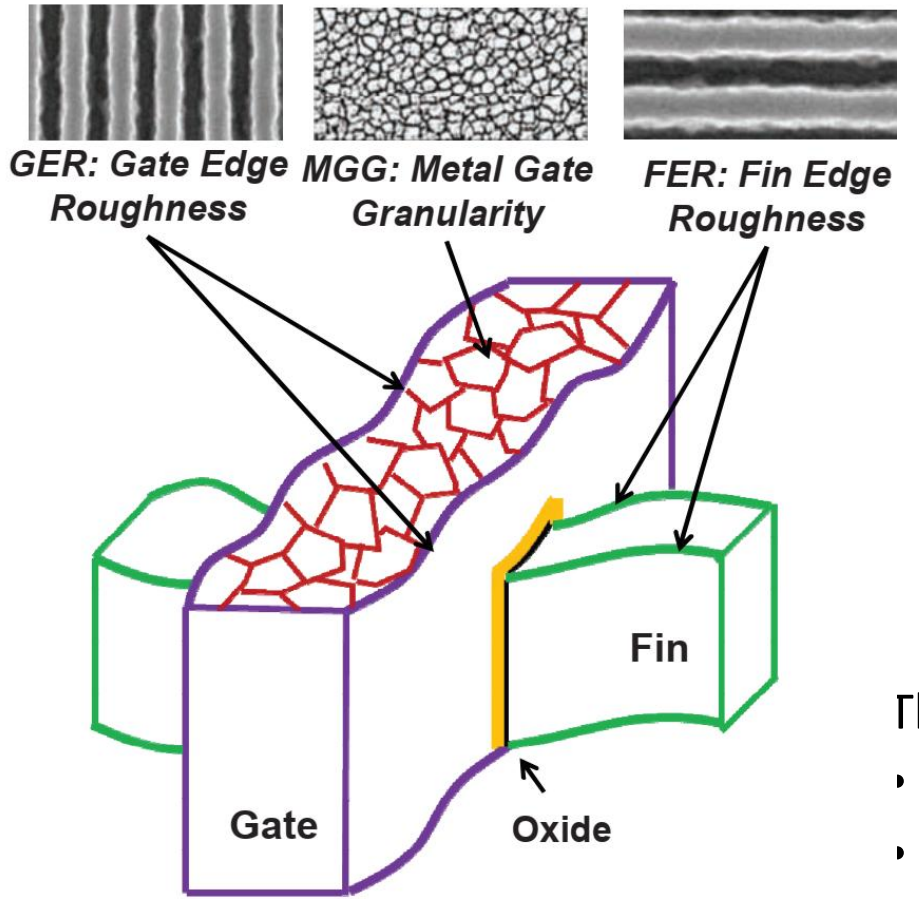


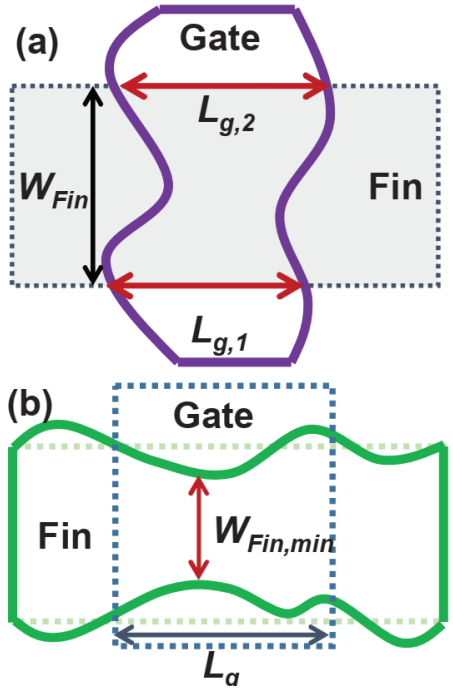
Measured σV_T=32mV
Simulated σV_T=31mV

The simulations based analysis suggest that the fin is heavily doped

[A. Asenov, IEDM, 2015, SC]

Random Variation Sources in FinFETs





Three major variation sources in FinFETs:

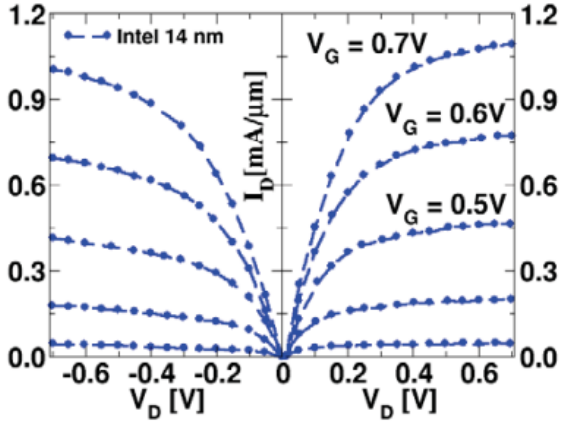
- metal gate granularity (MGG) / Work Function Variation
- gate edge roughness (GER)
- Fin edge roughness (FER)

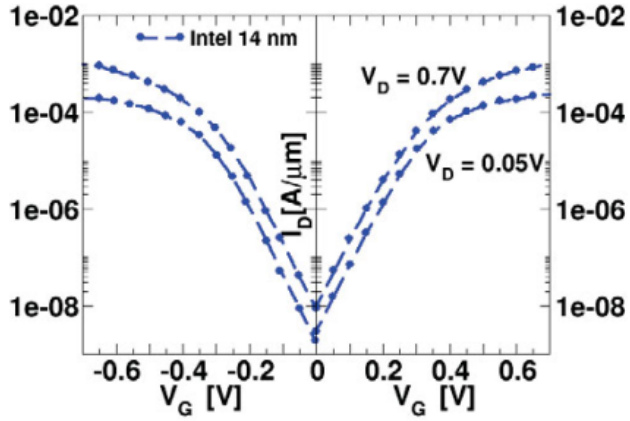
[X. Jiang et al., pp. 28.3.1-28.3.4, 2015 IEDM.]

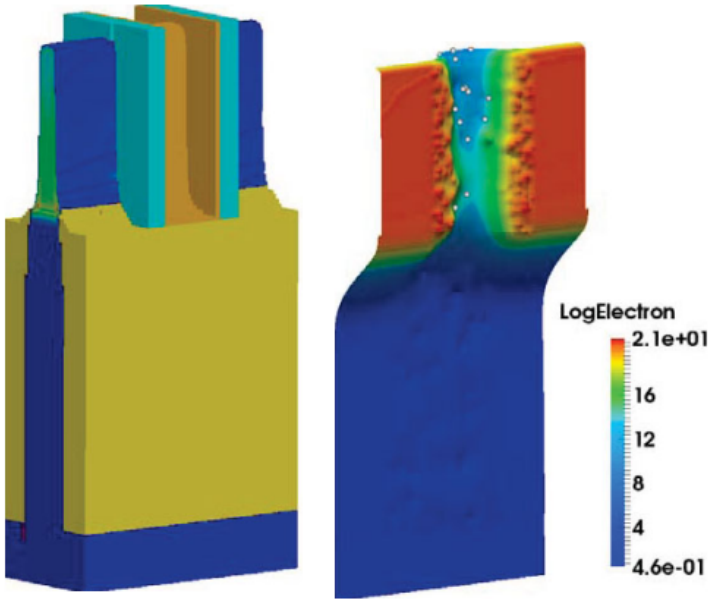
[A. Zimpeck et al., Mitigating process variability and soft errors at circuit-level for FinFETs, 2021]

Intel 14 nm FinFETs

Parameter	Value
H_{FIN}	42 nm
W_{FIN}	8 nm
L_G	20 nm
EOT	~ 0.9 nm
I_{DSAT}	1.04 mA/um
I_{DLIN}	0.237mA/um







Parameter		σV_T [mV]	A_{VT}
N_A [cm^{-3}]	$5x10^{17}$		
LER	2 nm		
MGG1	4 nm	18.0	1.07
MGG2	5 nm	19.5	1.18
MGG3	6 mm	21.7	1.27

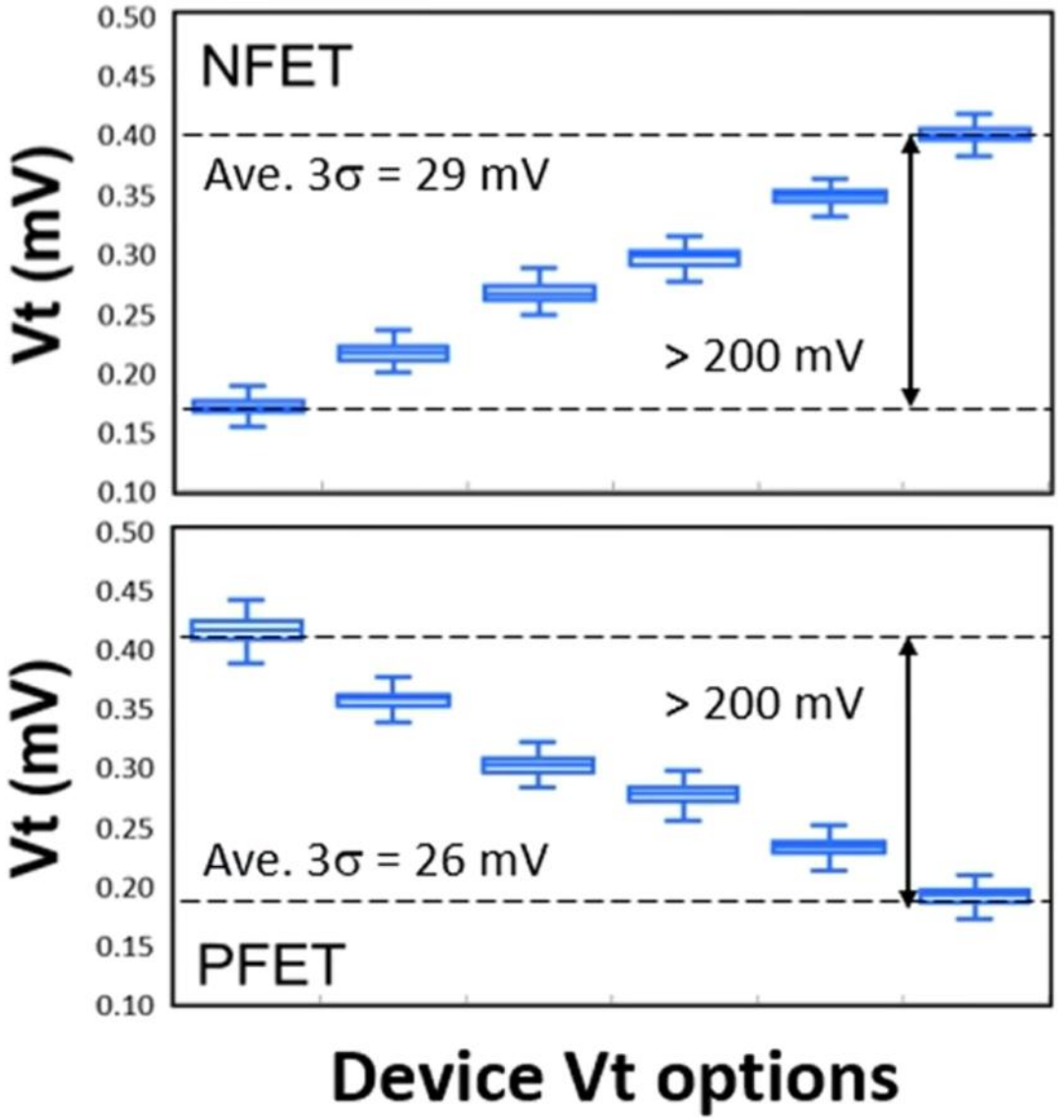
The simulation-based analysis suggests that the fin is lightly doped

- Lightly doped or undoped channel → reduced random dopant fluctuation

[A. Asenov, IEDM, 2015, SC]

TSMC 3nm FinFETs

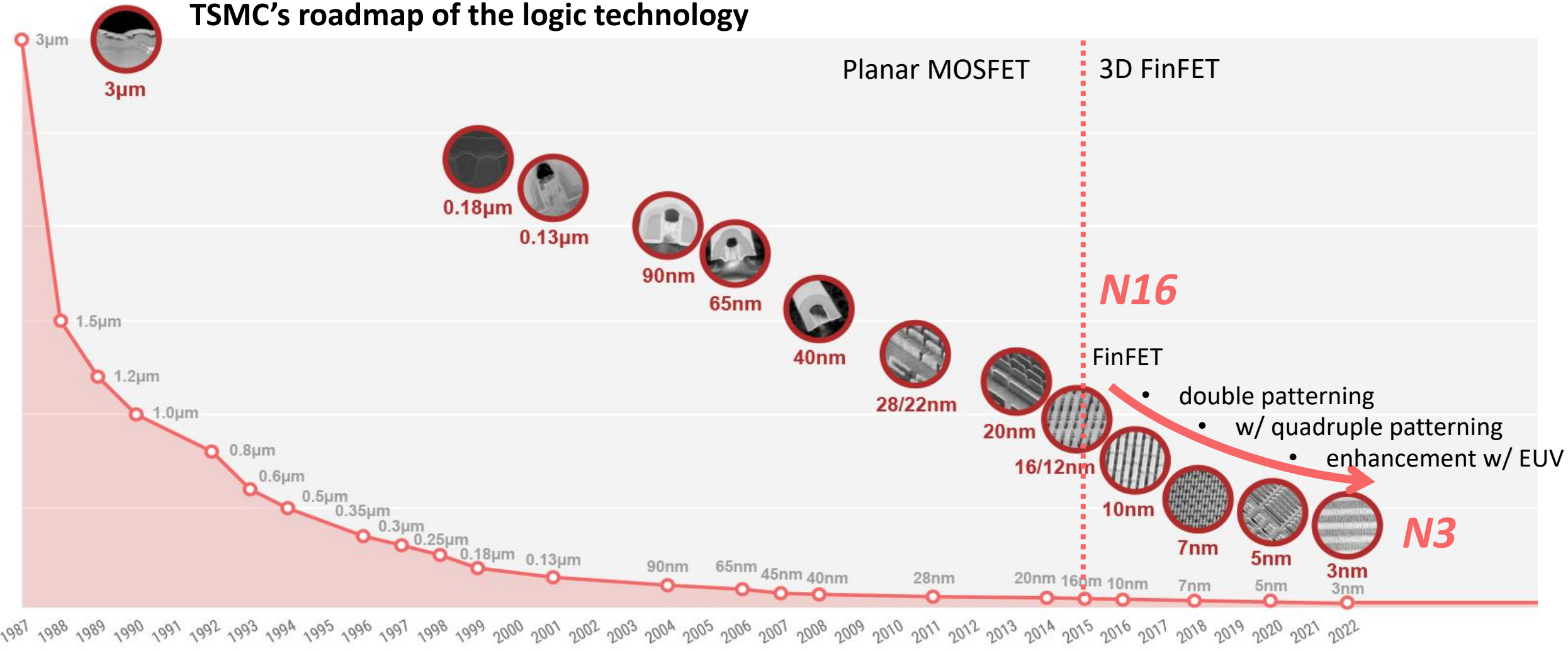
- CPP 45nm
- Multi-Vt device options
- Low Vt variations
  - 3σVt = 29mV for NFET
  - 3σVt = 26mV for PFET



[C.-H. Chang et al., 2022 IEDM, 27.1. (TSMC)]

Roadmap of Technology Nodes

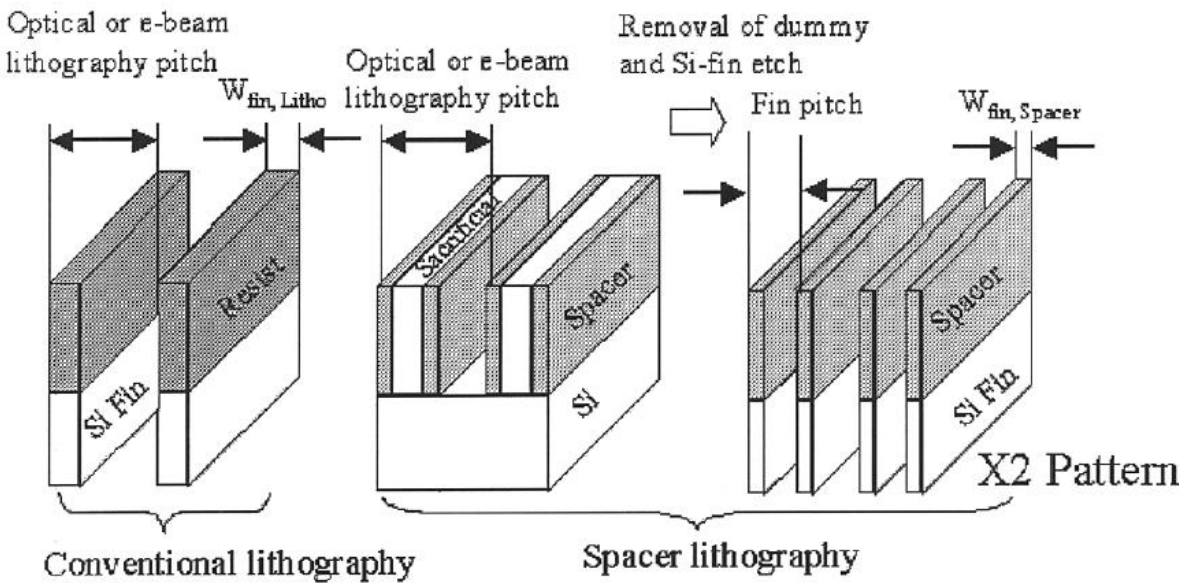
TSMC's roadmap of the logic technology



Ref: TSMC, link: https://www.tsmc.com/chinese/dedicatedFoundry/technology/logic

E. SICARD *et al.,* "Introducing 5-nm FinFET technology in Microwind," 2021.

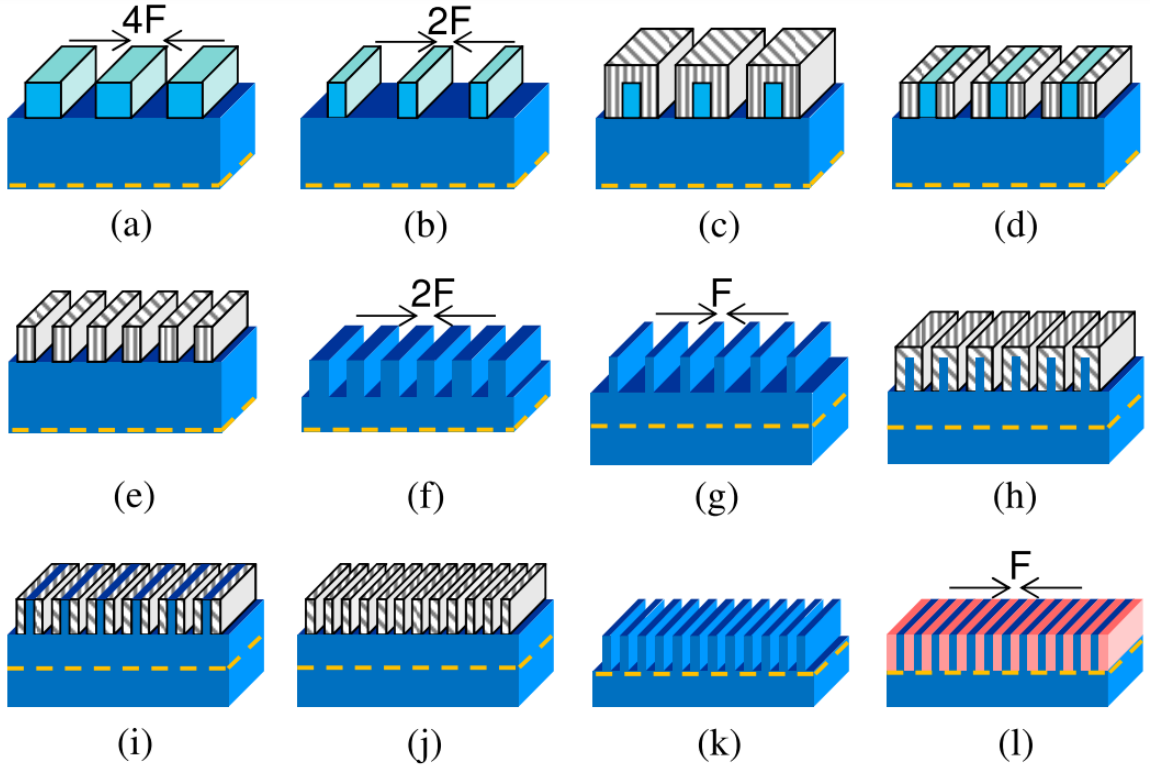
FEOL Technology: Gate Patterning



Self-aligned double patterning (SADP)

1. Dummy gate defined by conventional litho.
2. Thin layer deposited via CVD or ALD and etched back to form spacers
3. Spacers as a hard mask

Min.-sized feature down to 7 nm
(PR trimming technique only down to sub-20 nm)



Self-aligned quadruple patterning (SAQP)

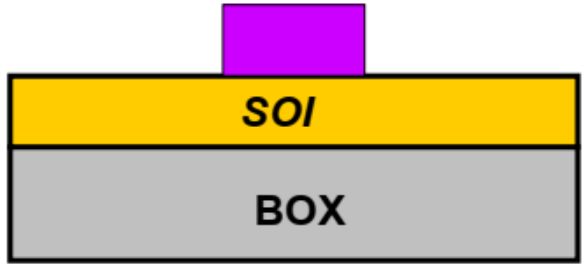
- 2 × SADP
- Better CD control
- 7-nm FinFET technology node

Ref: [L. Chang et al, Proceedings of the IEEE. 91. 1860 - 1873., 2003.] [S. Baudot et al., SISPAD, 2018.] [C. Auth et al., IEEE IEDM, pp. 29.1.1-29.1.4, 2017.] [C. Kodama et al., 18th IEEE ASP-DAC, pp. 267-272, 2013.]

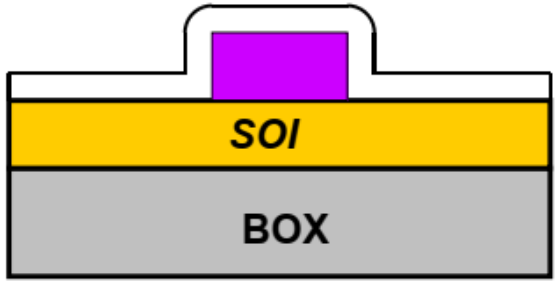
Fin Pattering: SADP (Self-aligned double patterning)

Spacer Lithography
a.k.a. Sidewall Image Transfer (SIT)
or Self-Aligned Double Patterning (SADP)

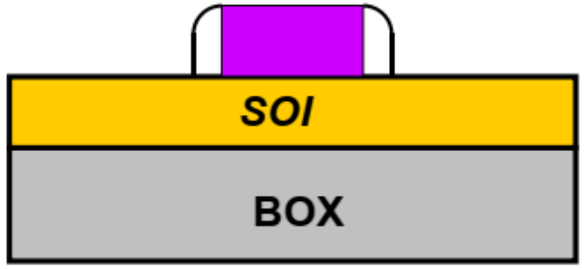
1. Deposit & pattern sacrificial layer



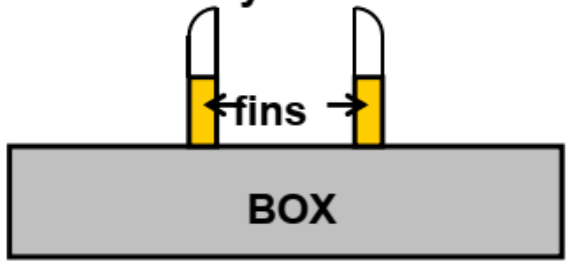
2. Deposit mask layer (SiO_2 or Si_3N_4)

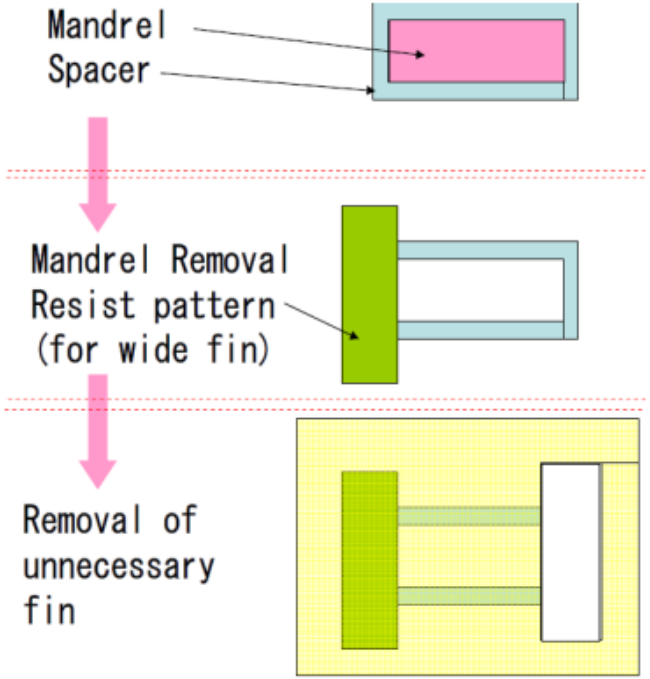


3. Etch back mask layer to form "spacers"



4. Remove sacrificial layer; etch SOI layer to form fins

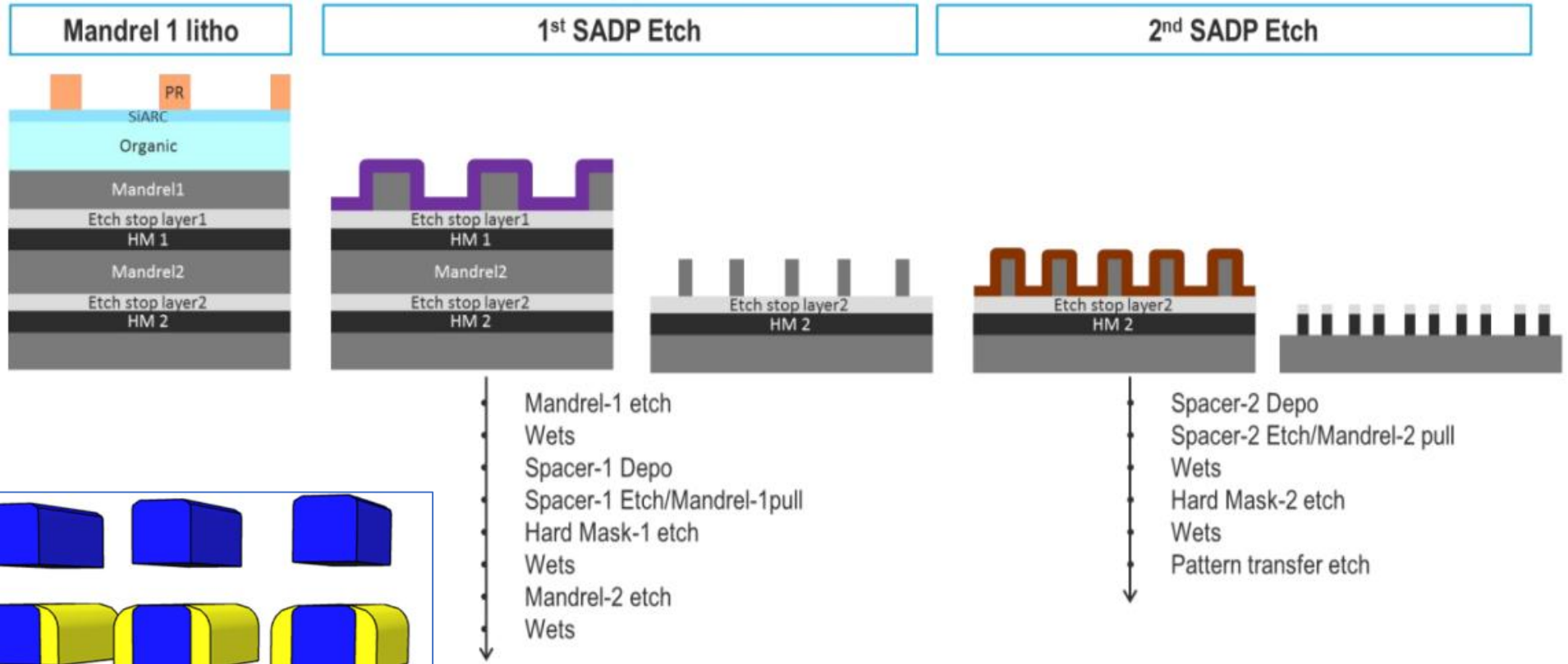


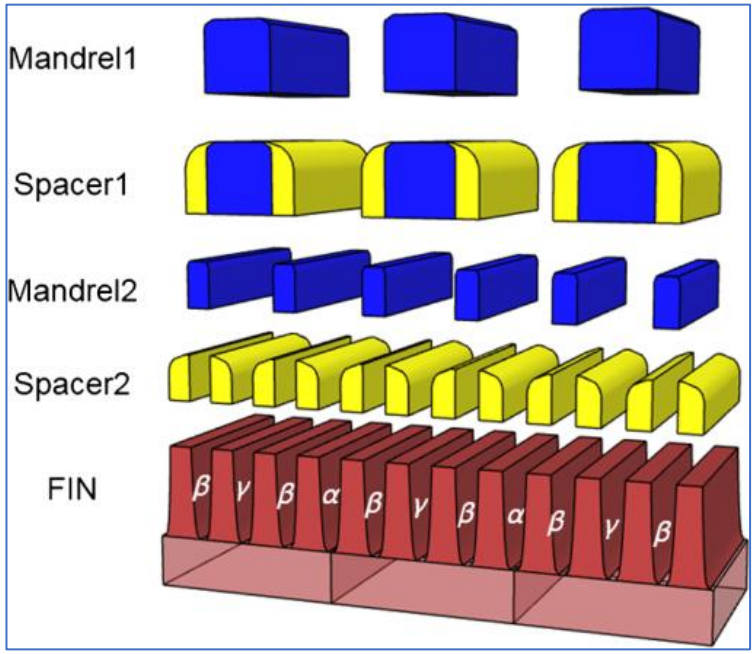


- **Note that fin pitch is 1/2× that of patterned layer**
- **Extra lithography steps required to etch the unused fins.**

[T.-J. K. Liu, VLSI, 2012, SC]

Self-Aligned Quadruple Patterning (SAQP)

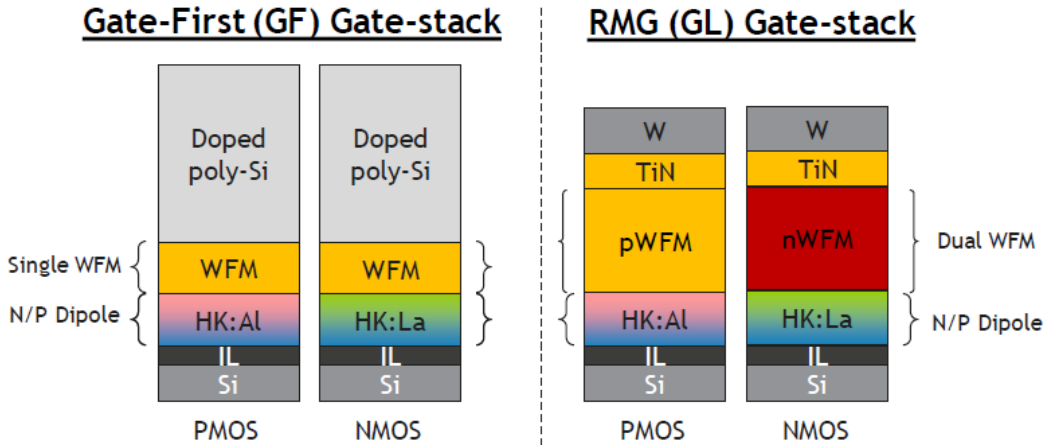




Self-aligned quadruple patterning (SAQP) is already the established process to be used for patterning fins for 7 nm and 5 nm FinFETs.

FEOL Technology: Gate Stack Formation

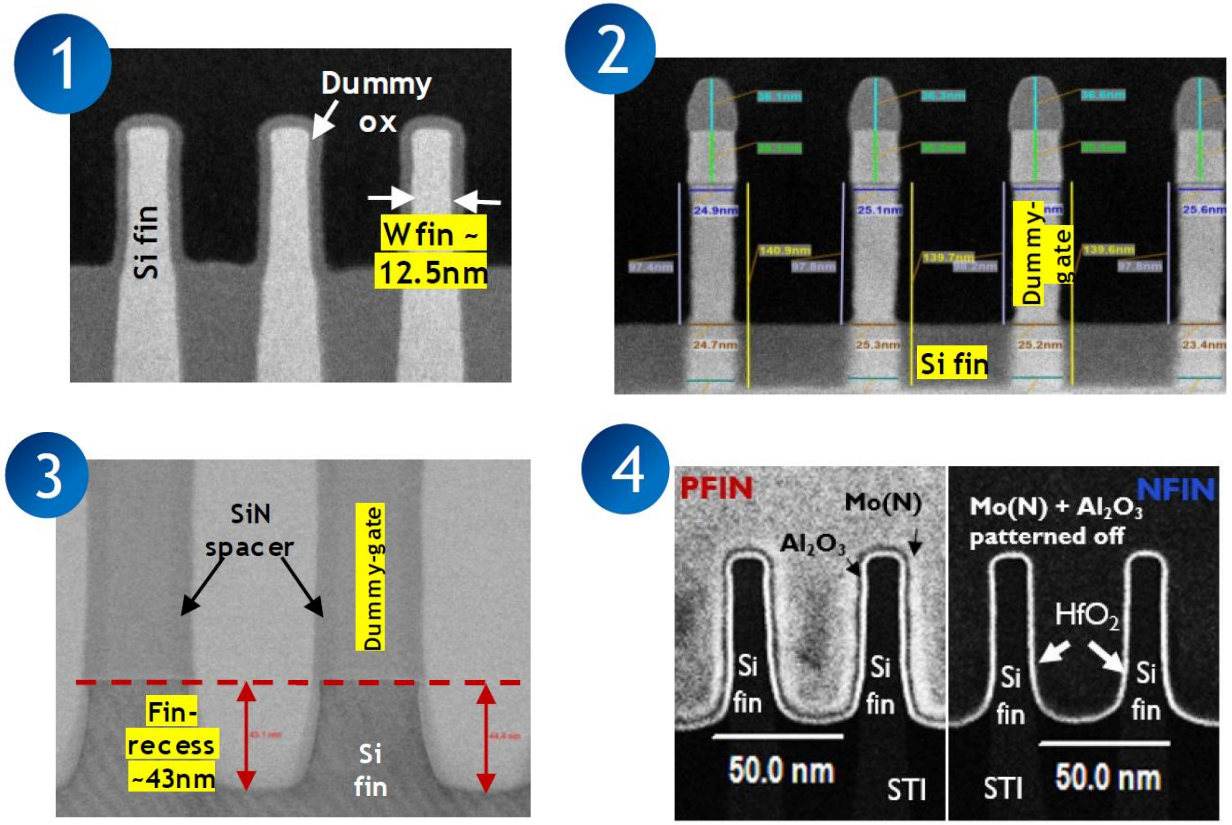
- **GF vs. GL**



GF: higher thermal budget on the gate from S/D activation, leading to stronger fermi-level pining

- **Steps of GL process**
  - Fin 1
  - Dummy-gate patterning 2
  - Gate-spacer (GSP) 3
  - Embedded epi → Thermal process for S/D activation
  - ILD-0
  - RMG 4 (replacement metal gate)

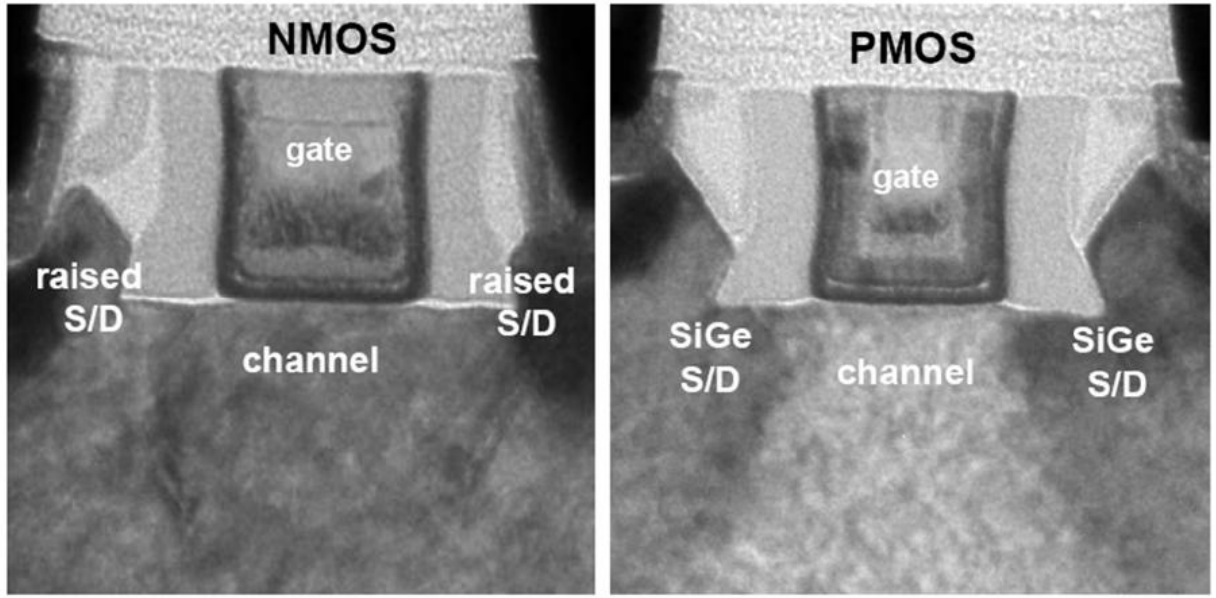
- **Process flow of GL**



Ref: [J. Ganguly et al., IEEE VLSI, T17-1, 2024.]

FEOL Technology: HKMG

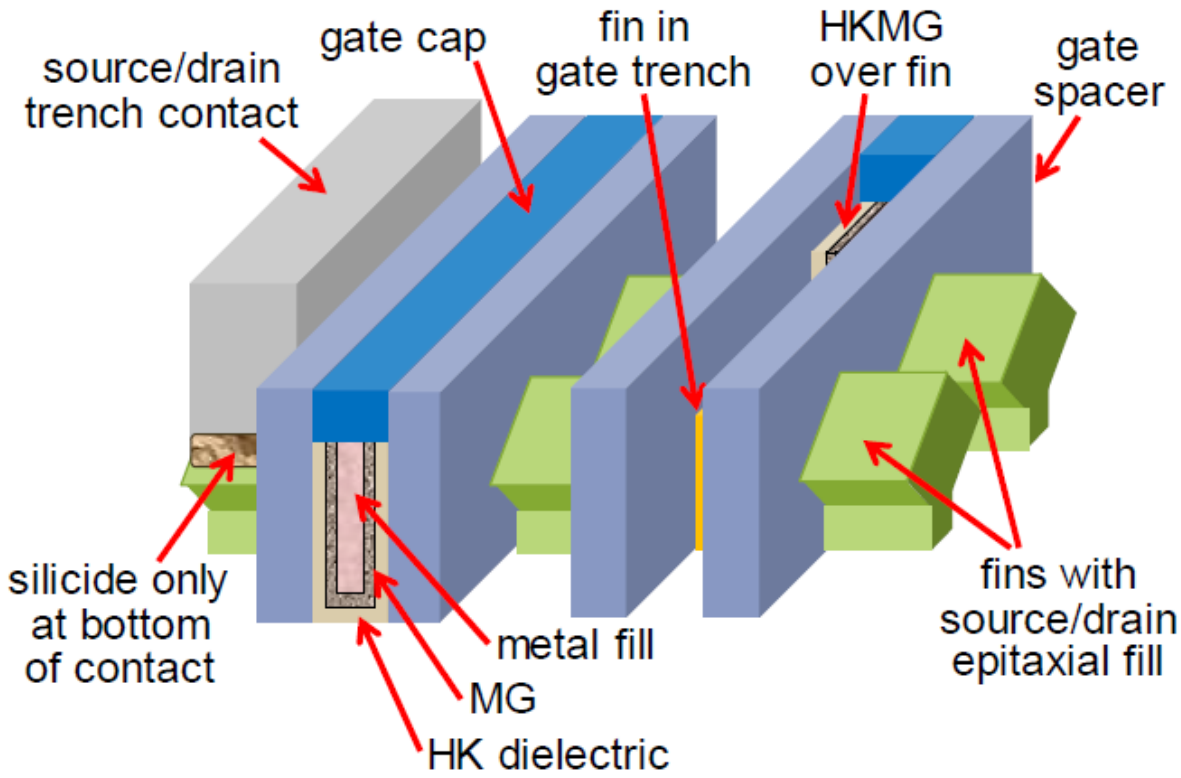
HKMG technology is adopted since 32 nm ~ 28 nm nodes to replace poly-Si/SiON stack.



Gate last process with HKMG technology:
better gate control with C_{ox} ↑

Ref: [P. Packan et al., IEEE IEDM, pp. 659-662, 2009.]
[A. L. S. Loke et al., IEEE Custom Integrated Circuits Conference (CICC), 2018.]

Cut view of FinFETs with HKMG



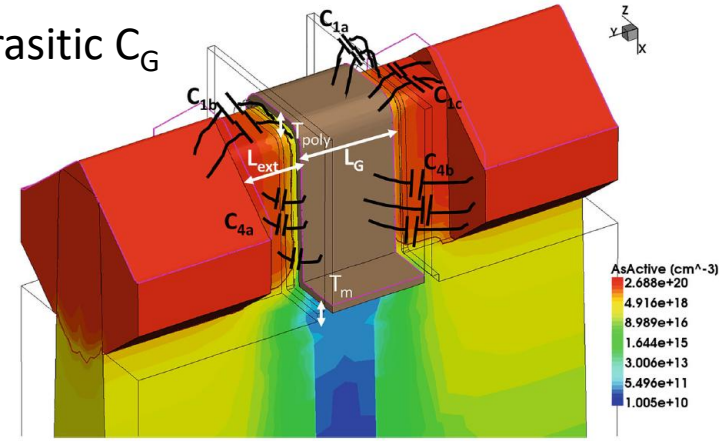
Through **ALD technology**

- Precise thickness and stoichiometric control
- V_{TH} tuned by various metal for NMOS and PMOS, respectively

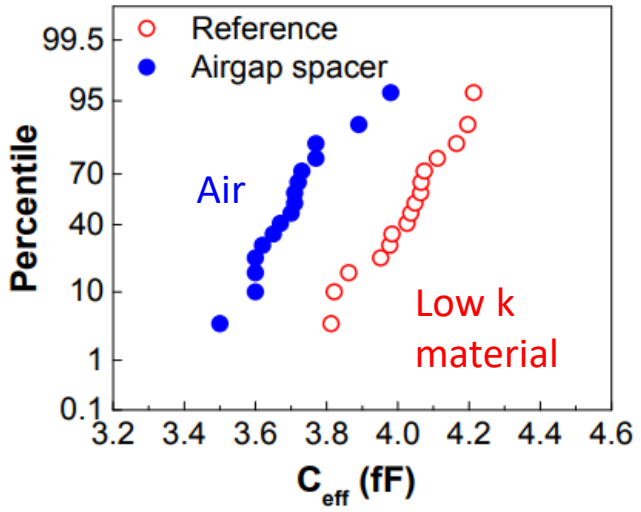
FEOL Technology: Spacer

- **Reduce C_{GS}: shorter delay**

Parasitic C_G

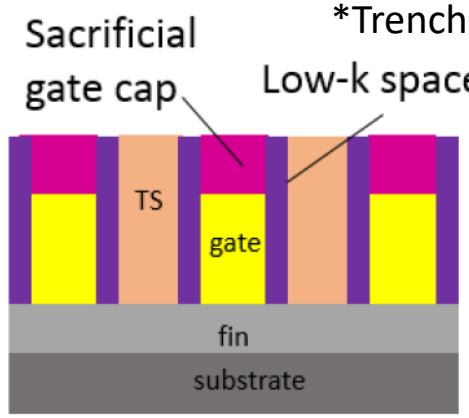


10 nm FinFET CMOS (IBM)

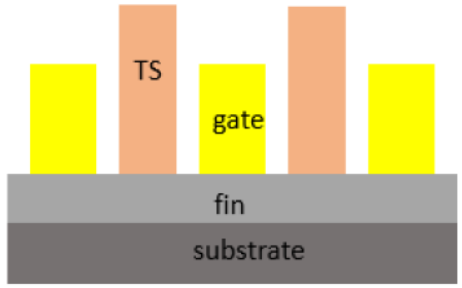


- **Process flow of air spacer**

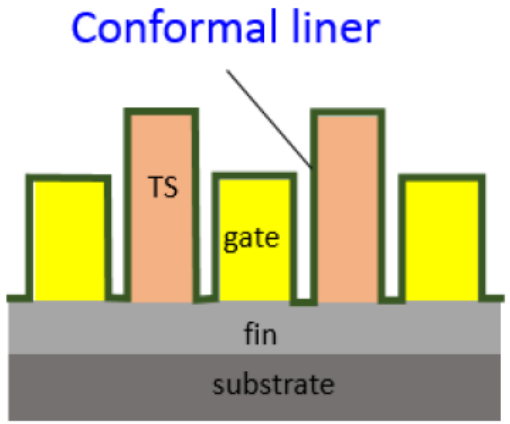
*Trench Contact (Trench siliciadation, TS)



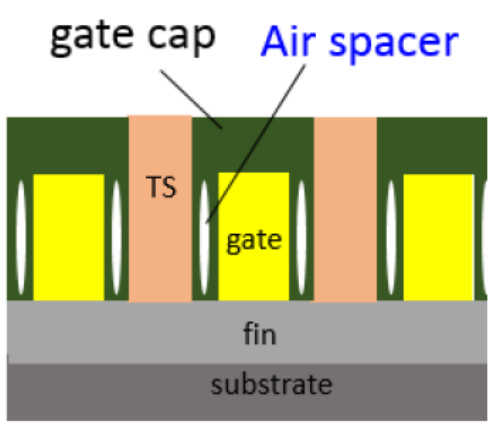
1) FinFET after RMG and trench silicidation (TS)



2) Remove sacrificial gate caps and spacers



3) Deposit conformal dielectric liners



4) Deposit non-conformal dielectric to pinch off

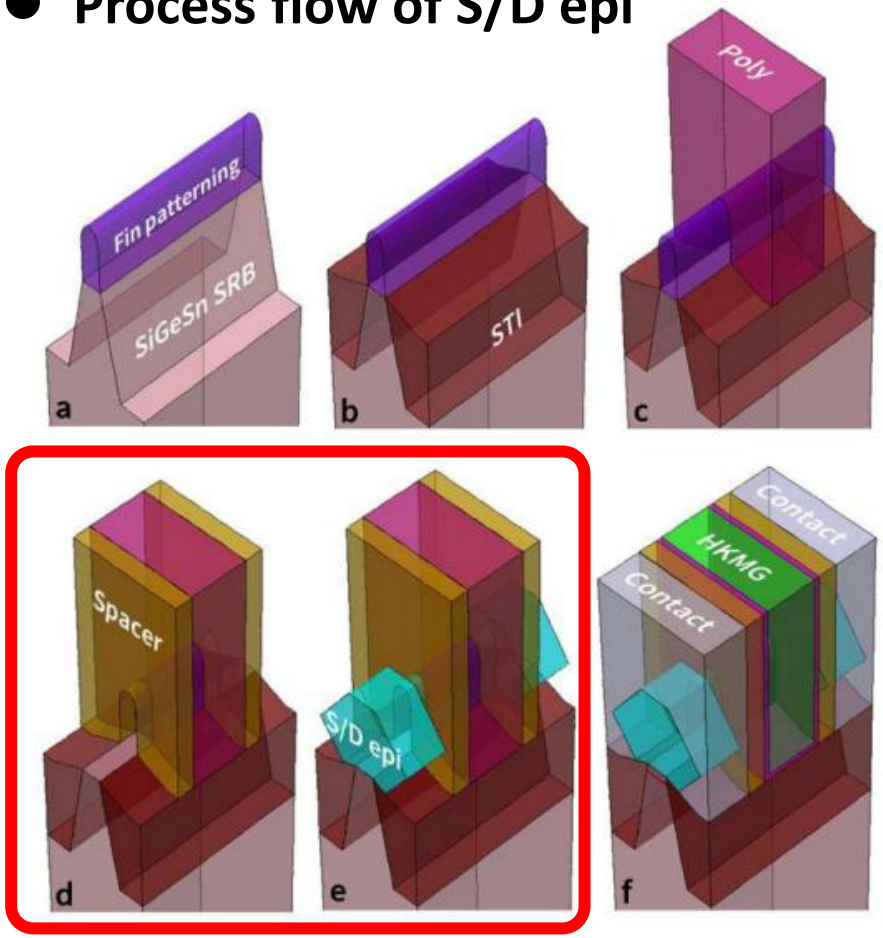
Critical process of the air spacer

1. Remove the gate cap and the low-k spacer
2. Deposit conformal liner
3. Deposit non-conformal dielectric to pinch off
4. Planarize

Ref: [K. Cheng et al., IEEE IEDM, pp. 17.1.1-17.1.4, 2016.] [S. M. Sharma et al., Silicon, 13, 4109–4122, 2021.]

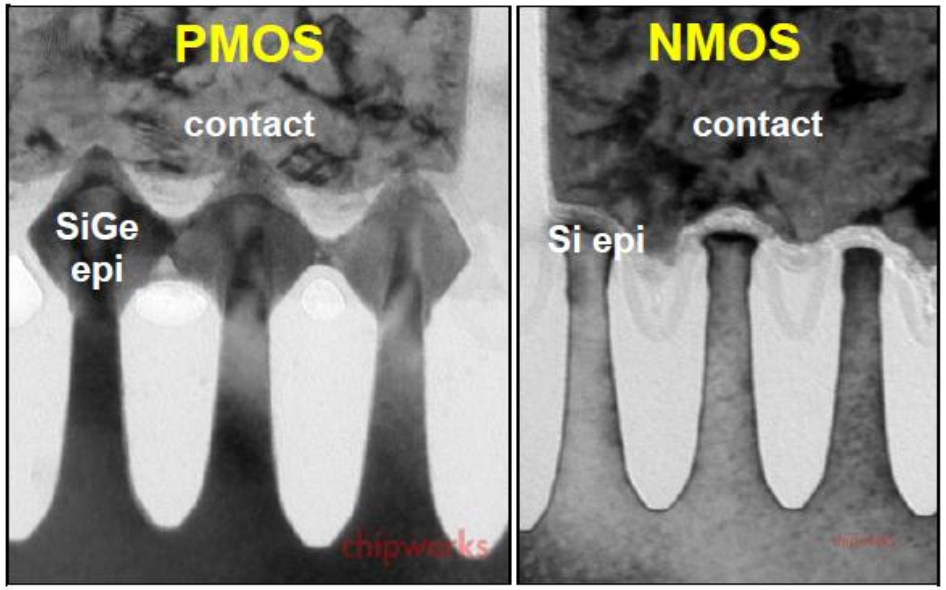
FEOL Technology: SD Epi

- **Process flow of S/D epi**



After spacer formation and S/D recess, *in situ* doped epi S/D were regrown.

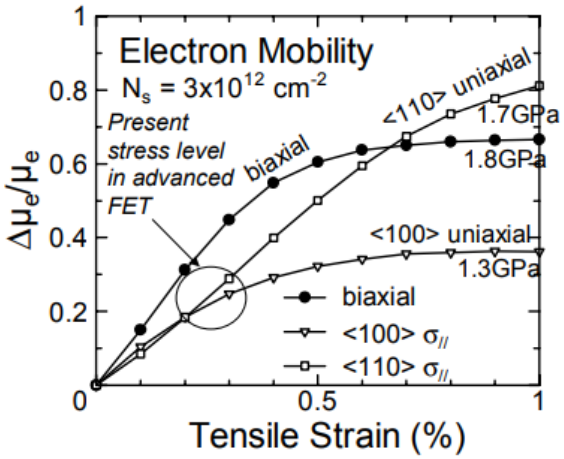
- **Various dopants for N and P FinFETs (TSMC, 16nm)**



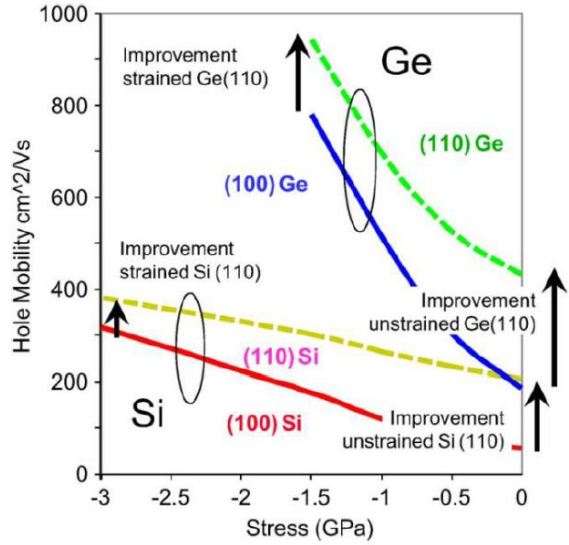
Compressive stress (SiGe): hole mobility ↑

Tensile stress (SiC): electron mobility ↑

- **Electron mobility**



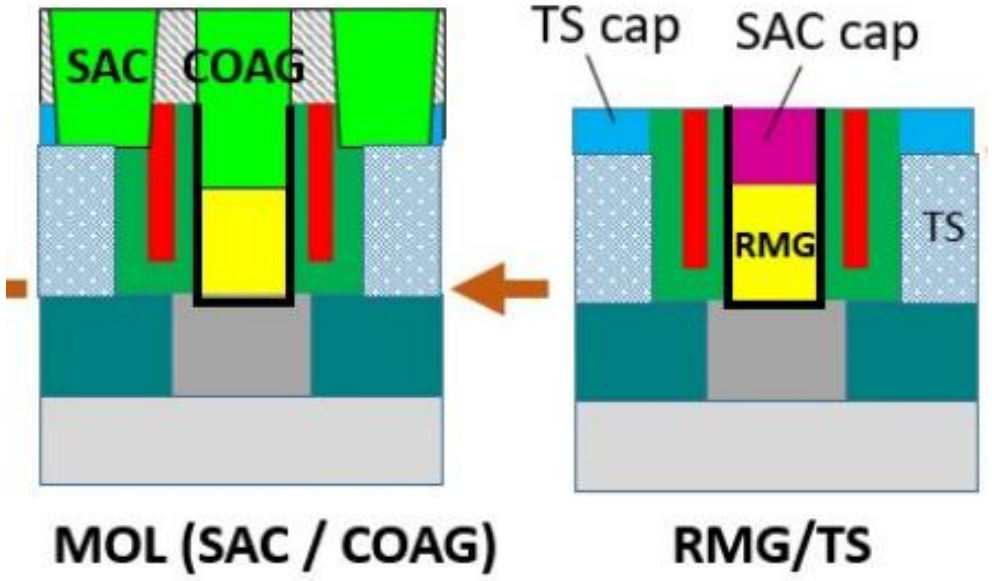
- **Hole mobility**



Ref: [S. Gupta et al., IEEE TED, 61, 5, 2014.] [D. James, 27th Annual SEMI Advanced Semiconductor Manufacturing Conference (ASMC), pp. 324-329, 2016.] [K. Uchida et al., IEEE IEDM, pp. 129-132, 2005.] [P. S. Goley et al., Materials, 7, 2301-2339, 2014.]

MEOL Technology: Self-aligned contacts

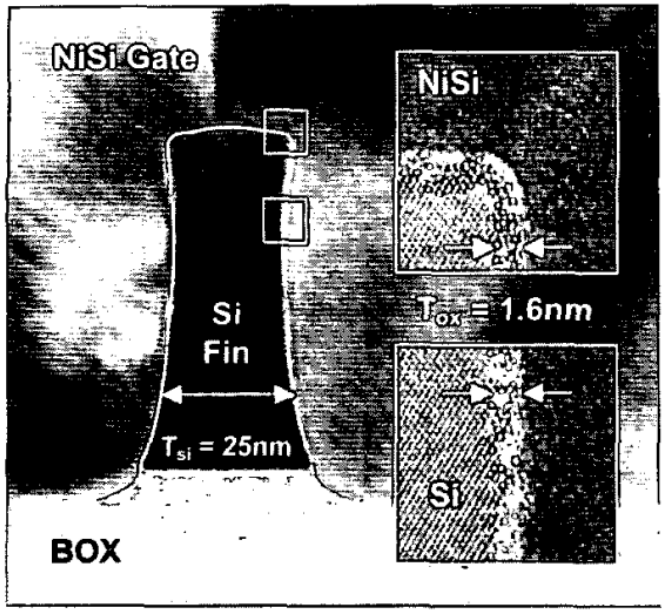
- **Self-aligned contacts**



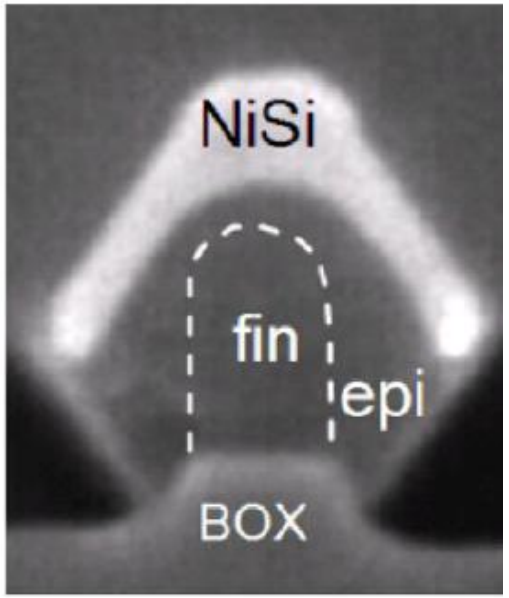
1. Various capping materials for gates and S/D
2. Selective etching to remove specific region
3. Self-aligned metal contact at target region

- **Contact: silicide (lower R_c)**

NiSi silicide on gates
(IBM, 2002)



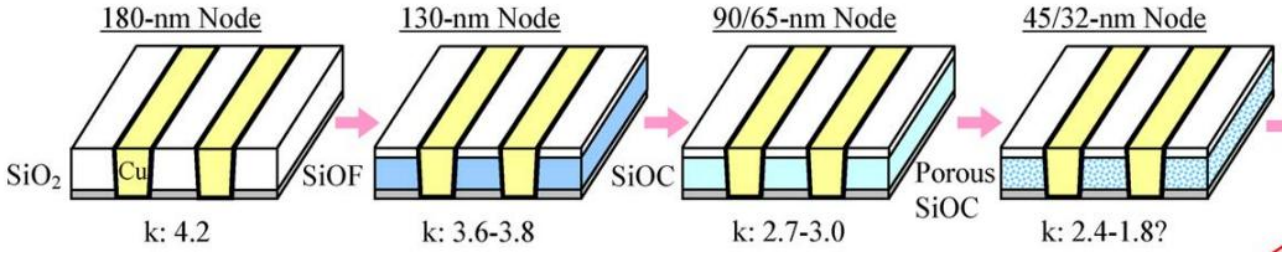
NiSi silicide on S/D
(IBM, 2015)



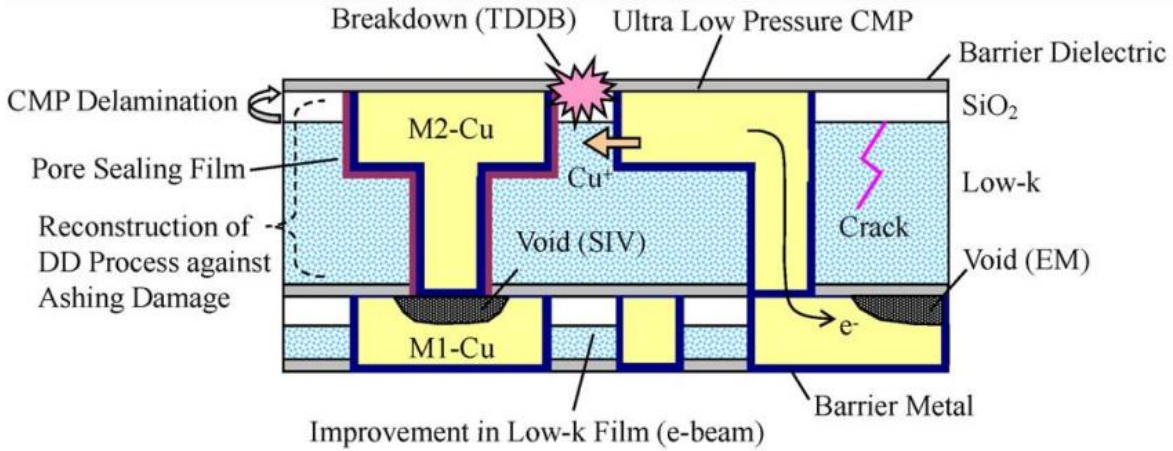
Ref: [K. Cheng et al., IEEE VLSI, pp. 1-2, 2020.] [J. Kedzierski et al., IEEE IEDM, pp. 247-250, 2002.] [N. Breil et al., Microelectronic Engineering, 137, 79–87, 2015.]

BEOL Technology: Low k insulator

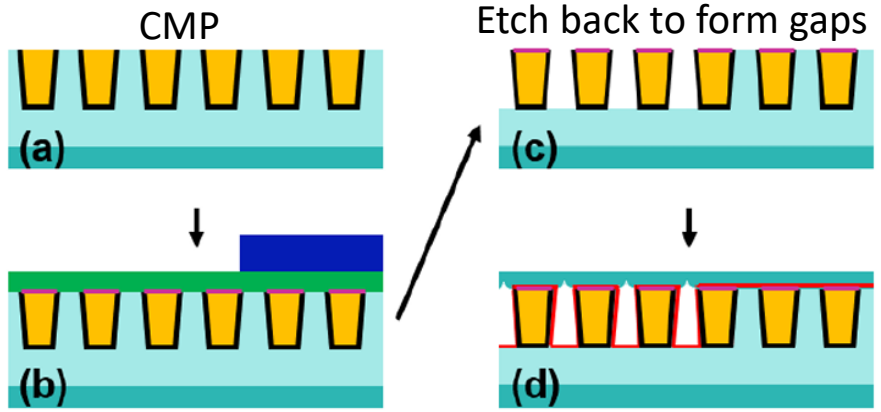
- **Low k materials**



- **Reliability issue of low k materials**

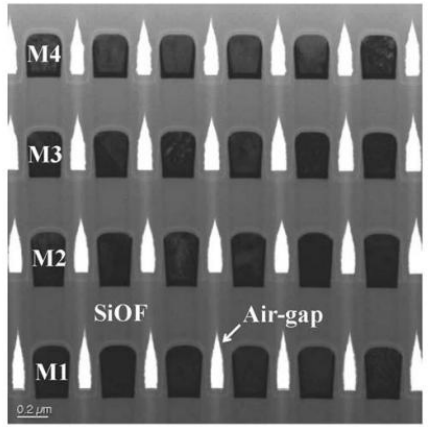


- **Air-gap interconnects**

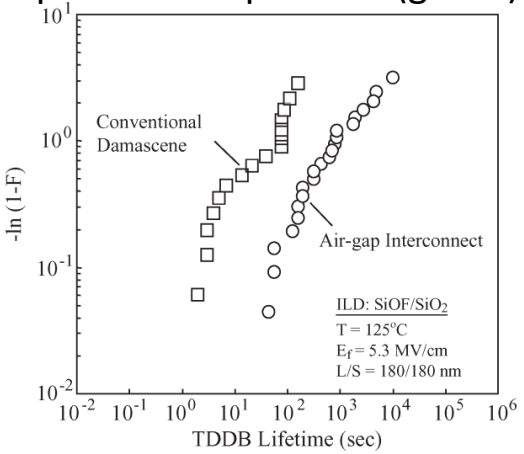


Define Region

Conformal low k (red) and dielectric pinch-off deposition (green)



TEM image

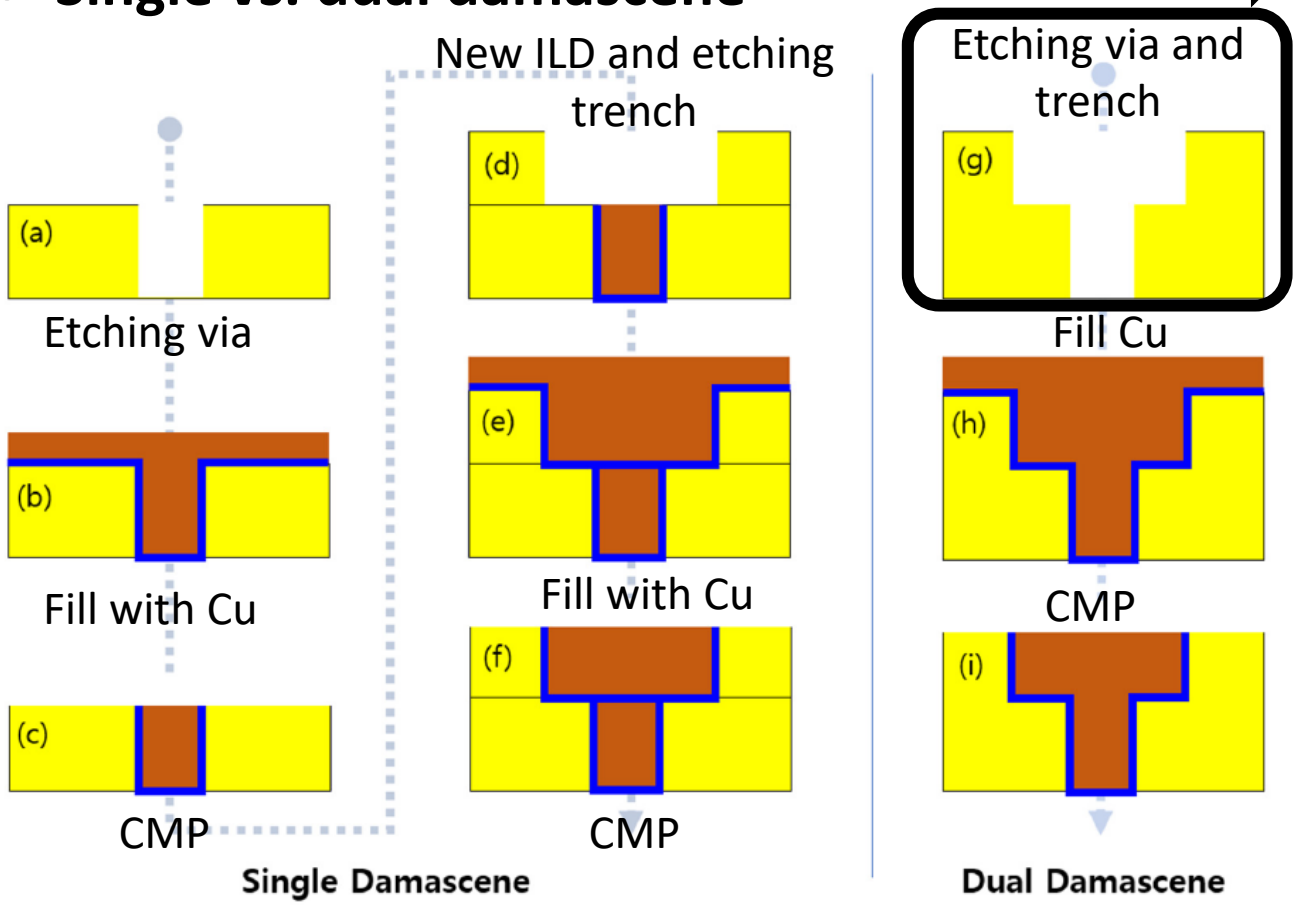


Enhanced life time with a lower C_{total}

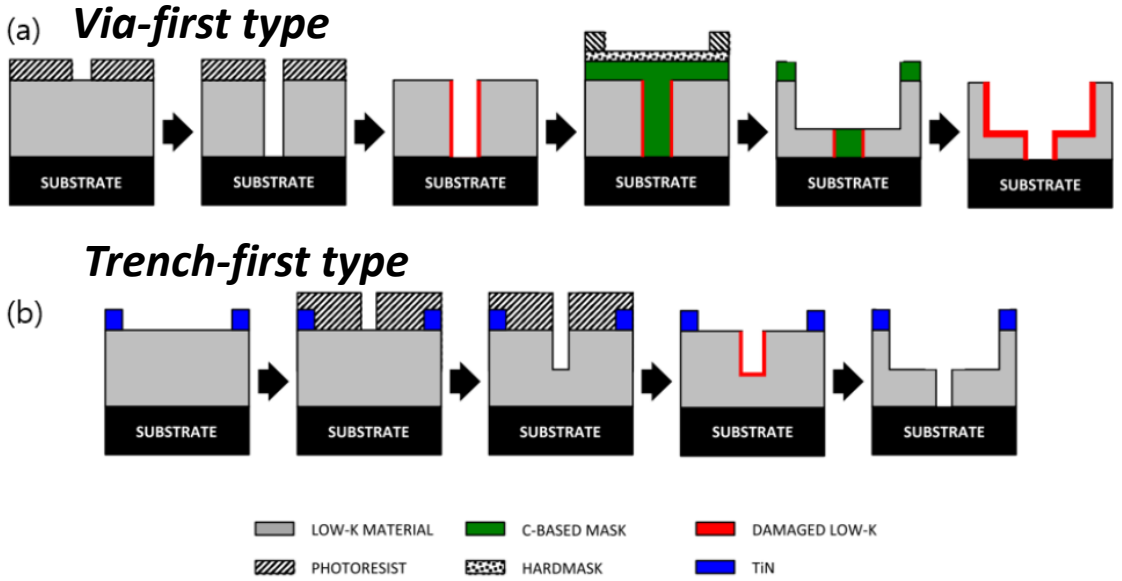
Ref: [24] S. Nguyen et al., ECS Trans., 85, 6, 25-39, 2018. [25] J. Nohuchi et al., IEEE TED, 56, 11, 2009.

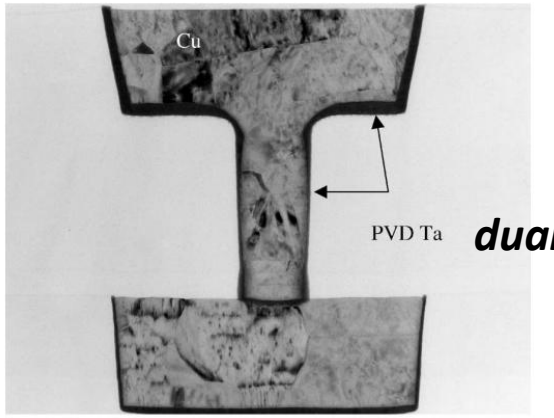
BEOL Technology: Dual Damascene

- **Single vs. dual damascene**



- **Key process of dual damascene**

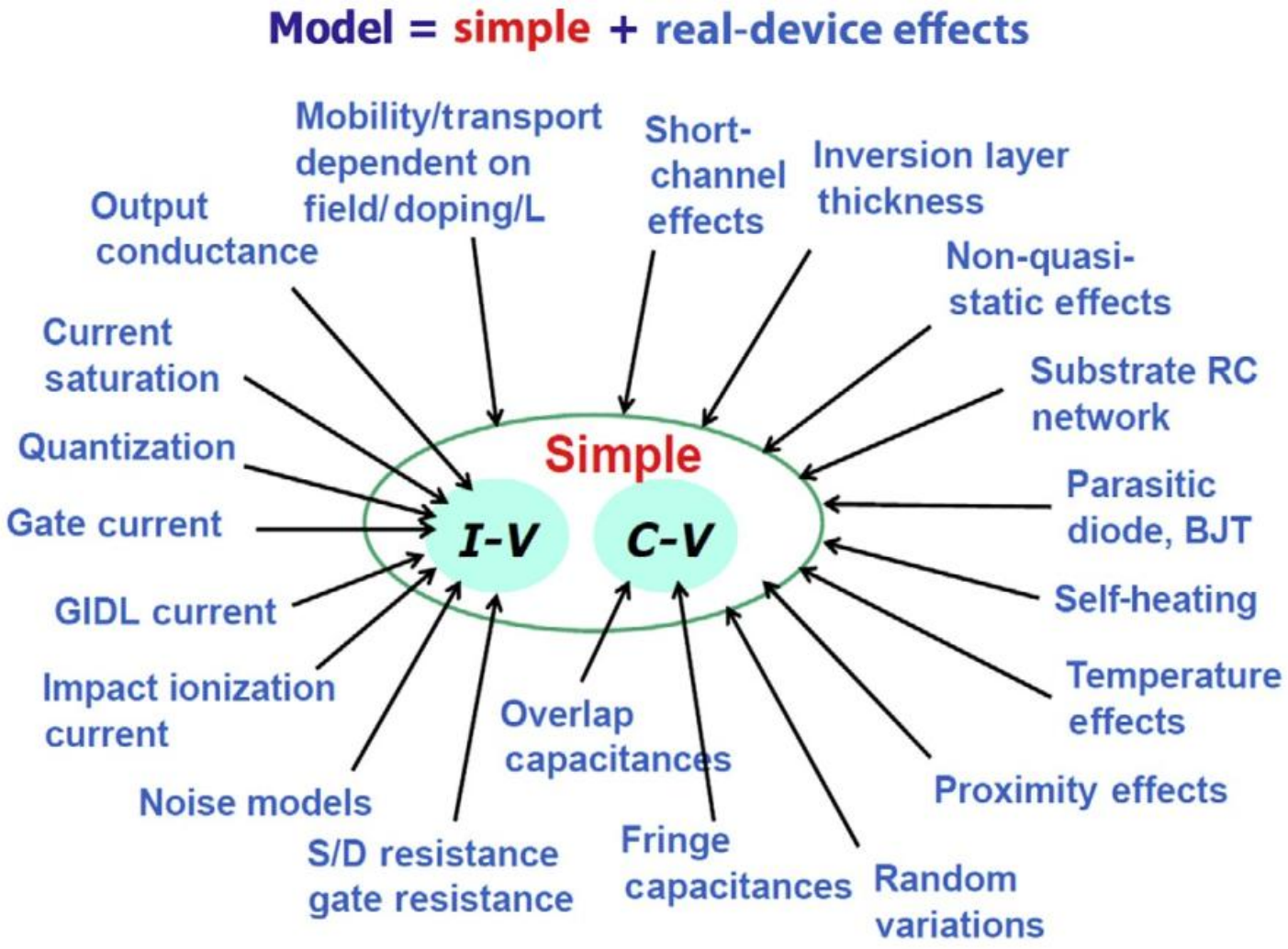




TEM image of Cu dual-damascene interconnects

Ref: [26] H.-W. Kim et al., Electronics, 11, 2914, 2022. [27] T. Usui et al., IEEE TED, 53, 10, 2006.

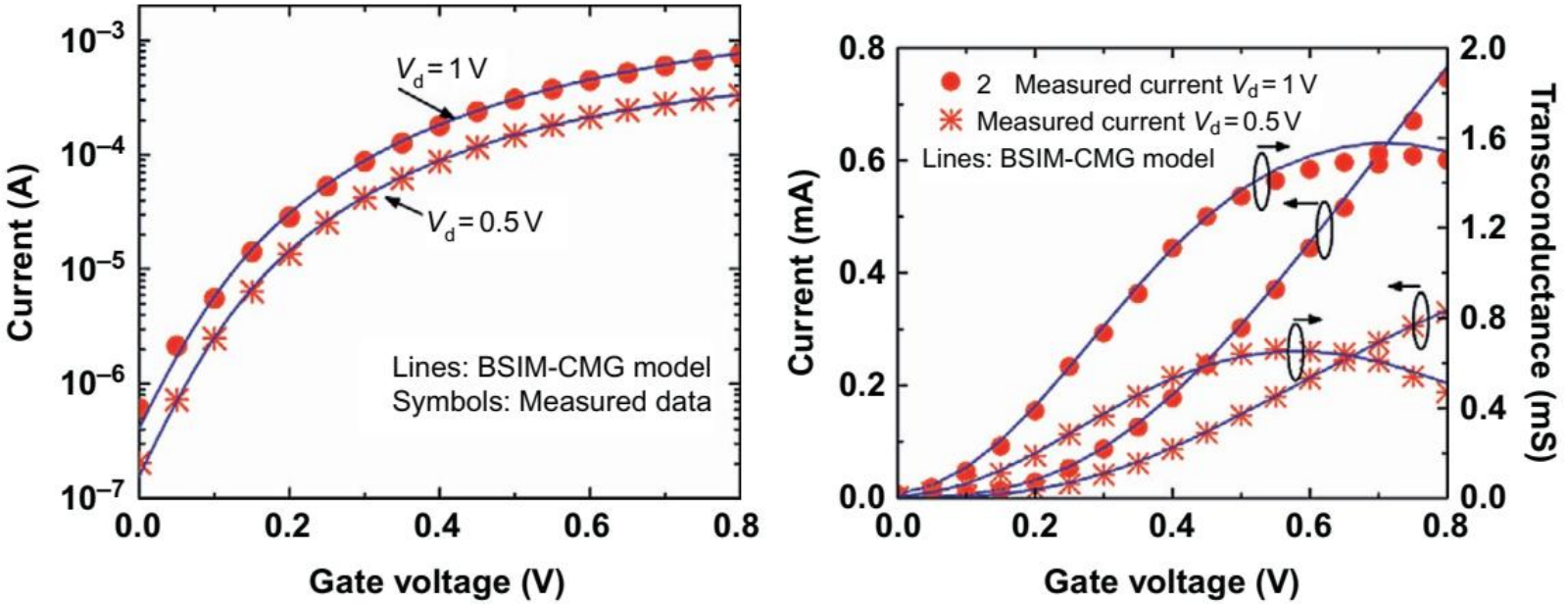
Compact model



A compact model is formed from a simple (long-channel) model and numerous real-device models. The latter constitute 90% of the model and are responsible for the global accuracy.

[YS Chauhan et al., "FinFET Modeling for IC Simulation and Design", 2015]

BSIM-CMG (common multigate)



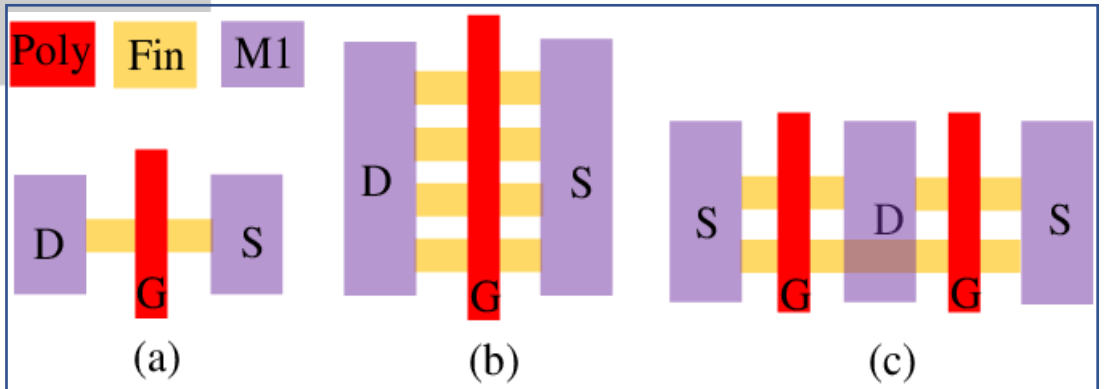
- Transfer characteristics of an FinFET modeled with the BSIM-CMG model showing the subthreshold and strong inversion region drain-current model.
- common multigate: all the multiple gates are electrically connected and share a common gate voltage.
- Multigate is a popular description for the FinFET in its various flavors—double gate, triple gate, quadruple gate. BSIM-CMG even models nanowire or pillar-shaped transistors with a cylindrical gate shape. These different FinFET flavors can be modeled by selecting the model switch **GEOMOD**.
- **BULKMOD**: FinFET on a bulk or an SOI substrate

[YS Chauhan et al., "FinFET Modeling for IC Simulation and Design", 2015]

BSIM-CMG Parameters

Table 11.1 Parameters that Should Be Set Before Starting the Extraction Flow

Parameter Name	Description
EOT	Gate oxide thickness
HFIN	Fin height
TFIN	Fin thickness
L	Fin length drawn
NFIN	Number of fins
NF	Number of fingers in parallel
NBODY	Channel doping concentration
BULKMOD	0: SOI; 1: bulk
GIDLMOD	0: off; I: on
GEOMOD	0: double gate; 1: triple gate; 2: quadruple gate
RDSMOD	0: internal; 1: external
DEVTYPE	0: PMOS; I: NMOS
NGATE	0: metal gate; >0: poly-gate doping



The layout of FinFET devices with

(a) 1 fin, 1 finger,

(b) 4 fins, 1 finger,

(c) 2 fins, 2 fingers.

[YS Chauhan et al., "FinFET Modeling for IC Simulation and Design", 2015]

BSIM-CMG Parameters

Name	Unit	Default	Min	Max	Description
TYPE	–	NMOS	PMOS	NMOS	NMOS = 1, PMOS = 0
BULKMOD	–	0	0	1	Substrate model selector; 0 = multi-gate on SOI substrate, 1 = multi-gate on bulk substrate.
COREMOD	–	0	0	1	Simplified surface potential selector; 0 = turn off, 1 = turn on (lightly-doped or undoped)
GEOMOD	–	1	0	3	Structure selector; 0 = double gate, 1 = triple gate, 2 = quadruple gate, 3 = cylindrical gate
CGEOMOD	–	0	0	1	For CGEOMOD = 1 only, GEO1SW = 1 enables the parameters COVS, COVD, CGSP, and CGDP to be in F per fin, per gate-finger, per unit channel width

[YS Chauhan et al., "FinFET Modeling for IC Simulation and Design", 2015]

BSIM-CMG Parameters

Name	Unit	Default	Min	Max	Description
$L^{(m)}$	m	30n	1n	–	Designed gate length
$D^{(m)}$	m	40n	1n	–	Diameter of cylinder (for GEOMOD = 3)
$TFIN^{(m)}$	m	15n	1n	–	Body (fin) thickness
$FPITCH^{(m)}$	m	80n	TFIN	–	Fin pitch
NF	–	1	1	–	Number of fingers
$NFIN^{(m)}$	–	1	>0	–	Number of fins per finger
$NGCON^{(m)}$	–	1	1	2	Number of gate contacts
$ASEO^{(m)}$	m^2	0	0	–	Source to substrate overlap area through oxide (all fingers)
$ADEO^{(m)}$	m^2	0	0	–	Drain to substrate overlap area through oxide (all fingers)
$PSEO^{(m)}$	m	0	0	–	Perimeter of source to substrate overlap region through oxide (all fingers)
$PDEO^{(m)}$	m	0	0	–	Perimeter of drain to substrate overlap region through oxide (all fingers)

[YS Chauhan et al., "FinFET Modeling for IC Simulation and Design", 2015]

BSIM-CMG Parameters

Name	Unit	Default	Min	Max	Description
ASEJ$^{(m)}$	m^2	0	0	–	Source junction area (all fingers; for bulk MuGFETs, BULKMOD = 1)
ADEJ$^{(m)}$	m^2	0	0	–	Drain junction area (all fingers; for bulk MuGFETs, BULKMOD = 1)
PSEJ$^{(m)}$	m	0	0	–	Source junction perimeter (all fingers; for bulk MuGFETs, BULKMOD = 1)
PDEJ$^{(m)}$	m	0	0	–	Drain junction perimeter (all fingers; for bulk MuGFETs, BULKMOD = 1)
COVS$^{(m)}$	F or F/m see CGEO1SW	0	0	–	Constant gate to source overlap capacitance (for CGEOMOD = 1)
COVD$^{(m)}$	F or F/m see CGEO1SW	CVOS	0	–	Constant gate to drain overlap capacitance (for CGEOMOD = 1)

[YS Chauhan et al., "FinFET Modeling for IC Simulation and Design", 2015]

BSIM-CMG Parameters

Name	Unit	Default	Min	Max	Description
$CGSP^{(m)}$	F or F/m see CGEO1SW	0	0	–	Constant gate to source fringe capacitance (for CGEOMOD = 1)
$CGDP^{(m)}$	F or F/m see CGEO1SW	0	0	–	Constant gate to drain fringe capacitance (for CGEOMOD = 1)
$CDSP^{(m)}$	F	0	0	–	Constant drain to source fringe capacitance
$NRS^{(m)}$	–	0	0	–	Number of source diffusion squares (for RGEOMOD = 0)
$NRD^{(m)}$	–	0	0	–	Number of drain diffusion squares (for RGEOMOD = 0)
$LRSD^{(m)}$	m	L	0	–	Length of the source/drain

Note: Instance parameters with superscript $^{(m)}$ are also model parameters.

[YS Chauhan et al., "FinFET Modeling for IC Simulation and Design", 2015]